VIX Futures Term-Structure and Currency Returns

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Abstract

I use the VIX futures term-structure to build currency risk factors and show that they are priced in the cross-section of currency returns and help predict currency returns. Currencies with greater exposure to the level and curvature factors earn a lower risk premium, acting as insurance, while those more exposed to the slope factor earn a higher premium, and the premia sit in the variance-risk-premium component, not in news about expected future volatility. A Bayesian factor-selection procedure places the term-structure factors in the stochastic discount factor with high posterior probability. I rationalize these results with a two-agent preferred-habitat model in which experts have horizon-specific volatility hedging demands and arbitrageurs face funding and margin costs, and currency premia depend on each country's share of global hedging demand relative to the US.

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1. Introduction

I build currency risk factors from the VIX futures term-structure and show that they are priced in the cross-section of currency excess returns and help predict currency returns. A VIX futures contract settles in cash on the value of the VIX, the 30-day implied volatility of the S&P500, on its expiry date. The buyer gains when the VIX at expiry is above the futures price at which they bought the contract, and loses when it is below. Contracts expire from one to nine months ahead, so investors can buy protection against a rise in implied volatility at each of these horizons, and the term-structure of futures prices shows how investors price volatility risk at each horizon. I summarize the term-structure with its first three principal components, which explain about 99% of its variation. The first is the level, which moves all maturities together and jumps in periods of stress such as 2008, 2011 and 2020. The second is the slope, which falls when short-term volatility rises relative to long-term volatility and rises when stress fades. The third is the curvature, which moves the middle of the curve against its two ends.

Currencies differ in how much they move with innovations in these three components. Each month I estimate each currency's exposure (beta) to the innovations over the previous 48 months, controlling for the Dollar factor, sort currencies into five portfolios on these betas, and hold the portfolios for one month. The high-minus-low portfolio of each sort is a tradable currency risk factor, and the three factors proxy for the global component of the volatility term-structure. Over the baseline sample, the level and curvature factors earn about -7% per year, with Sharpe ratios of -0.9 and -1.0. The slope factor earns about 7% per year, with a Sharpe ratio of 0.9. The return spread lines up with the betas. Currencies with the highest level and curvature betas earn the lowest returns, and currencies with the lowest slope betas earn the lowest returns. Transaction costs take 0.2% to 0.3% a year off these returns.

In Fama and MacBeth (1973) regressions, each factor carries a significant price of risk in both anomaly portfolio and country-level cross-sections. A currency with a beta of one on the level factor earns about 6% to 8% less per year, and one with a beta of one on the curvature factor about 4% to 7% less, so these two factors act as insurance. A currency with a beta of one on the slope factor earns about 7% to 9% more per year. The factors are priced alongside the Carry and Dollar factors of Lustig et al. (2011), and they add to them. In the country cross-section, adding a term-structure factor to Carry and Dollar raises the cross-sectional R^2 from 0.67 to as much as 0.90, and in the portfolio and advanced-country cross-sections the GRS test does not reject that the pricing errors are zero. The factors are not the carry trade in another form. Their loadings on the Carry factor are well below

one, and both legs of every factor hold a mix of advanced and emerging-market currencies in almost every month.

These premia pay for the cost of hedging volatility, not for news about volatility itself. A move in the VIX futures curve can reflect news about how volatile the market will be, or news about the premium investors pay to hedge that volatility. I separate the two by forecasting S&P500 realized volatility with a heterogeneous autoregressive model, in the spirit of Corsi (2009), and splitting each factor into the part that the revisions in these forecasts explain and the part that they leave. News about expected volatility explains only about 6% of the variance of each factor. I then build currency portfolios that mimic each part. Portfolios that mimic the premium news earn almost all of the factors' premium, while portfolios that mimic the news about expected volatility earn at most 0.6% per year and are never distinguishable from zero. Currency investors are therefore paid for exposure to the price of volatility protection at each horizon.

I develop a model with two types of agents to rationalize why currencies carry this risk. Experts in every country, such as pension funds, insurers and asset managers, hold assets whose returns move with shocks to global volatility. They hedge with volatility claims at three horizons, and these claims trade only in one country, the Home economy. Their mandates fix how much hedging they need at each horizon, as for preferred-habitat investors in the bond market. Arbitrageurs take the other side, but these positions use up their capital and margin, and the cost depends on the horizon. The price of volatility risk at each horizon is the compensation arbitrageurs need for holding these positions. Countries differ in their share of global hedging demand, and the difference sets how much each currency moves with volatility shocks. When a shock hurts the Home investor, the Home discount factor responds strongly, while the discount factor of a country that hedges less than Home responds less, so that country's currency loses value against the Home currency in exactly those states. Holding the currency leaves the Home investor with volatility risk that the foreign country does not share, and the currency pays a higher expected excess return in compensation. The model gives each currency's expected excess return directly, and at every horizon it rises with the gap between the Home share and the country's share of hedging demand, whatever the sign of the price of risk. The model also gives the estimated signs a meaning. The sign of each price of risk tells whether hedgers are buying or selling protection at that horizon, which no one observes directly. Read through the model, the three estimated signs say that experts buy protection at all three horizons.

The data on the VIX futures market match these assumptions. Institutional investors

whose liabilities and portfolios are sensitive to volatility, such as pension funds, insurers and multi-asset managers, use VIX futures to hedge volatility at specific horizons. VIX futures are by far the largest futures market for volatility. In 2025, open positions in VIX futures were worth about $7.7 billion per day on average, against about $0.3 billion for VSTOXX futures, the largest volatility futures market outside the US. Trading shows the same gap, with about $3.9 billion of VIX futures traded per day, or about $970 billion over the year, against about $0.1 billion per day and $27 billion over the year for VSTOXX futures. A deep market for volatility claims exists only in the US, so the US plays the role of the Home economy, and I take the perspective of a US investor throughout. The positions also match the roles of the two types of agents. Figure 1 plots net positions from the CFTC Traders in Financial Futures report. Between July 2010 and September 2026, asset managers are net long in 82% of the months and dealers in 86%, while leveraged funds are net short in 93% of the months, in line with Cheng (2019). Dealers appear to buy VIX futures to hedge the long-volatility products they sell to customers, such as VIX exchange-traded notes and options, in which case their position passes end-investor demand on to the futures market. Hedging demand is also concentrated at particular horizons. Figure 2 shows that the two nearest contracts hold about two thirds of open interest, and contracts beyond six months about 3%. Demand concentrated at specific horizons and absorbed by a different group of traders, mainly leveraged funds, is the preferred-habitat structure of the model, in which experts' mandates create pressure at particular horizons and arbitrageurs are paid for absorbing it.

The same innovations also help predict currency returns. Following Londono and Zhou (2017), I regress next-month currency excess returns on the innovations, controlling for the interest rate differential and the lagged return, in pooled and currency fixed-effects regressions. All three innovations predict higher returns, and the slope gives the strongest effect. A one standard deviation steepening of the curve predicts returns higher by about 26 basis points over the next month, or 3.2% annualized. In April 2020, as pandemic stress left the front of the curve faster than the back, the slope rose by four standard deviations, and the estimates imply an appreciation of the average currency against the dollar of about 1.1% over the following month. As usual for exchange rates at short horizons, the share of the variation the innovations explain is small.

Several further results support these findings. A Bayesian model selection exercise, following Bryzgalova et al. (2023), lets the term-structure factors compete with Carry and Dollar for a place in the stochastic discount factor. The probability that at least one of them belongs there is 0.98 to 0.99, and the probability that two or more belong

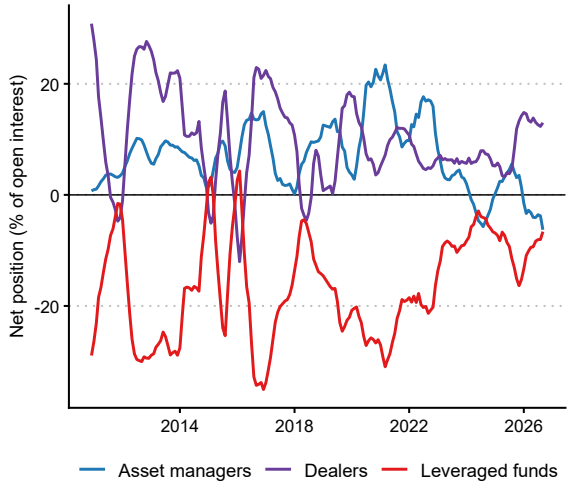


Figure 1: Net positions in VIX futures by trader type. Net long minus short futures positions of asset managers, dealers and leveraged funds as a percentage of total open interest, six-month moving average of month-end reports, CFTC Traders in Financial Futures, July 2010 to September 2026.

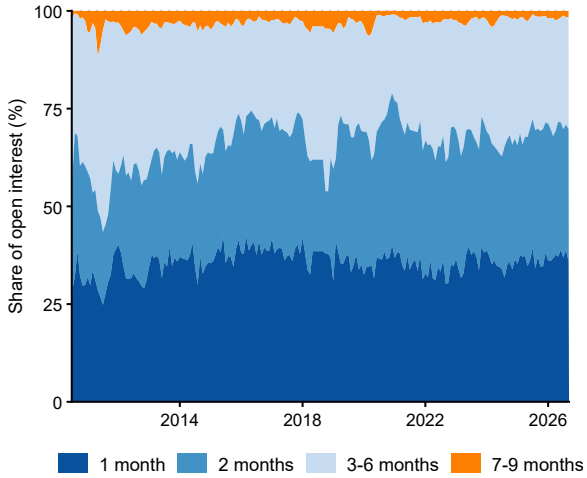


Figure 2: VIX futures open interest by expiry. Share of total open interest held in the front contract, the second contract, contracts three to six and contracts seven to nine, monthly averages of daily shares, LSEG Datastream, July 2010 to September 2026.

there is about 0.8. Using the US curve as a proxy for a global volatility term-structure is reasonable. Option-implied curves for the Euro Stoxx 50, the FTSE 100 and the SMI show that the level of the US curve is almost entirely global and its slope mostly global, while its curvature is mostly local. The results are similar with three or ten portfolios instead of five. On a longer sample of synthetic VIX futures, which I build from S&P500 options available from 1996, the level and slope factors are still priced, while the curvature result and the predictive results hold only in the traded futures sample.

The paper adds to the work on risk factors in the cross-section of currency returns. Lustig and Verdelhan (2007) sort currencies into portfolios on their interest rates and show that high interest rate currencies earn higher returns because they are more exposed to US consumption growth risk. Lustig et al. (2011) show that a Carry factor and a Dollar factor explain most of the cross-section of currency portfolio returns, and sorts on momentum (Menkhoff et al., 2012b) and value (Menkhoff et al., 2017) add further dimensions. Verdelhan (2018) shows that the Carry and Dollar factors account for a large share of the systematic variation in bilateral exchange rates, and Hassan and Zhang (2021) review the economics behind these premia and argue that they come from persistent differences in how exposed countries are to global shocks, which follow from country characteristics such as size, centrality in world trade and reliance on commodity exports. I add three tradable factors built from equity volatility derivatives, which are priced alongside Carry and Dollar, and I test them on country excess returns as well as on

portfolios.

The closest papers link currency returns to volatility and crash risk. Menkhoff et al. (2012a) show that global FX volatility risk is priced in the cross-section of currency returns and explains the returns to carry trades. Brunnermeier et al. (2008) show that carry trades are exposed to crash risk and lose money when risk appetite falls, Farhi et al. (2015) use currency options to show that crash risk accounts for a large part of carry trade returns, and Chernov et al. (2018) show that jumps are an important part of the risk in currency returns. Farhi and Gabaix (2016) explain exchange rates with rare disasters, in the spirit of Barro (2006). On the time-series side, Della Corte et al. (2016) show that currency volatility risk premia predict exchange rates, and Londono and Zhou (2017) show that the US equity variance risk premium predicts currency returns and helps explain the forward premium puzzle. Equity tail and downside risk also help explain the cross-section of currency returns. Lettau et al. (2014) and Dobrynskaya (2014) show that high-interest currencies lose more when the stock market falls, and Fan et al. (2022) build an equity tail risk factor from S&P500 options that is priced in the cross-section of currency returns. These papers use one measure of volatility or tail risk at a time. I build on them by using the whole term-structure of volatility, so that volatility risk can carry a different price at each horizon, and by showing that the currency premia come from news about the price of hedging volatility rather than from news about volatility.

The paper also relies on the evidence that volatility is a priced risk. Drechsler and Yaron (2010) show how a variance premium arises in equilibrium, and Bollerslev et al. (2009) show that the variance risk premium predicts aggregate stock market returns. Volatility claims carry premia that change over time and differ across maturities. Egloff et al. (2010) show that more than one factor drives the term-structure of variance swap rates, Johnson (2017) shows that movements in the shape of the VIX term-structure mostly reflect compensation for volatility risk, Dew-Becker et al. (2017) study how the price of variance risk differs across maturities, and Aït-Sahalia et al. (2020) show that investors' willingness to insure against volatility depends on the maturity and rises after market downturns. Zhou (2018) reviews this evidence and shows that the variance risk premium predicts short-horizon returns in equity, bond, credit and currency markets, and links it to changes in macroeconomic uncertainty. The model brings these premia to currencies. Hedging demand follows the preferred-habitat model of Vayanos and Vila (2021), and the arbitrageurs who supply volatility claims face capital and margin costs, as in Gârleanu and Pedersen (2011) and He and Krishnamurthy (2013). As in Lustig et al. (2011) and Verdelhan (2018), currency premia come from differences in how countries' discount factors

load on global risk, and the premium takes the form of Backus et al. (2001). In Gourinchas et al. (2025), preferred-habitat demand for bonds moves exchange rates. In my model, the demand is for volatility hedges, and currency data alone reveal its direction at each horizon.

The rest of the paper is organized as follows. Section 2 presents the model, Section 3 describes the data and the construction of the factors, Section 4 tests whether the factors are priced, Section 5 splits them into news about expected volatility and news about the premium, Section 6 studies return predictability, and Section 7 concludes.

2. Model

I develop a simple stylized two-agent model linking horizon-specific volatility hedging demand to currency risk premia in a world with a large Home economy and smaller Foreign economies. Experts in all countries act as preferred-habitat investors with institutional hedging mandates for volatility at multiple horizons, while arbitrageurs are risk-neutral intermediaries who absorb residual volatility positions subject to maturity-dependent balance-sheet and margin costs. Only the Home economy has a deep market for tradable volatility-linked contracts. These contracts load on the global component of volatility term-structure risk, allowing Foreign experts to access global volatility shocks through positions in Home claims, accepting the local Home component as unavoidable basis risk. When expert hedging demand must be absorbed by arbitrageurs, scarcity in volatility claims arises and generates equilibrium prices of risk for level, slope, and curvature shocks. Differences in countries' institutional hedging mandates then translate into heterogeneous stochastic discount factor (SDF) loadings and, ultimately, into currency risk premia. The market for volatility claims and the market for currencies are segmented, allowing the arbitrageur price of a Home claim to differ from the value a country kernel would assign the same payoff, generating the cross-section of currency returns. This is similar to the investor clientele perspective of Vayanos and Vila (2021) and Gourinchas et al. (2025).

The modeling framework builds on several strands of the asset-pricing literature. First, I adopt the preferred-habitat and demand-based equilibrium logic of Vayanos and Vila (2021), in which investor demand concentrated in particular segments interacts with intermediary balance-sheet constraints to generate risk premia. Second, the supply side of the model follows the intermediary-based asset-pricing tradition of Gârleanu and Pedersen (2011) and He and Krishnamurthy (2013), where arbitrageurs face maturity-dependent capital costs and absorb investor demand at a price that reflects their risk-bearing capacity. Investors' desire to hedge volatility shocks across horizons is consistent with the broader literature

documenting that volatility is an important priced risk, such as Drechsler and Yaron (2010). Finally, in linking volatility hedging to exchange rates, I follow the international asset-pricing logic of Lustig et al. (2011) and Verdelhan (2018), where currency premia arise from differences in countries' SDF loadings on global risk factors.

Taken together, this is a preferred-habitat account of the volatility term-structure, in which horizon specific hedging mandates are intermediated by arbitrageurs at a balance-sheet cost, embedded in a SDF framework for exchange rates, and in which segmentation between the two markets turns cross-country heterogeneity in those mandates into currency risk premia.

2.1. Intuition

Experts, such as institutional investors, insurers, and pension funds, face volatility-sensitive liabilities and portfolio exposures that give rise to hedging demand at specific horizon buckets. They desire insurance against unexpected movements in volatility across different horizons. In the model, these hedging needs are represented by three maturity buckets of volatility-linked claims, corresponding to shifts in the level, slope, and curvature of the volatility term-structure. Because the Home economy hosts the only deep tradable volatility term-structure market, experts in all countries use Home claims to hedge their exposure to the global volatility factor. Home claims load on both the global factor and a local Home component. Experts target the global component and accept the local Home noise as basis risk, since it is the only available hedging vehicle. Local volatility components specific to each Foreign country are unhedgeable, as no foreign volatility-linked claims are traded.

Arbitrageurs intermediate these trades but face maturity-dependent funding and margin costs. When hedging demand is concentrated at specific horizons, those volatility buckets become scarce and command nonzero prices of risk. Because the claims are in zero net supply, the arbitrageurs stand on the opposite side of the experts in every bucket, and the price of risk is the compensation they require for the position they are left holding. Crucially, countries differ in the share of aggregate global hedging demand their experts contribute at each volatility horizon, generating heterogeneous SDF loadings on the global volatility factor across countries. Exchange rates inherit these differences. Two countries whose institutional investors hold different shares of aggregate hedging demand at a given horizon are exposed to global volatility shocks in different degrees, and this wedge is compensated in the foreign exchange market through currency risk premia.

Both parts of the mechanism move over time. Intermediary tightness varies with funding conditions, and hedging mandates vary as institutional portfolios and the liabilities

behind them change.

2.2. Assets

Time is discrete and the one-period economy is repeated, with the date-t state observed by all agents at t and market clearing holding period by period. Agents solve their problems period by period without regard to future states, similar to the instantaneous mean-variance preferences imposed by Vayanos and Vila (2021). There is a large Home economy, denoted by H, and $N-1$ smaller Foreign economies, so that there are N countries in total. Each country c has a domestic risky asset and a domestic riskless bond paying an interest rate known at t. The riskless bond is also traded across countries, so that a position in a foreign currency is a position in the foreign riskless bond financed at the Home riskless rate. I assume that the payoff space that can be reached by trading currencies and bonds internationally is complete, as in the asset market view of the exchange rate in Backus et al. (2001). Alongside these sits a third and smaller market, in volatility-linked claims, which only the Home economy hosts.

Each country c has a volatility term-structure, and movements in it are summarised by three components, corresponding to a parallel shift in the whole curve, a rotation, and a change in its convexity. I index these buckets by $k \in \{L, S, C\}$ for level, slope and curvature, and collect the one-period innovations in the vector $\Delta TS^{(c)}_{t+1} = (\Delta TS^{(c)}_{L,t+1}, \Delta TS^{(c)}_{S,t+1}, \Delta TS^{(c)}_{C,t+1})' \in \mathbb{R}^3$, so that $\Delta TS^{(c)}_{k,t+1}$ is the date-$t+1$ innovation in bucket k of country c's curve. Three buckets rather than a continuum is a simplification made by the model. The innovations in each country's term-structure can be decomposed into a global and a local component,

$$\Delta TS^{(c)}_{t+1} = \varphi_c \, \Delta TS^{g}_{t+1} + \psi_c \, \Delta TS^{\mathrm{loc},(c)}_{t+1}, \tag{1}$$

where $\varphi_c \geq 0$ measures how much country c loads on the global volatility term-structure shock vector $\Delta TS^g \in \mathbb{R}^3$ and $\psi_c \geq 0$ the loading on the local component, with $\varphi_H > 0$ so that Home claims carry global exposure at all. All three vectors have mean zero conditional on date-t information, the decomposition holds bucket by bucket, and the global and local components are mutually orthogonal. The curve loadings φ_c and ψ_c describe the factor structure of the volatility term structures rather than preferences or institutional structure, so they are constant.

The covariance structure of the global component of the volatility-linked claims is characterized by $\Sigma_{\Delta TS^g} = \mathrm{Var}_t(\Delta TS^g_{t+1})$ and Σ_{loc} for that of the local component. Both are considered to be diagonal, with entries σ^2_k and $\sigma^2_{\mathrm{loc},k}$, so that a shock to one bucket is

uncorrelated with a shock to another. This allows the shocks to be read one at a time rather than a single composite measure.

The Home economy hosts the only deep liquid market for traded volatility-linked claims, and there are three of them, one per bucket. The claim on a bucket k is a one-period contract, entered at t and settled at $t+1$ in Home currency, and in zero net supply, so that for every unit held long there is a unit held short. It is a zero-cost contract in the sense that no upfront payment is required, but it is not free to carry, since margin must be posted against the position, as for a futures contract.

The claim's excess payoff is

$$R^{TS}_{k,t+1} = \lambda_{k,t} + \Delta TS^{(H)}_{k,t+1}, \qquad \lambda_{k,t} \equiv E_t\left[R^{TS}_{k,t+1}\right], \tag{2}$$

the sum of a conditional mean known at t and the mean-zero innovation to bucket k of the Home curve. Therefore, the payoff is proportional to the curve innovation, so a position in the claim is a position in the curve and nothing else, which is what makes these contracts a vehicle for volatility exposure. And $\lambda_{k,t}$, defined here simply as the expected excess payoff, is the compensation the market requires for carrying the position. Its value is not free, and it is determined in equilibrium. The drift and the innovation are kept apart throughout, since the drift is what anyone holding the claim weighs against the cost of carrying it, while the innovation is what the kernels load on and what the betas are measured against.

Positions in the claims are measured in units of global exposure rather than in contracts. Since each claim delivers φ_H units of global exposure by (1), a position of $q \in \mathbb{R}^3$ global units is held through q/φ_H claims and delivers an expected excess payoff of $\lambda'_{\Delta TS,t} q/\varphi_H$, where $\lambda_{\Delta TS,t}$ collects the three $\lambda_{k,t}$.

2.3. Agents

The model is composed by preferred-habitat experts and by arbitrageurs. Experts in each country perform asset allocation in risky and riskless assets, and wish to hedge their volatility exposures. Arbitrageurs in the Home economy supply the volatility claims to experts, but face capital and margin costs. In the VIX futures market, hedging demand comes from the long side, which pays the premium, and the arbitrageurs correspond to the short side, which supplies liquidity and earns it (Cheng, 2019).

Experts in the Home and Foreign economies begin by allocating wealth between risky assets and the domestic riskless asset using standard mean–variance principles. This first-stage portfolio choice, which I do not model explicitly, generates overall portfolio exposure to global volatility term-structure shocks, because the returns of risky assets co-move with

innovations in the volatility term-structure. Their liability structures and institutional constraints impose horizon-specific hedging mandates $h_t^{(c)} = (h_{L,t}^{(c)}, h_{S,t}^{(c)}, h_{C,t}^{(c)})' \in \mathbb{R}^3$ for the level, slope and curvature of the global component of the volatility term-structure, where each country's mandate reflects its own institutional structure and varies over time as those portfolios and liabilities change. The mandates are stated in units of exposure to the global component ΔTS^g, and not to a country's own curve, and the reason is the asset structure rather than a preference. The global component is the only part of the volatility term structure that every expert can reach, since Home claims are the only traded vehicle. For a Foreign expert the local component of its own curve, $\psi_c \Delta TS^{\text{loc},(c)}$, is unhedgeable, because no foreign volatility-linked claims exist. For every expert the local Home component $\psi_H \Delta TS^{\text{loc},(H)}$ arrives alongside the global factor whenever Home claims are held, but it is unintended basis risk rather than a hedging target.

In this respect, experts play the role of preferred-habitat investors in the volatility term-structure, analogous to the pension funds and insurance companies that Vayanos and Vila (2021) place in specific maturity segments of the bond term-structure. The habitat itself is the horizon: an expert's demand is concentrated in particular volatility buckets rather than spread across the curve, and that concentration is what makes horizon-specific scarcity possible. The elasticity of that demand is a separate matter, and in the baseline I take it to be zero, so that mandates are inelastic with respect to the price of protection[2].

Experts therefore choose $q_{TS,t}^{(c)} \in \mathbb{R}^3$ to align their global volatility term-structure exposure with the mandate by solving a tracking-error minimisation problem,

$$\min_{q_{TS,t}^{(c)}} \left\| q_{TS,t}^{(c)} - h_t^{(c)} \right\|^2, \tag{3}$$

whose solution is

$$q_{TS,t}^{(c)*} = h_t^{(c)}. \tag{4}$$

The mandate $h_t^{(c)}$ is therefore the expert's equilibrium position in global volatility term-structure space, and it is held whatever the claims cost, given the inelastic demand assumption. Minimising tracking error against a mandate is the delegated-risk-management logic of Gârleanu and Pedersen (2011), in which the agent who holds the position is not the agent whose marginal utility is at stake, and the preferred-habitat interpretation of Vayanos and Vila (2021), in which portfolio exposures induce hedging needs concentrated

[2]Hedging demand is assumed inelastic for simplicity, but the model does not require it. In Appendix A I develop the elastic case in full.

in particular directions[3].

Arbitrageurs are risk-neutral intermediaries operating in the Home economy who take net positions in TS to absorb demand imbalances, but face maturity-dependent capital and margin costs

$$C_t(x) = \frac{\kappa_t}{2} x'Wx, \tag{5}$$

where $x \in \mathbb{R}^3$ is the vector of arbitrageur positions in the three volatility-linked claims. The scalar $\kappa_t > 0$ is intermediary tightness, which scales the overall cost of supplying volatility exposure and captures the balance-sheet, margin and capital constraints that make doing so expensive, and it varies with funding conditions and with the risk-bearing capacity of the arbitrageurs who supply that exposure. A high κ_t means arbitrageurs are capital or liquidity constrained, while a low κ_t means that volatility hedging is cheaply absorbed. The matrix $W \succ 0$ encodes maturity-specific supply frictions, with W_{kk} governing the cost of supplying bucket k individually. I take W to be diagonal, so that there are no cross-horizon cost interactions. With cross-horizon frictions the price of risk in a bucket would depend on the composition of hedging demand across all buckets, so the sign of a bucket's price would be the sign of $(WH_t)_k$ rather than of the demand in that bucket alone, and the level, slope and curvature estimates could not be read one at a time.

The quadratic specification yields the linear supply function

$$x_t = \kappa_t^{-1} W^{-1} \lambda_{\Delta TS,t}, \tag{6}$$

where $\lambda_{\Delta TS,t} \in \mathbb{R}^3$ is the equilibrium price of risk vector for the Home volatility-linked claims. The cost is symmetric in the sign of the position, so arbitrageurs require compensation for holding a claim and for being short it alike, as in the margin structure of Gârleanu and Pedersen (2011), where both long and short positions consume capital.

2.4. Market Clearing and Stochastic Discount Factor (SDF)

Aggregate expert demand across all countries is the elementwise sum of country mandates,

$$H_t = \sum_c h_t^{(c)} \in \mathbb{R}^3, \tag{7}$$

where $H_{k,t} > 0$ in a bucket corresponds to experts collectively wanting to be long that volatility bucket and $H_{k,t} < 0$ to experts collectively wanting to be short. Since mandates are expressed in global volatility term-structure exposure units and each Home claim

[3]The gap between this objective and a household's own first-order condition is the delegation friction, and Appendix A makes the delegation explicit and derives what the baseline here assumes.

delivers φ_H units of global exposure, the aggregate demand in Home claim units is H_t/φ_H.

The volatility-linked claims are derivative contracts in zero net supply[4]. Market clearing therefore requires that the positions of experts and arbitrageurs sum to zero, since the two are on opposite sides of the same contract,

$$x_t + \frac{H_t}{\varphi_H} = 0, \qquad x_t = -\frac{H_t}{\varphi_H}. \tag{8}$$

This is the clearing convention of the preferred-habitat literature the model draws on. Vayanos and Vila (2021) clear each maturity by requiring that the positions of arbitrageurs and habitat investors sum to zero, and Gourinchas et al. (2025) do the same in a two-country setting, with the consequence in both that the price of risk is proportional to the arbitrageurs' own exposure and therefore carries the opposite sign to habitat demand.

Arbitrageur positions are constrained by balance-sheet costs. Using the arbitrageur supply function in Equation 6, market clearing implies

$$\kappa_t^{-1} W^{-1} \lambda_{\Delta TS,t} = -\frac{H_t}{\varphi_H}. \tag{9}$$

Solving yields the equilibrium price of risk for the Home volatility-linked claims,

$$\lambda_{\Delta TS,t} = -\kappa_t \, W \left(\frac{H_t}{\varphi_H} \right) \in \mathbb{R}^3, \qquad \lambda_{k,t} = -\frac{\kappa_t \, W_{kk} \, H_{k,t}}{\varphi_H}. \tag{10}$$

The minus sign comes from the clearing condition. Arbitrageurs stand opposite the experts, so a bucket in which experts are net buyers of protection is one whose claim has to carry a negative expected payoff before an arbitrageur will supply it. The Home loading φ_H governs how efficiently Home claims serve as a vehicle for global volatility term-structure hedging. A higher φ_H means each claim delivers more global exposure per unit, so fewer claims are needed to satisfy the aggregate mandate, reducing the demand pressure on arbitrageur balance sheets and lowering the equilibrium price of risk[5].

This price of risk vector $\lambda_{\Delta TS}$ is common to all countries. It is the single global

[4] In bond markets a nonzero exogenous net supply is standard, representing the outstanding stock issued by governments or corporations. Volatility-linked claims are derivatives, so they are zero-sum by construction: every long position is matched by an equal short position and aggregate net supply is zero. The distinction matters here rather than being a technicality, since it is what fixes the direction of (8) and with it the sign of the price of risk.

[5] Setting $\varphi_H = 1$ is a normalisation available at any time, by rescaling ΔTS^g so that the Home country has unit loading, and it reduces Equation 10 to $\lambda_{\Delta TS,t} = -\kappa_t W H_t$ with claim units coinciding with global exposure units. I do not impose it, because φ_H is carried through the attenuation argument in later sections and it is clearer to see where it enters.

price of volatility term-structure risk, determined by the arbitrageur, who is the marginal investor in the Home volatility claims market. An increase in the absolute size of aggregate hedging demand raises the amount of risk arbitrageurs must absorb and therefore raises the absolute price of risk. The relation is structurally analogous to the preferred-habitat and intermediary-based pricing mechanisms of Vayanos and Vila (2021), Gârleanu and Pedersen (2011), and related literature, in which prices of risk are set by investor demand imbalances relative to intermediary risk-bearing capacity.

While $\lambda_{\Delta TS,t}$ is a single global price vector, the SDF loading on volatility term-structure risk differs across countries because experts hold heterogeneous positions determined by their mandates. Country c's SDF loading on global volatility term-structure innovations reflects the country's contribution to the aggregate global hedging demand[6]. Define the relative mandate vector

$$\omega_{c,t} = h_t^{(c)} \oslash H_t \in \mathbb{R}^3, \qquad \omega_{c,k,t} = \frac{h_{k,t}^{(c)}}{H_{k,t}}, \tag{11}$$

where $\oslash$ denotes elementwise division, so that $\omega_{c,k,t}$ is country c's share of aggregate hedging demand in each bucket and sums to one across countries at every date.[7] Country c's SDF loading on global volatility term-structure is then

$$\theta_{c,t} = \omega_{c,t} \odot \lambda_{\Delta TS,t} \in \mathbb{R}^3, \qquad \theta_{c,k,t} = \omega_{c,k,t}\, \lambda_{k,t}, \tag{12}$$

where $\odot$ denotes the elementwise (Hadamard) product. One global price of risk per bucket, common to all countries, scaled by one country-specific relative mandate per bucket. This separates global pricing from local institutional structure, and both move, the price with intermediary conditions and aggregate demand, and the shares because both the numerator and the denominator of Equation 11 move. Holding the shares fixed would require every country's mandate to move in exact proportion to the aggregate, which would rule out the reallocation of hedging demand across countries. A country whose institutions

[6] Appendix A derives $\theta_{c,t}$ from a household that delegates its hedging to a constrained manager, and recovers Equation 12 up to a positive scale. Since $\lambda_{\Delta TS,t}$ carries κ_t, (12) puts intermediary tightness into every country kernel by construction. In the appendix, I show that if hedging demand responds to price, then intermediary tightness is naturally transmitted to the loadings.

[7] Two conditions on Equation 11 are imposed here and used throughout. Every country's mandate in a bucket has the same sign as the aggregate, so the shares are genuinely positive shares, and the aggregate $H_{k,t}$ is bounded away from zero, so they are defined. The first does real work and is what makes the ordering the beta sort produces agree with the ordering of the premium. The second is a condition on the exposition rather than on the economics, since as $H_{k,t}$ approaches zero the shares diverge but the price of risk vanishes at the same rate, the product of the two that reaches the country kernels stays finite, and the bucket simply ceases to be priced.

demand a larger share of a given bucket than an equal split would give, $\omega_{c,k,t} > 1/N$ across the N countries, has a kernel more sensitive to innovations in that bucket than the average. If mandates were homogeneous across countries then $\omega_{c,t} = \mathbf{1}/N$ for every c, every loading would collapse to the same constant, and the model would reduce to a single-agent benchmark with no cross-section of currency premia at all.

Equation 12 is a reduced-form bridge rather than a first-order condition of the agents behind the country SDFs, so it is worth being explicit about what carries the shock across. A country's discount factor has to load on the shocks that country is exposed to, and the exposure here runs through its experts. Their first-stage portfolio choice already leaves them exposed to global volatility term-structure innovations, and the mandate $h_t^{(c)}$ is the position in that space they hold as a result, carried through Home country volatility claims. What fixes how much of the shock reaches the country is therefore the mandate, since a country contributing a larger share of aggregate hedging demand carries more of it, and that is why the loading is the share times the price. Heterogeneous mandates therefore generate heterogeneous SDF loadings, and exchange rates inherit the difference, since two countries whose institutions hold different shares at a given horizon are exposed to the global shock in different degrees and that wedge turns into a currency risk premium. Exposure and pricing come apart here, and this is where an assumption about market access does its work. The arbitrageur is the marginal investor in the volatility claims and sets $\lambda_{\Delta TS,t}$, while the experts take that price as given and hold the quantity their mandate dictates, so each market has its own clientele. That is the structure of Vayanos and Vila (2021), where a habitat investor holds the bond of its own maturity and no other, and of Gourinchas et al. (2025), whose currency and bond markets are populated by different investor clienteles and bridged only partly by arbitrageurs.

Volatility term-structure risk is not the only shock a country is exposed to. I therefore include two standard reduced-form sources of discount-factor volatility alongside it. A global shock $u_{g,t+1}$ captures worldwide fluctuations in economic activity, financial conditions or global marginal utility, and a country-specific shock $u_{c,t+1}$ captures domestic variation in consumption, productivity or financial frictions. Neither is microfounded within the hedging mechanism above. They are the usual non-volatility factors of international SDF specifications such as Lustig et al. (2011), and they are here so that the volatility block is one component of the kernel rather than the whole of it.

Country c's log nominal stochastic discount factor is then

$$-m_{c,t+1} = i_{c,t} + a_{c,t} + \delta_{c,t}\, u_{c,t+1} + \gamma_{c,t}\, u_{g,t+1} + \theta'_{c,t}\, \Delta TS^{g}_{t+1}, \tag{13}$$

where $i_{c,t}$ is the riskless rate, $a_{c,t} = \frac{1}{2}\mathrm{Var}_t(m_{c,t+1})$ the convexity term that makes it so under conditional lognormality, $\delta_{c,t}$ and $\gamma_{c,t}$ the scalar loadings on the country-specific and global shocks, and $\theta_{c,t}$ the loading on global volatility term-structure innovations from Equation 12. All loadings are known at t and the shocks realise at $t+1$, so the SDF is conditionally affine in the state, which is the form used in the literature. The three shocks are mutually orthogonal conditional on date-t information, which keeps the volatility block additively separable from the rest of the kernel. The local Home component is treated as noise in the observable proxy.

2.5. Currency Pricing, Conditional Betas, and Risk Factors

Following the asset market view of exchange rates, as in Backus et al. (2001), the log change in the nominal exchange rate (s_t) between the Foreign and Home currency is equal to the difference of the log nominal SDFs of the two countries, where the exchange rate is expressed in Foreign currency units per one unit of Home currency,

$$\Delta s_{F,t+1} = m_{H,t+1} - m_{F,t+1}, \tag{14}$$

which requires markets to be complete within the currency and bond space that the agents behind the kernels actually trade. It is not completeness over every payoff in the economy. The volatility claims lie outside that space, since they are priced by the arbitrageur and the experts' position in them is fixed by mandate rather than chosen against the price.

The excess return to a Home investor who holds foreign currency is therefore

$$\begin{aligned} rx_{F,t+1} &= -\Delta s_{F,t+1} + i_{F,t} - i_{H,t} \\ &= a_{H,t} - a_{F,t} - \overbrace{\delta_{F,t}\, u_{F,t+1}}^{\text{Foreign country shocks}} + \overbrace{\delta_{H,t}\, u_{H,t+1}}^{\text{Home country shocks}} \\ &\quad + \underbrace{\big(\theta_{H,t} - \theta_{F,t}\big)' \Delta TS^{g}_{t+1}}_{\text{Global volatility term-structure shock}} + \underbrace{\big(\gamma_{H,t} - \gamma_{F,t}\big) u_{g,t+1}}_{\text{Other global shocks}}. \end{aligned} \tag{15}$$

The currency excess return for a Home investor who invests in foreign currency F is driven by four components. The country-specific shocks capture all domestic risk not mediated by the global volatility term-structure, including each country's own local volatility component $\psi_c\, \Delta TS^{\mathrm{loc},(c)}$, which no traded claim reaches outside the Home country and therefore enters here rather than in the volatility block. The country-specific components are diversifiable in the cross-section of currencies. The global volatility term-structure shock is the component driven by heterogeneous institutional hedging mandates and is the focus of this paper.

The remaining global shocks $(\gamma_{H,t} - \gamma_{F,t})\, u_{g,t+1}$ capture non-volatility sources of global risk. Substituting Equation 12, the global volatility term-structure component becomes:

$$\Big(\theta_{H,t} - \theta_{F,t}\Big)' \Delta TS^{g}_{t+1} = \Big[\Big(\omega_{H,t} - \omega_{F,t}\Big) \odot \lambda_{\Delta TS,t}\Big]' \Delta TS^{g}_{t+1}. \tag{16}$$

The k-th element of the loading vector reads $\lambda_{k,t}(\omega_{H,k,t} - \omega_{F,k,t})$: one global price of risk per bucket, multiplied by the cross-country mandate wedge. Two countries with different institutional mandates earn different currency risk premia proportional to the difference in their relative contribution to global hedging demand. The volatility-driven contribution to the excess return is therefore proportional to this mandate wedge. From here I index foreign countries by $c \in \{1, \ldots, N\}$, since the object of interest is the cross-section, and the bilateral case is $c = F$.

The cross-section the model delivers lives in the mandate wedge, and what carries it into returns is each currency's exposure to the global volatility factor. The object of interest is therefore the conditional beta of the currency excess return on that factor,

$$\boldsymbol{\beta}_{\Delta TS^{g},c,t} = \mathrm{Cov}_t\Big(rx_{c,t+1}, \Delta TS^{g}_{t+1}\Big)\Big[\mathrm{Var}_t\Big(\Delta TS^{g}_{t+1}\Big)\Big]^{-1}, \tag{17}$$

a row vector with one entry per bucket. Write the difference in SDF loadings as $b_{c,t} \equiv \theta_{H,t} - \theta_{c,t} \in \mathbb{R}^3$, so that the volatility block of Equation 15 is $b'_{c,t}\, \Delta TS^{g}_{t+1}$.

Most of (15) drops out of the covariance. The constants $a_{H,t} - a_{c,t}$ are known at t and so covary with nothing. The three non-volatility terms have loadings known at t and shocks mutually orthogonal conditional on date-t information, so each contributes nothing either. Only the volatility block survives, and since ΔTS^{g}_{t+1} is conditionally mean zero, the j-th entry of the covariance is

$$\mathrm{Cov}_t\Big(b'_{c,t}\Delta TS^{g}_{t+1},\ \Delta TS^{g}_{j,t+1}\Big) = \sum_i b_{c,i,t}\left(\Sigma_{\Delta TS^{g}}\right)_{ij} = \left(b'_{c,t}\,\Sigma_{\Delta TS^{g}}\right)_j, \tag{18}$$

so that

$$\boldsymbol{\beta}_{\Delta TS^{g},c,t} = b'_{c,t}\,\Sigma_{\Delta TS^{g}}\Big[\Sigma_{\Delta TS^{g}}\Big]^{-1} = b'_{c,t} = \Big(\theta_{H,t} - \theta_{c,t}\Big)'. \tag{19}$$

This means that the conditional beta captures exactly the SDF loading wedge. Substituting Equation 12 bucket by bucket,

$$\beta_{\Delta TS^{g},c,k,t} = \lambda_{k,t}\Big(\omega_{H,k,t} - \omega_{c,k,t}\Big), \tag{20}$$

one global price of risk multiplied by one mandate wedge, which is the exposure the

cross-section is built on.

However, the global factor ΔTS^{g} is not directly observable. What is observed is the Home curve innovation $\Delta TS^{(H)}$, which loads on the global factor through the decomposition in Equation 1. Since $\varphi_H > 0$, it carries the global factor, but it also carries the Home component alongside it, so a beta estimated against the Home curve is a beta on a noisy proxy.

Take the same two moments against the Home curve, which is Equation 17 with $\Delta TS^{(H)}_{t+1}$ in place of ΔTS^{g}_{t+1}. Using the orthogonality of the global and local components,

$$\mathrm{Cov}_t\Big(rx_{c,t+1}, \Delta TS^{(H)}_{t+1}\Big) = \varphi_H\, b'_{c,t}\, \Sigma_{\Delta TS^{g}}, \qquad \mathrm{Var}_t\Big(\Delta TS^{(H)}_{t+1}\Big) = \varphi_H^2\, \Sigma_{\Delta TS^{g}} + \psi_H^2\, \Sigma_{\mathrm{loc}}, \tag{21}$$

the local component dropping from the covariance because the currency excess return does not load on it. The observable beta is therefore

$$\boldsymbol{\beta}_{\Delta TS,c,t} = \varphi_H\, b'_{c,t}\, \Sigma_{\Delta TS^{g}} \Big[\varphi_H^2\, \Sigma_{\Delta TS^{g}} + \psi_H^2\, \Sigma_{\mathrm{loc}}\Big]^{-1}, \tag{22}$$

where, unlike in Equation 19, the inverse does not cancel. Under the diagonality assumption of $\big[\varphi_H^2\, \Sigma_{\Delta TS^{g}} + \psi_H^2\, \Sigma_{\mathrm{loc}}\big]$, Equation 22 separates bucket by bucket into

$$\beta_{\Delta TS,c,k,t} = A_k\, \beta_{\Delta TS^{g},c,k,t}, \qquad A_k = \frac{\varphi_H\, \sigma_k^2}{\varphi_H^2\, \sigma_k^2 + \psi_H^2\, \sigma^2_{\mathrm{loc},k}} > 0, \tag{23}$$

where the attenuation A_k is built from φ_H, ψ_H, σ_k^2 and $\sigma^2_{\mathrm{loc},k}$, none of which carries a country subscript, so it is the same positive number for every currency in bucket k. The observable beta is therefore the global beta scaled by a common factor to the whole cross-section. Therefore, ordering currencies based on the observable beta recovers the same ordering that would be formed on the unobservable one[8].

The beta has a direct reading in terms of mandates, and that reading is what makes a sort the natural design. By Equation 20 it is a common shift and scaling of the country's own share $\omega_{c,k,t}$, so where a currency sits in the cross-section of betas is a statement about how much its institutions contribute to aggregate hedging demand in bucket k relative to Home. A country contributing a large share has an SDF that already moves with the

[8]Two features are worth clarifying. Diagonality is doing real work here and was not needed for (19): with dense covariances the inverse in (22) would be dense too, bucket k's observable beta would mix all three elements of $b_{c,t}$, and a sort on bucket k would no longer order currencies on bucket k mandates. The exactness of the global beta and the bucket-by-bucket form of the observable one therefore rest on different assumptions. And the same attenuation that leaves the ordering intact makes the level of an estimated beta uninformative, since A_k is not measured.

global shock, so its currency offers the Home investor little exposure he does not already carry, while a country contributing little has an SDF that barely moves and leaves him carrying the difference. Ranking currencies on the beta is therefore a way of ranking countries on the mandate wedge, which is the object that gives a premium.

Both terms multiplying the country's own share in Equation 20, the price of risk $\lambda_{k,t}$ and the Home share $\omega_{H,k,t}$, are common to every currency at a given date. A ranking is invariant to a common shift and to a common positive scaling, so the ordering of betas in bucket k is the ordering of the mandate wedges when $\lambda_{k,t} > 0$ and its exact reverse when $\lambda_{k,t} < 0$. Movements in intermediary tightness or in aggregate hedging demand therefore shift every currency's beta together and cannot change which currencies a sort on that beta separates, only the direction in which it orders them.

With multiple Foreign countries the model generates a cross-sectional spread in exposures, and the traded factor is the return on a long-short portfolio of beta-sorted currencies,

$$fTS_{k,t+1} = \frac{1}{N_T} \sum_{c \in T_k} rx_{c,t+1} - \frac{1}{N_B} \sum_{c \in B_k} rx_{c,t+1}, \tag{24}$$

where T_k and B_k are the top and bottom baskets of a sort on bucket k, holding N_T and N_B currencies, making $fTS_{k,t+1}$ a traded proxy for a factor no country observes directly. Averaging within a leg diversifies the country-specific shocks, and with them the local volatility components $\psi_c \Delta TS^{\text{loc},(c)}$ that Equation 15 carries inside them, since those are independent across countries. Differencing the legs then removes everything the two baskets share, which is every Home term, none of which carries a country subscript. What survives is the exposure the sort ranked currencies on, so the realised return on the long-short portfolio is dominated by the global volatility term-structure component. Diversification also works on realisations rather than on expected returns, so it makes the factor a cleaner measure of the shock without making the premium it earns a volatility term-structure premium alone.

Forming a long-short portfolio on sorted exposures in order to extract a factor that no country observes directly is the global-factor extraction logic of Verdelhan (2018), Fan et al. (2022), among others.

2.6. Conditional currency risk premium

The model has so far produced one expected return, the price of risk $\lambda_{\Delta TS,t}$ on the volatility claims, and everything it has said about currencies has been about exposure. Equation 15 says what a currency excess return loads on, Equation 20 says how much, and Equation 24 builds a traded factor out of the spread in those loadings, but none of them

says what a currency earns for carrying the exposure. Cross-sectional tests estimate a price of risk, which is a statement about expected returns, so the model needs an expected return of its own, and this is the step that carries the hedging mechanism into the estimated prices of risk.

It takes five standard assumptions beyond the structure already imposed. The first is conditional lognormality (A1), that the log kernel and the log excess return are conditionally jointly normal at every date. I maintain it rather than approximate it, because the higher-cumulant corrections that would otherwise replace it are country specific and would not drop out of the cross-sectional comparison the paper rests on. The second is the completeness of the internationally traded payoff space (A2), already assumed in Subsection 2.2, which is what delivers Equation 14. Without it the exchange rate equals the difference of the log kernels only up to a wedge orthogonal to traded payoffs, and every premium below would carry that wedge.

The third and fourth extend conditions already recorded with Equation 13. Conditional orthogonality (A3) makes $u_{c,t+1}$, $u_{g,t+1}$ and ΔTS^{g}_{t+1} mutually orthogonal at every date, and adds that country-specific shocks are orthogonal across countries as well. Measurability (A4) keeps the loadings known at t and the shocks realising at $t+1$. The fifth is constant conditional covariances (A5), which requires that $\Sigma_{\Delta TS^{g}}$ and Σ_{loc}, introduced in Subsection 2.2 and already taken to be diagonal there, do not vary with the date-t state. The consequence is that all time variation in premia runs through the price of risk rather than through the quantity of risk.

The object to be priced is the expected log excess return. Under (A1) the log Euler equation gives[9]

$$E_t[rx_{c,t+1}] = \operatorname{Cov}_t(-m_{H,t+1},\, rx_{c,t+1}) - \tfrac{1}{2}\operatorname{Var}_t(rx_{c,t+1})\,, \tag{25}$$

where the Jensen term is retained rather than discarded, since it depends on the mandate wedge and is not common across currencies. Evaluating both terms under (A3) and (A4) and combining them, the expected log excess return is the half-variance difference of the two kernels taken shock by shock. Writing

$$\Pi^{\text{other}}_{c,t} \;\equiv\; \tfrac{1}{2}\Big(\delta^2_{H,t}\operatorname{Var}_t(u_{H,t+1}) - \delta^2_{c,t}\operatorname{Var}_t(u_{c,t+1})\Big) + \tfrac{1}{2}\Big(\gamma^2_{H,t} - \gamma^2_{c,t}\Big)\operatorname{Var}_t(u_{g,t+1})$$

for the contribution of the non-volatility shocks, which the model takes as given, the

[9] Appendix B shows the full derivation of the conditional currency risk premium.

premium in full is

$$
\begin{aligned}
E_t[rx_{c,t+1}] &= \Pi_{c,t}^{\text{other}} + \underbrace{\tfrac{1}{2}\Big(\theta'_{H,t}\Sigma_{\Delta TS^g}\theta_{H,t} - \theta'_{c,t}\Sigma_{\Delta TS^g}\theta_{c,t}\Big)}_{\text{volatility term structure}} \\
&= \Pi_{c,t}^{\text{other}} + \tfrac{1}{2}\sum_k \sigma_k^2\,\lambda_{k,t}^2\,\underbrace{\big(\omega_{H,k,t} - \omega_{c,k,t}\big)}_{\text{mandate wedge}}\underbrace{\big(\omega_{H,k,t} + \omega_{c,k,t}\big)}_{\text{combined share}},
\end{aligned}
\tag{26}
$$

the half-variance-difference form of the currency premium in Backus et al. (2001). Three objects multiply. The squared price of risk $\lambda_{k,t}^2$ is common to every currency and scales the cross-section without generating it. The mandate wedge $\omega_{H,k,t} - \omega_{c,k,t}$ carries the country subscript and is the cross-sectional driver. The combined share $\omega_{H,k,t} + \omega_{c,k,t}$ is positive under the maintained sign restriction, so the premium is increasing in the wedge in every bucket, whatever the sign of $\lambda_{k,t}$. A currency earns a higher premium when its country contributes a smaller share of aggregate hedging demand than Home does.

The combined share adds two shares and is not the cross-sectional sum, which is one by construction. It leaves out every country other than Home and c, so under the maintained sign restriction it lies strictly in $(0, 1)$ whenever there are more than two countries, and it attenuates the wedge rather than amplifying it. It equals one only in a two-country economy, where the premium would be linear in the wedge and the squared shares would not appear at all. The quadratic form of Equation 26 therefore exists precisely because there is a cross-section to price.

The economics of Equation 26 is worth stating in words. A country with a low share $\omega_{c,k,t}$ has institutional investors that contribute little to aggregate hedging demand in bucket k, so by Equation 12 its kernel loads weakly on $\Delta TS_{k,t+1}^g$ while Home's loads more heavily. When a shock to that bucket arrives, Home's SDF moves and country c's barely does, so by Equation 14 the exchange rate moves against the Home investor in exactly the states his own SDF treats as bad. The currency's excess return carries $\lambda_{k,t}(\omega_{H,k,t} - \omega_{c,k,t})$ per unit of the shock, and Home's bad states in that bucket are those in which $\lambda_{k,t}\Delta TS_{k,t+1}^g < 0$, so with a positive wedge the currency loses precisely there, whatever the sign of $\lambda_{k,t}$. The Home investor holding it is bearing the part of Home's volatility term-structure exposure that the foreign kernel does not share, which is what Equation 26 measures as the difference between the two kernel variances. The larger the wedge, the larger that uninsured part and the higher the compensation.

Conversely, a country whose share is close to Home's has a kernel that responds to the same shock in much the same way, the two largely offset in the exchange rate, and the

premium is small. A country that hedges more than Home, $\omega_{c,k,t} > \omega_{H,k,t}$, reverses the comparison and earns a negative premium, since holding its currency insures the Home investor rather than exposing him. The reading is deliberately stated in terms of the kernel rather than in terms of volatility rising or falling, because which direction of the curve shock counts as Home's bad state depends on the sign of $\lambda_{k,t}$, and that sign is not the same in all three buckets.

Nothing restricts the wedge to be positive, and the case in which it is not deserves a word, since it is the ordinary situation for the countries that hedge most. A negative wedge is not the same thing as a negative share. A negative share is the country hedging against the aggregate direction, which the sign restriction above rules out because it would break the correspondence between the ordering the sort produces and the ordering of the premium. A negative wedge with both shares positive breaks nothing, because the Home share is common to every currency. In the beta it is a shift, and a ranking is invariant to a shift. In the premium it enters only through $\omega^2_{H,k,t}$, so the premium stays monotone decreasing in $\omega_{c,k,t}$ whatever Home's share happens to be. What the Home share fixes is the level at which premia cross zero, and not the ordering of the cross-section. The traded factor does not see it at all, since $\omega_{H,k,t}$ cancels from (27), so the sign of the wedge is visible in the premium on an individual currency and nowhere else.

The price of risk enters squared because it appears in both kernels, in Home's through $\theta_{H,t}$ and in country c's through $\theta_{c,t}$, and the premium is a difference of two kernel variances. Two channels of time variation follow. The cross-section widens when hedging pressure is intense and narrows when it is not, and the ordering of premia moves as mandates are reallocated across countries.

Applying Equation 26 to each basket and differencing, the expected return on the traded factor is

$$E_t[fTS_{k,t+1}] = \tfrac{1}{2}\sigma_k^2\lambda_{k,t}^2\left[\overline{\omega^2}^{B_k}_{k,t} - \overline{\omega^2}^{T_k}_{k,t}\right] + \tfrac{1}{2}\sum_{j\neq k}\sigma_j^2\lambda_{j,t}^2\left[\overline{\omega^2}^{B_k}_{j,t} - \overline{\omega^2}^{T_k}_{j,t}\right] + \eta^{\text{other}}_{k,t}, \qquad (27)$$

where $\overline{\omega^2}^{B_k}_{j,t}$ and $\overline{\omega^2}^{T_k}_{j,t}$ are the basket averages of squared bucket-j shares, the second term collects incidental exposure to the other buckets, and $\eta^{\text{other}}_{k,t}$ is the top-minus-bottom spread in expected-return components other than the volatility term structure. The Home share cancels from the difference, since it enters both baskets identically.

The own-bucket term carries the sign of $\lambda_{k,t}$ by construction, since the top basket always holds the higher beta: with $\lambda_{k,t} > 0$ those are the low-share currencies and the bracket is positive, and with $\lambda_{k,t} < 0$ they are the high-share currencies and it is negative.

Provided the incidental and non-term-structure terms do not dominate, the shares are positive,[10] and $\lambda_{k,t}$ keeps one sign over the sample. Therefore, the estimated price of risk, the average return on the traded factor and the price of risk on the claims all share a sign, $\text{sign}(\hat{\lambda}_k) = \text{sign}(E[fTS_k]) = \text{sign}(\lambda_{k,t})$.

The premium (26) carries $\lambda_{k,t}$ squared and is therefore silent about the sign of the price of risk, while the beta (20) is linear in it and carries the sign the premium discards. Because the sort selects on the beta, it is the selection and not the pricing that returns the sign to the traded factor.

Home country is the US in everything that follows, and the empirical work takes the perspective of a US investor. Nothing in the model requires that choice, and any reference currency could play the role, but the Home economy has to be the one hosting the traded volatility curve.

3. Data, risk factor formation and portfolio composition

3.1. Data

I obtain spot (s_t, in log) and 1-month forward exchange rates (f_t, in log) with respect to USD from Barclays Bank International (BBI), World Markets Company/Refinitiv (WMR) and Refinitiv, all via Datastream. I set the order of preference as WMR being the most preferable source and BBI as the least preferable, and I change the source of the data whenever it becomes available from a preferred source, as is commonly done in the literature. I also get data on real effective exchange rates from the IMF International Financial Statistics.

The exchange rate data is for 48 countries[11]: Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Croatia, Cyprus, Czech, Denmark, Egypt, Euro, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Ireland, Italy, Japan, Kuwait, Malaysia, Mexico, Netherlands, New Zealand, Norway, Philippines, Poland, Portugal, Russia, Saudi Arabia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Kingdom, Ukraine. Appendix C

[10] The beta is linear in the share and the premium quadratic, so when all shares carry one sign, squaring is monotone and the two orderings agree. A country whose mandate runs against the aggregate has $\omega_{c,k,t} < 0$ and breaks that agreement, since the sort ranks on the share while the premium depends on its absolute value. It then enters an extreme basket while earning a premium inconsistent with that position, which attenuates the factor. The condition is imposed bucket by bucket.

[11] The set of currencies effectively included depends on the rolling window length specified in Subsection 3.2, which determines the starting point at which the risk factor becomes available. The list shown here represents the full set of currencies that can potentially enter the construction. Additionally, I follow Kroencke et al. (2014) and Corte et al. (2016), among others, and leave out indicated countries due to large deviations of covered interest rate parity. See Table C3 in Appendix C for details.

provides further details on the data, such as start and end date for each currency, development group (advanced or emerging), Datastream mnemonics and periods of covered interest parity violations.

I use daily VIX futures contracts with maturities from one to nine months to capture the volatility term-structure. VIX futures are contracts on S&P500 implied volatility and the data are sourced from Datastream. Trading in VIX futures begins in March 2004 with maturities up to four months, while contracts with longer maturities are introduced gradually through 2006. To ensure a balanced maturity structure, the baseline sample begins in October 2006 and ends in March 2024. I use the US curve as a proxy for the global volatility term-structure. Using index options from three European markets, Appendix H shows that the US curve contains a global component.

Given the shorter availability of exchange-traded VIX futures, I also build a longer robustness sample. In Appendix F I replicate VIX futures from S&P500 index options, applying the CBOE VIX methodology to the OptionMetrics volatility surface, which gives a daily series of synthetic VIX futures at horizons of one, two, three, six and nine months from January 1996 to March 2024.

Figure 3 displays selected VIX futures term-structures, and Figure 4 plots the complete time-series across maturities. Under normal conditions, the term-structure is upward sloping, with longer maturities priced above shorter ones, reflecting the mean-reverting nature of volatility and the premium investors demand for holding long-term volatility exposure. During periods of market stress the entire curve shifts upward and flattens or inverts as short-term contracts surge relative to longer maturities. These episodes signal elevated near-term uncertainty and a compression of volatility risk premia, as short-term protection becomes disproportionately expensive and the usual compensation for bearing long-horizon volatility risk temporarily vanishes.

3.2. Construction of VIX term-structure currency risk factors

In order to extract the common components that drive variations in the VIX futures term-structure, I compute an expanding-window Principal Component Analysis (PCA)[12] on daily VIX futures term-structures, and use the first three principal components sampled

[12]Following the term-structure literature (Joslin et al., 2011), let Y contain n maturities and let W be the $n \times 3$ matrix of the first three eigenvectors of $\text{cov}(Y)$. Rows correspond to maturities ordered from shortest ($i = 1$) to longest ($i = n$). Signs are fixed by enforcing positive loadings on the longest maturity. The components are then scaled into canonical level, slope, and curvature factors using Level scale $= \sum_{i=1}^{n} W_{i1}$, Slope scale $= W_{n,2} - W_{1,2}$ and Curvature scale $= W_{n,3} - 2W_{m,3} + W_{1,3}$, where m indexes the central maturity. The principal components are divided by these scalars so that PC1 corresponds to a parallel shift, PC2 to a tilt, and PC3 to a second-difference movement.

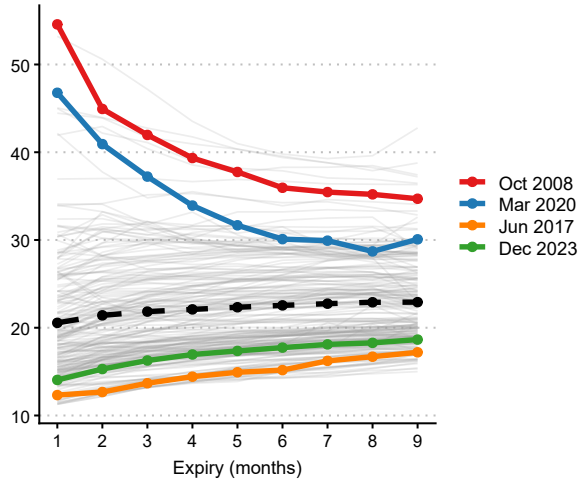


Figure 3: Selected end-of-month VIX futures term-structures. Grey lines are the term-structures at all other month-ends in the sample, the coloured lines are the selected dates and the dashed black line is the sample average.

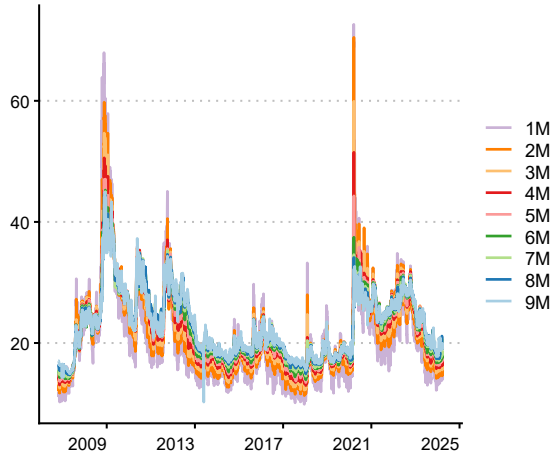


Figure 4: Time-series of VIX futures, daily

at month-end. Signs of the factor loadings are chosen such that increases in PC1 and PC3 and decreases in PC2 correspond to the standard stress-period pattern of higher overall volatility (shift of the curve), front-end spikes (flattening or inversion of the curve), and greater curvature in the term-structure. In order to ensure enough data for the principal components, I start with a window of one year. Then at each following date I use all available past data, so that after the initial startup period the procedure has no look-ahead bias.

Figure 5 shows the factor loadings of each contract on the first three principal components using the full-sample (last-date) decomposition. Together, they explain about 99% of the variation in the curve. All expirations load positively and by similar amounts on PC1, identifying it as the level factor. PC2 displays increasing loadings across maturities, capturing the slope. PC3 loads positively on the short and long ends and negatively in the middle, identifying it as the curvature factor. Figure 6 shows that PC1 exhibits sharp upward spikes during stress periods (2008–2009, 2011, 2020), consistent with its interpretation as the level of the term-structure. Outside those episodes, PC1 remains near a lower, more stable baseline. PC2 steps down markedly in the same stress periods, reflecting short-end implied volatility rising relative to the long end, and then gradually reverts upward as conditions stabilize. PC3 shows brief, high-amplitude movements only around the major dislocations, indicating temporary increases in curvature when the short and long maturities move jointly against the middle.

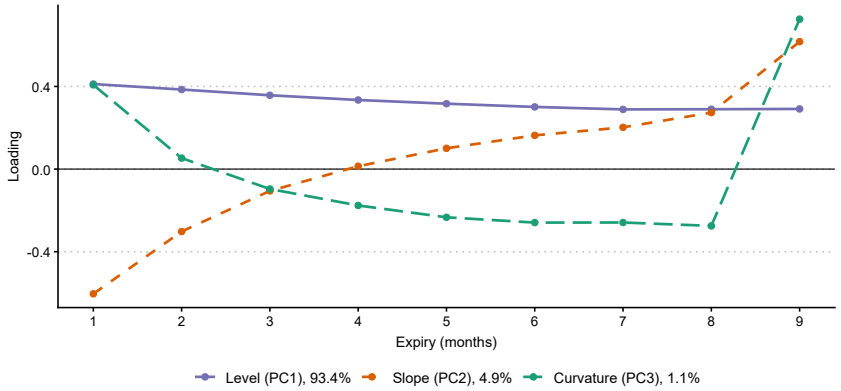


Figure 5: Factor loadings of VIX futures PCA. This figure shows the factor loadings of the full-sample principal component analysis on the level of VIX futures. Loadings are scaled such that they correspond to common measures of level, slope, and curvature. Percentage of variance explained by each principal component is between parenthesis.

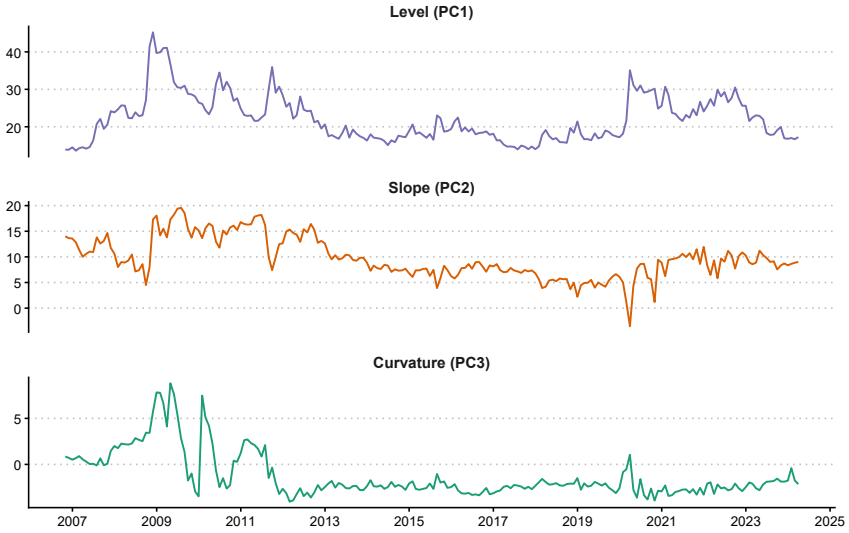


Figure 6: Principal components of expanding window PCA. This figure shows the time-series of the first three principal components extracted from VIX futures using an expanding window PCA, with a startup period of 252 days. PCA is performed on daily data, with the resulting factors sampled at month-end. The signs of the loadings are such that PC1 captures the level of the curve, PC2 captures the slope, and PC3 captures curvature. Increases in PC1 reflect upward shifts in the entire curve, typically associated with deteriorating market conditions, while increases in PC2 reflect steepening, often interpreted as improving conditions. PC3 captures changes in curvature, though its interpretation is less directly linked to macro-financial states.

To evaluate whether these common volatility term-structure factors are priced in the cross-section of currency returns, I sort currencies into five[13] portfolios based on their lagged exposure to innovations in the term-structure principal components, $\beta_{TS^j,i}$. For

[13] In Subsection E.1 I repeat the same analysis but with currencies sorted into three portfolios and in Subsection E.2 with currencies sorted into ten portfolios. Results are analogous.

that, I compute each currency's monthly log excess return of buying a foreign currency in the forward market and then selling it in the spot market after one month, $rx_t = f_{t-1} - s_{t-1} - \Delta s_t$, and estimate the following rolling 48-month[14] regression:

$$rx_{i,t} = \beta_{DOL,i} DOL_t + \beta_{TS^j,i} \Delta TS_t^j + \varepsilon_{i,t}, \qquad j \in TS \tag{28}$$

where DOL is the Dollar factor proposed by Lustig et al. (2011), used to control for the overall variation in currency excess returns, and $TS = \{\text{PC1}, \text{PC2}, \text{PC3}\}$ represents the principal components extracted from VIX futures term-structure and Δ denotes the monthly variation.

Then, at the end of month t, I sort currencies based on $\beta_{TS^j,i}$ into five portfolios and compute the equal-weighted currency excess return of each portfolio in the following month. $\beta_{TS^j,i}$-sorts provide tradable diversified currency portfolios that proxy for the global component of volatility term-structure risk. In terms of the model, these portfolios group currencies with similar exposures to the priced global volatility term-structure factor. The proposed currency risk factor for each VIX futures term-structure common component is thus the high-minus-low (HML) beta-sorted portfolio, denoted as $fPC1$ for the level factor, $fPC2$ for slope and $fPC3$ for curvature.

Why the factors are portfolios of currencies rather than positions in the volatility claims themselves is worth settling here. The claims deliver the Home curve, which by Equation 1 carries the local component alongside the global one, while the mechanism runs on the global component alone, and a long-short portfolio of currencies diversifies the country-specific pieces away. A position in the claims also has a fixed loading, so the spread in exposures to it is whatever the data happen to supply, whereas a sort manufactures that spread by construction, which is what identifying a price of risk requires. The third reason is the one the model supplies. Under the market access assumed in Section 2 no agent stands in both markets, so the value a country's discount factor would assign a Home claim differs from the arbitrageur's price by the shadow price of the mandate, as Appendix A shows. Using the claims as factors for currency test assets would impose a single discount factor across two markets that no agent bridges, which is a restriction the model denies and whose failure is the mechanism rather than a problem.

Figure 7 shows wide and persistent dispersion in exposures across the five currency portfolios, with a clear and rather stable ranking for each component. For level (PC1), the spread between high- and low-beta portfolios remains broadly steady over most of the

[14]Results are robust to changes in the rolling window and are available upon request.

sample, narrowing only temporarily around major volatility events. Slope (PC2) betas exhibit similar patterns, with a noticeable compression of betas during the 2014–2016 period and again around 2020, after which the dispersion gradually re-emerges. Curvature (PC3) shows the largest swings, but even there the overall pattern is similar.

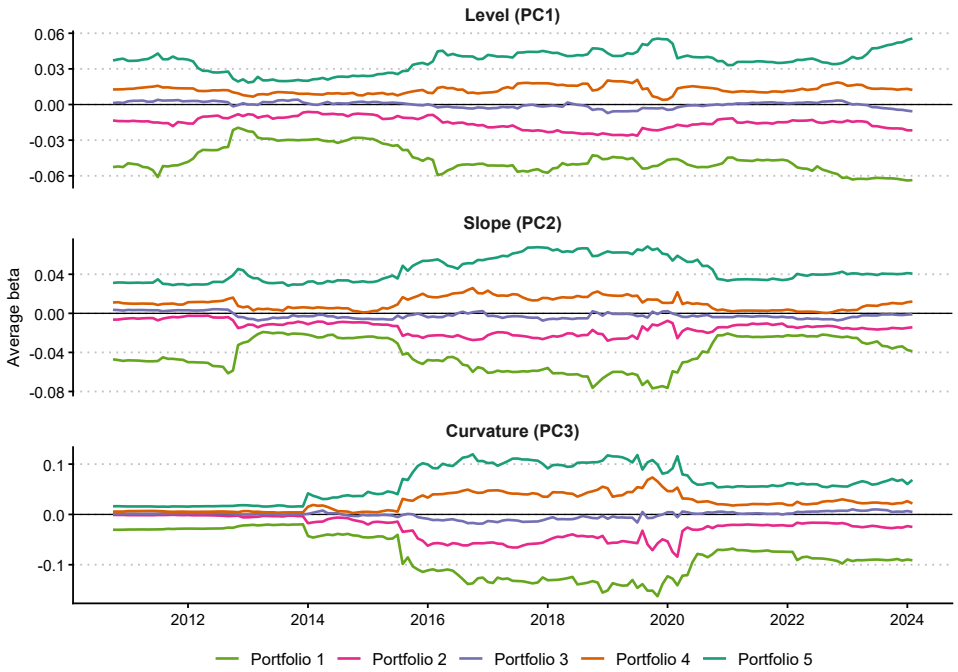


Figure 7: Rolling betas to VIX principal component innovations by currency portfolio. This figure shows the time-series of average betas for five currency portfolios sorted each month by their estimated exposure to innovations in VIX term-structure components. The principal components, capturing the level (PC1), slope (PC2), and curvature (PC3) of the VIX futures curve, are extracted monthly using an expanding-window PCA. Innovations in each component are computed as the monthly change relative to the previous observation. For each currency, rolling 48-month regressions of excess returns on these innovations are estimated following Equation 28, controlling for the Dollar factor. The resulting betas measure time-varying sensitivities to shocks in the VIX term-structure. The figure reports portfolio-average betas across the five beta-sorted groups, illustrating persistent cross-sectional heterogeneity and cyclical variation in exposures to volatility-level (PC1), slope (PC2), and curvature (PC3) innovations.

Summary statistics for the beta-sorted currency portfolios are shown in Table 1. In all three dimensions, average excess returns and Sharpe ratios line up with the underlying beta rankings, though not portfolio by portfolio. For the level (PC1) and curvature (PC3) sorts, the low-beta portfolio earns the highest return and Sharpe ratio and the high-beta portfolio the lowest, while the slope (PC2) sort shows the opposite pattern. These patterns indicate that currencies with stronger exposure to each volatility term-structure factor earn systematically different excess returns, consistent with compensation for exposure to priced shocks. In each case, the return differential is concentrated in the extremes: the highest-beta portfolios for level (PC1) and curvature (PC3), and the lowest-beta portfolio

for slope (PC2), generate the bulk of the cross-sectional spread. The resulting HML portfolios deliver large average excess returns of around 7% annualized in absolute value, with statistically significant t-statistics and strong Sharpe ratios.[15] Portfolio skewness is negative across most portfolios, consistent with downside tail risk typical of currency strategies, but the HML slope (PC2) portfolio exhibits positive skewness.

Table 1: VIX futures beta-sorted portfolios summary statistics. This table reports summary statistics for monthly currency excess returns of portfolios sorted by estimated betas with respect to VIX futures innovations. Panel A sorts currencies on β_{PC1} (level), Panel B on β_{PC2} (slope), and Panel C on β_{PC3} (curvature). P1 contains currencies with the lowest beta exposures, while P5 contains those with the highest. HML denotes the high-minus-low portfolio (P5 – P1). The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. *t-stat* refers to the t-statistic testing whether the average monthly excess return is significantly different from zero.

Portfolio	P1	P2	P3	P4	P5	HML
Panel A: sorts on β_{PC1}						
Avg Ann. Return	0.34	-1.54	-0.85	-2.12	-6.56	-6.91
t-stat	[0.16]	[-0.82]	[-0.56]	[-1.10]	[-2.77]	[-3.37]
Ann. Volatility	7.73	6.92	5.59	7.05	8.67	7.50
Sharpe Ratio	**0.04**	**-0.22**	**-0.15**	**-0.30**	**-0.76**	**-0.92**
Skewness	-0.01	-0.56	-0.49	-0.40	-0.63	-0.69
Panel B: sorts on β_{PC2}						
Avg Ann. Return	-7.43	-1.44	-1.17	-1.39	-0.47	6.96
t-stat	[-3.09]	[-0.72]	[-0.83]	[-0.78]	[-0.20]	[3.16]
Ann. Volatility	8.81	7.33	5.19	6.56	8.78	8.06
Sharpe Ratio	**-0.84**	**-0.20**	**-0.23**	**-0.21**	**-0.05**	**0.86**
Skewness	-0.96	-0.35	-0.18	-0.98	-0.53	0.63
Panel C: sorts on β_{PC3}						
Avg Ann. Return	0.11	-1.90	-1.33	-1.18	-6.57	-6.67
t-stat	[0.05]	[-0.99]	[-0.93]	[-0.62]	[-2.83]	[-3.61]
Ann. Volatility	8.52	7.00	5.22	7.00	8.50	6.77
Sharpe Ratio	**0.01**	**-0.27**	**-0.25**	**-0.17**	**-0.77**	**-0.99**
Skewness	-0.27	-0.20	-0.84	-0.28	-0.60	-0.80

The results in Table 2 show that the returns of the beta-sorted portfolios load on the Carry factor of Lustig et al. (2011), and several of these loadings are statistically significant. However, the estimated coefficients are uniformly well below one in magnitude. This means that even when the Carry factor helps explain part of the variation in these

[15]These returns are before transaction costs. Using bid and ask quotes on spot and one-month forward rates, and charging the half-spread only when a currency enters or leaves a leg, costs average 0.2% to 0.3% a year. For an investor trading each factor in the direction of its premium, the net returns are 6.7%, 6.7% and 6.4% a year for the level, slope and curvature factors (t-statistics 2.5, 2.4 and 2.6).

portfolios, the movements are far from one-to-one, and the factor does not account for the dominant share of their returns. The HML portfolios also load significantly on Carry, yet the coefficients remain modest. Taken together, these results show that the beta-sorted portfolios are not simply repackaging the traditional carry trade, they contain variation linked to exposures to volatility term-structure rather than being driven primarily by interest rate differentials.

Table 2: Time-series regression of VIX futures beta-sorted portfolios on the Carry factor. This table reports the results of time-series regressions of monthly excess returns of VIX futures beta-sorted currency portfolios on the Carry factor of Lustig et al. (2011). Panels A–C correspond to portfolios sorted on betas with respect to innovations to the first three principal components of VIX futures: level (PC1), slope (PC2), and curvature (PC3), respectively. P1 includes currencies with the lowest beta exposure and P5 includes those with the highest. HML denotes the high-minus-low portfolio (P5 – P1). Newey-West t-statistics are reported in brackets.

Portfolio	P1	P2	P3	P4	P5	HML
Panel A: sorts on β_{PC1}						
Intercept	-0.03	-0.03	-0.02	-0.03	-0.04	-0.01
	[-1.70]	[-1.66]	[-1.13]	[-1.38]	[-1.63]	[-0.58]
Carry Factor	0.32	0.17	0.10	0.09	-0.21	-0.54
	[3.38]	[1.86]	[1.31]	[1.00]	[-1.14]	[-5.89]
R^2	0.147	0.053	0.024	0.014	0.052	0.433
Panel B: sorts on β_{PC2}						
Intercept	-0.05	-0.02	-0.02	-0.03	-0.04	0.02
	[-1.97]	[-0.83]	[-1.06]	[-1.65]	[-1.60]	[0.66]
Carry Factor	-0.20	0.04	0.05	0.19	0.30	0.50
	[-0.75]	[0.43]	[0.96]	[1.87]	[2.84]	[3.15]
R^2	0.043	0.002	0.008	0.071	0.096	0.319
Panel C: sorts on β_{PC3}						
Intercept	-0.03	-0.03	-0.03	-0.02	-0.05	-0.02
	[-1.31]	[-1.31]	[-1.91]	[-1.05]	[-1.87]	[-1.24]
Carry Factor	0.28	0.08	0.15	0.08	-0.13	-0.41
	[2.48]	[1.06]	[2.21]	[0.86]	[-0.58]	[-3.81]
R^2	0.088	0.011	0.068	0.010	0.020	0.307

3.3. Composition of currency risk factors

It is important to understand the underlying portfolio compositions over time in order to see how the portfolios behave and what type of information they capture. Figure 8 shows the portfolio assignment for each beta-sorted portfolio over time. There is substantial time-variation in how individual currencies are ranked by their exposures to innovations

in the three VIX term-structure components. Although some broad patterns appear, for example, traditional safe-haven currencies such as JPY and CHF often sit in the high-beta portfolios for level (PC1), while higher-yielding emerging-market currencies more frequently populate the low-beta groups, the assignments are far from static. ISK is a striking example: it alternates between low-, mid-, and high-beta portfolios for level (PC1) and slope (PC2), but appear more consistently in higher-beta portfolios for curvature (PC3). JPY, CHF, and DKK, by contrast, show stable rankings for level (PC1) but much more variation in their slope (PC2) and curvature (PC3) rankings, indicating distinct sensitivities to different parts of the VIX futures curve. These patterns demonstrate that the portfolio ranks trace evolving sensitivities to VIX term-structure innovations rather than fixed "safe" or "risky" labels, and that currencies load differently on level, slope, and curvature components rather than following a stable macro-financial grouping.

The model gives that ranking a reading. The level beta is $\lambda_{L,t}(\omega_{H,L,t} - \omega_{c,L,t})$ and the estimated level price of risk is negative, so at any date the high-beta level basket holds the countries whose own level-hedging share exceeds Home's. Membership therefore moves as hedging demand is reallocated across countries, which is what the time variation in the figure shows, and the currencies sitting there at a given date are those whose institutions carry the larger share of aggregate long-horizon hedging demand at that date. Where JPY and CHF appear in that basket, the model reads it as Japanese and Swiss institutions contributing more of that demand than their US counterparts, which is plausible given the long-dated guarantees they carry, and by Equation 26 those currencies earn a negative premium, which is what a safe-haven currency does without the label having to be permanent. The shares are not observed, so this is what the model implies the sorting reveals rather than a measurement of hedging demand.

Figure 9 makes the underlying factor structure even clearer by showing whether each currency enters the long, neutral, or short leg of the proposed HML risk factors. The heatmaps show that the composition of the long–short factors is neither constant nor mechanically aligned with carry-trade positions: safe-haven currencies do not always appear in the long leg, high-yielders do not always appear in the short leg, and many currencies move across legs as volatility conditions change. GBP, ZAR, and ISK frequently switch sides in all three HML risk factors, entering the long leg when their slope betas spike and falling into the short or neutral leg when exposures compress as their underlying risk sensitivities shift.

To assess whether the factors embed a systematic country-type bias, Figure 10 shows, for each month, the share of emerging-market currencies in the long and short legs of

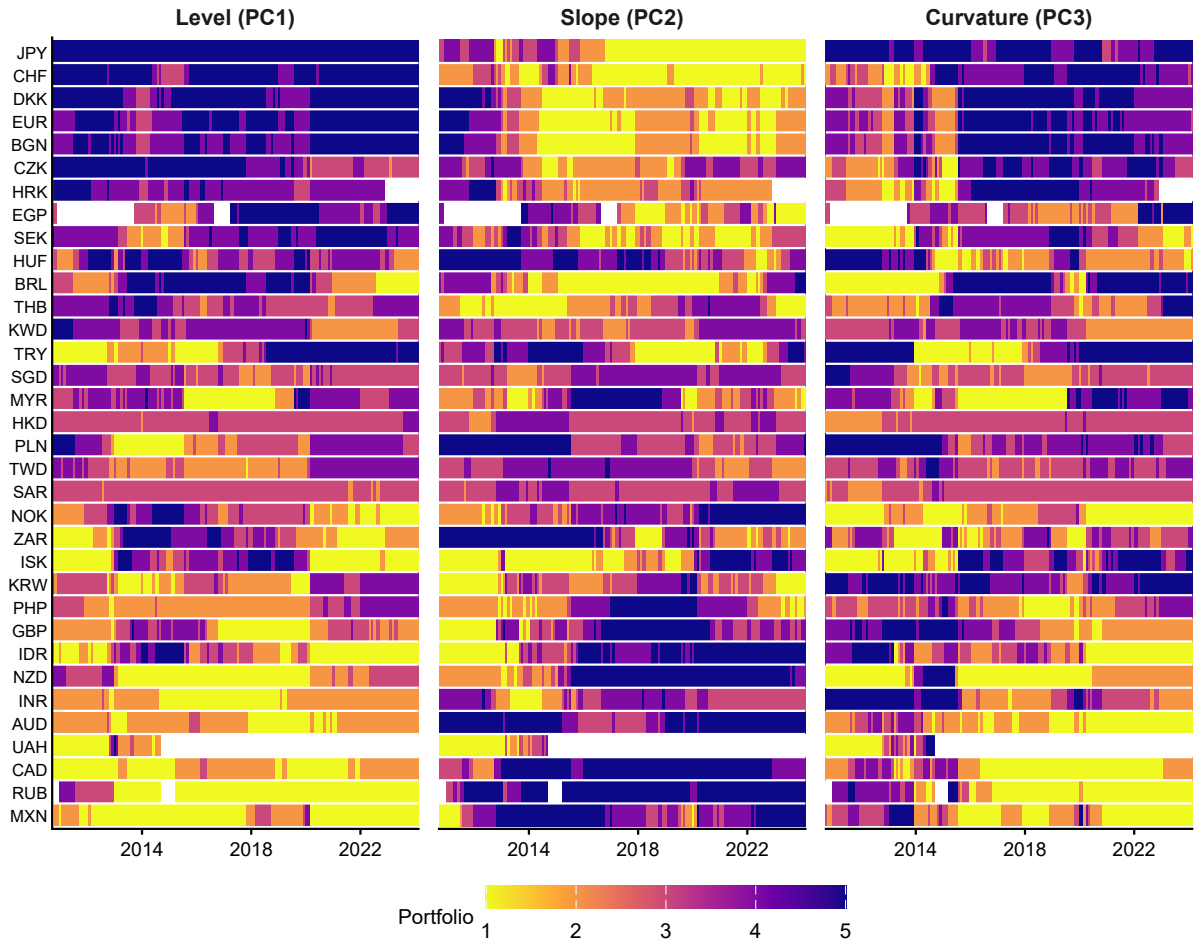


Figure 8: Composition of currency portfolios sorted on VIX principal component betas. This figure displays the time-varying portfolio assignments of individual currencies based on their rolling 48-month betas with respect to innovations in the first three VIX term-structure principal components: level (PC1), slope (PC2), and curvature (PC3). Each color represents the portfolio (from 1 to 5) to which a currency belongs in a given month, determined by its beta ranking from the previous month's estimation. Left corresponds to portfolios sorted on PC1 betas, middle panel on PC2, and right on PC3. Lighter shades indicate lower-exposure portfolios, while darker shades represent higher exposures. The figure highlights persistent clustering of currencies with similar sensitivities to volatility term-structure shocks, as well as transitions across portfolios reflecting evolving exposure patterns over time.

the three HML factors, against the average share of emerging-market currencies in the full cross-section, which is about 60%. Both legs of every factor combine advanced and emerging-market currencies in almost every month, and their shares move above and below the cross-sectional share over time, so no factor is driven by one country group. The level factor leans somewhat toward advanced currencies in its long leg and toward emerging-market currencies in its short leg, which hold on average 45% and 63% emerging-market currencies, while the legs of the slope and curvature factors stay close to balanced, at 56% and 51% for slope and 62% and 52% for curvature. None of the factors is therefore simply

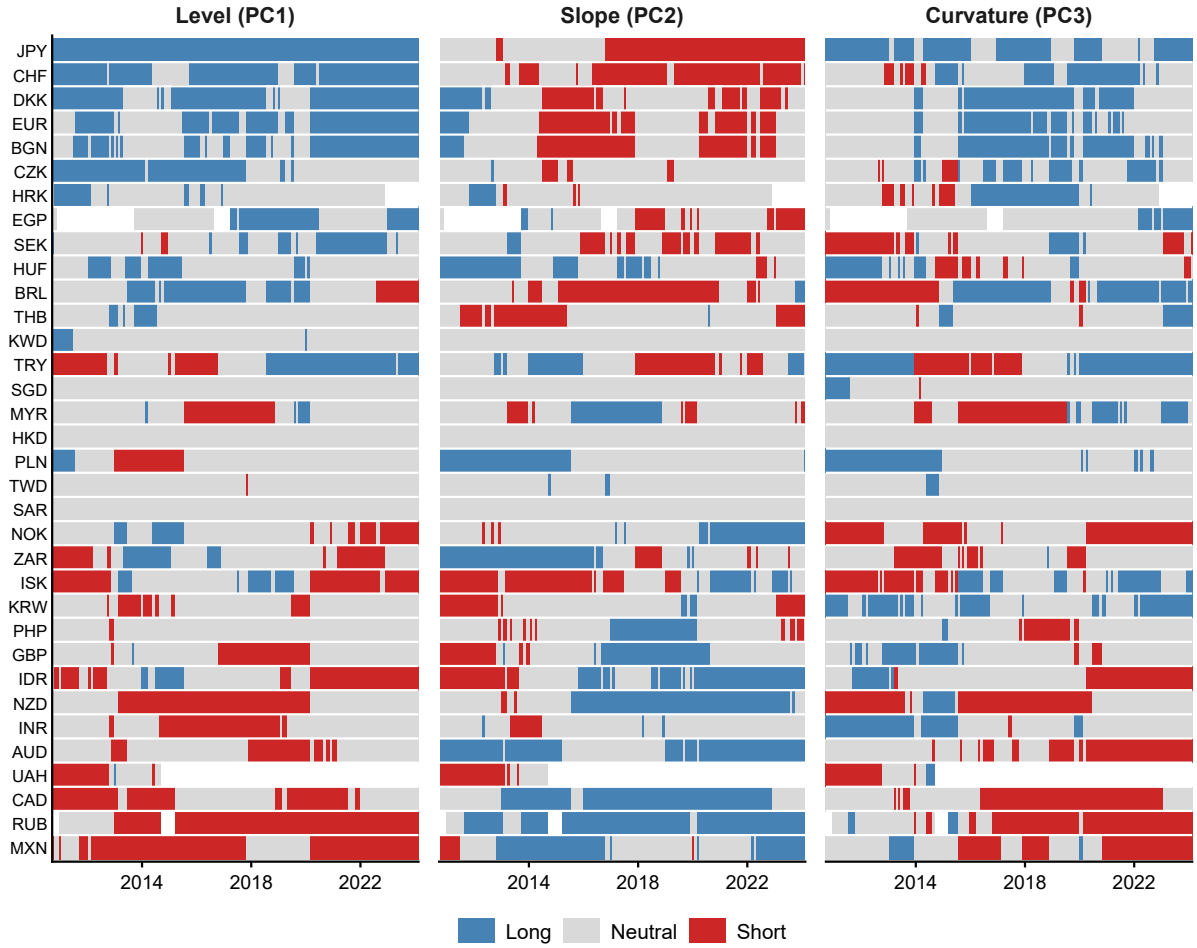


Figure 9: Composition of currency risk factors constructed from VIX principal component betas. This figure shows the time-varying composition of the high-minus-low (HML) currency risk factors associated with exposures to innovations in the first three VIX term-structure principal components: level (PC1), slope (PC2), and curvature (PC3). For each month, currencies are sorted into five portfolios based on their rolling 48-month betas with respect to the respective principal component innovation, and the factor is constructed as the return difference between the highest- and lowest-beta portfolios. The heatmap indicates, for each currency and month, whether it enters the factor with a long position (blue), short position (red), or is excluded from the factor (gray). Each panel corresponds to one VIX component. The figure highlights persistent cross-sectional asymmetries in how currencies load onto the VIX futures term-structure level, slope, and curvature shocks, as well as shifts in factor composition reflecting changing sources of volatility risk exposure across time.

financing positions in advanced low-yield currencies to invest in high-yield emerging-market currencies, and their composition shifts with volatility-risk exposures rather than following a fixed divide between advanced and emerging markets.

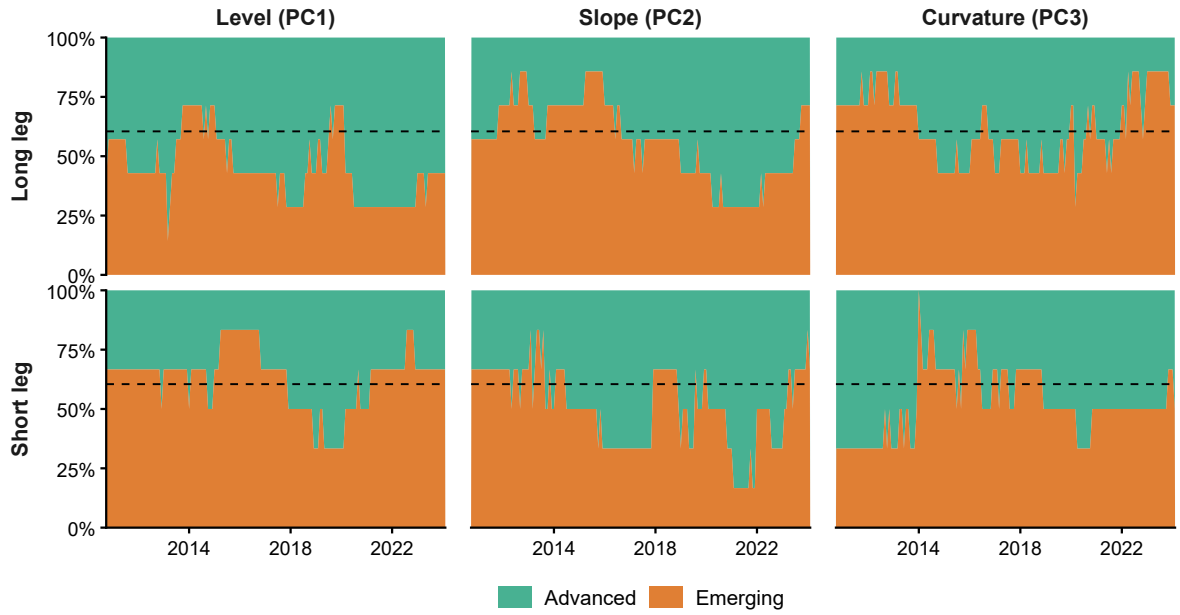


Figure 10: Emerging-market composition of the VIX term-structure factors. Share of emerging-market currencies (orange) and advanced-market currencies (green) among the currencies in the long leg (top row) and the short leg (bottom row) of the high-minus-low factors built on betas to innovations in the level, slope and curvature of the VIX futures term-structure, monthly. The dashed line is the average share of emerging-market currencies among all currencies in the cross-section over the sample (about 60%), so a leg whose orange area lies above (below) the line holds more (fewer) emerging-market currencies than the cross-section as a whole.

4. The price of global volatility term-structure risk in the cross section of currency returns

4.1. Explaining currency excess returns with VIX term-structure currency risk factors

As suggested by the stylized model in Section 2, the return of the long-short beta-sorted portfolio can capture the global component in the volatility term-structure factor. Therefore, with the volatility term-structure risk factors in hand, I can then evaluate if they are priced in the cross-section of currency returns by estimating their associated price of risk and their ability to explain the cross-section of currency excess returns. For that, I use the two-pass methodology put forward by Fama and MacBeth (1973). Since the influential study of Lustig and Verdelhan (2007), this method has been widely employed in currency asset pricing precisely because it links asset returns to risk factors through the SDF. Following Cochrane (2005), the fundamental pricing equation states that the price

of any asset equals the expected discounted payoff:

$$p_t \;=\; \mathbb{E}_t[\, m_{t+1}\, x_{t+1}\,]\,, \tag{29}$$

where p_t is the asset price at time t, x_{t+1} its payoff, and m_{t+1} the SDF. In the absence of arbitrage opportunities, risk-adjusted excess returns must have zero mean and satisfy the pricing condition:

$$\mathbb{E}_t[\, m_{t+1}\, rx_{i,t+1}\,] \;=\; 0. \tag{30}$$

A linear SDF can be represented as

$$m_t = \xi\Big[1 - (f_t - \mathbb{E}[f_t])b^\top\Big], \tag{31}$$

where ξ is a scalar intercept, f_t is a $1 \times K$ vector of risk factors observed at time t, $\mathbb{E}[f_t]$ denotes the vector of their expected values, and b collects the factor loadings of the SDF. The element b_k measures how much the k-th factor contributes to the SDF, conditional on the presence of the other factors.

Combining the SDF representation with the pricing condition leads to the standard beta-pricing relation,

$$\mathbb{E}[rx_t] = \underbrace{\mathrm{cov}(rx_t, f_t)\Sigma_f^{-1}}_{\beta}\,\underbrace{\Sigma_f b^\top}_{\lambda^\top}, \tag{32}$$

which shows that expected excess returns are determined by two components: the $1 \times K$ vector of risk prices, λ, and the $N \times K$ matrix of risk exposures, β. Therefore, the expected excess return for asset i is

$$\mathbb{E}[rx_i] = \beta_i \lambda^\top. \tag{33}$$

In empirical work, the risk exposures (β_i) and risk prices (λ) are typically estimated using the two-pass Fama–MacBeth procedure. The first pass estimates the risk exposures (or factor loadings), and the second pass recovers the prices of risk.

The first pass involves running time-series regressions of each asset's excess return on the factors:

$$rx_{i,t} = \alpha_i + f_t\beta_i^\top + \epsilon_{i,t}, \quad i = 1, \ldots, N, \quad t = 1, \ldots, T, \tag{34}$$

where α_i captures the risk-adjusted average excess return of asset i.

The second pass then runs a cross-sectional regression of average asset returns on the estimated betas. Since expected excess returns $\mathbb{E}[rx_i]$ are not observable, they are

proxied by average realized excess returns, $\bar{rx}_i = \frac{1}{T}\sum_{t=1}^{T} rx_{i,t}$. The cross-sectional regression becomes

$$\bar{rx}_i = \hat{\beta}_i \lambda^\top + a_i, \quad i = 1, \ldots, N, \tag{35}$$

where a_i is the pricing error for asset i, and $\hat{\beta}_i\hat{\lambda}^\top$ is its model-implied risk premium. Therefore, the vector of risk prices is estimated as

$$\hat{\lambda} = \bar{rx}\hat{\beta}(\hat{\beta}^\top\hat{\beta})^{-1}, \tag{36}$$

where $\hat{\beta}$ is the $N \times K$ matrix of estimated factor loadings, and $\bar{rx}$ is the vector of average realized excess returns across assets. Then, the cross-sectional R^2 is computed as

$$R^2 = 1 - \frac{\frac{1}{N}\sum_{i=1}^{N}\hat{a}_i^2}{\text{var}\left(\bar{rx}_i\right)}. \tag{37}$$

The covariance matrix of the estimated risk prices, however, must be adjusted to account for the fact that the factor loadings are estimated rather than observed. I apply the Shanken (1992) correction to obtain consistent standard errors for the estimated prices of risk, in which the standard errors are inflated to account for the fact that $\hat{\beta}_i$ are estimated rather than known. This yields robust inference for the estimated risk premia under the two-pass Fama–MacBeth procedure.

Additionally, the overall pricing accuracy of the model is evaluated with the Gibbons et al. (1989) statistic, which tests whether all first-stage alphas are jointly zero. The test uses the time-series regression residuals and the factor covariance structure to assess whether the model leaves any systematic mispricing. Failure to reject the null means the alphas are statistically indistinguishable from zero, so the model is not leaving predictable errors. Rejection of the null means at least one asset shows a nonzero alpha, indicating the model cannot fully eliminate pricing errors.

4.2. Test assets: style-sorted portfolios and country excess returns

I use four different sets of currency test-assets, all of which I also include the factors themselves as test-assets. The first set includes the five Carry, five Momentum and five Value portfolios, while the second set includes only the five Momentum and five Value portfolios. The third and fourth cross-sections are country-level excess returns[16], where the third includes all available countries and the fourth restricts the sample to advanced

[16]Currencies with fewer than five years of data are excluded. Missing observations are replaced by the monthly cross-sectional mean.

countries only. The Carry portfolios are the five forward premium sorted portfolios, following Lustig et al. (2011). The Momentum portfolios are built by sorting currencies on previous-month excess return, as in Menkhoff et al. (2012b), and the Value portfolios are built by sorting currencies on the 5-year log-return of the real exchange rate, as in Menkhoff et al. (2017). The foreign exchange literature tends to focus more on explaining the Carry sorted portfolios, I go a step further and add additional portfolio and country cross-sections to test the hypothesis.

4.3. Cross-sectional pricing results: estimated risk premia

The second stage results[17] for portfolio cross-sections are shown in Table 3, where columns (1) to (5) are for the Carry, Momentum and Value test assets, and columns (6) to (10) are for the Momentum and Value test assets. Table 4 show the second stage results for the country level analysis. The first noticeable thing is that there is a jump in the cross-sectional R^2 when any of the VIX term-structure currency risk factors are included, relative to the model only with the Carry and Dollar factors, showing that these additional factors provide valuable information to pricing the cross-section of returns. This pattern is even more pronounced in the country level cross-sections. The GRS test behaves differently across the two types of test assets. In the portfolio cross-sections and among advanced countries I cannot reject the null hypothesis that all pricing errors are zero at the 5% significance level in any specification. In the all-country cross-section the null is rejected in every specification, including the benchmark that contains only the Carry and Dollar factors, so no model I consider prices the full country cross-section exactly, even though adding the term-structure factors raises the cross-sectional R^2 from 0.67 to as much as 0.90.

I find that the level (PC1) and curvature (PC3) factors have a negative price of risk, while the slope (PC2) factor has a positive price of risk. A higher exposure to the level (PC1) and curvature (PC3) factors leads to a lower average excess return across assets, where an asset with a beta of one pays a risk premium of about -6% to -8% and -4% to -7% per year, respectively, acting as insurance. The slope (PC2) factor pays a positive risk premium of about 7% to 9% per year for an asset with a beta of one.

In terms of the model in Section 2, the equilibrium price of risk is $\lambda_{\Delta TS,t} = -\kappa_t W \frac{H_t}{\varphi_H}$, where $H_t = \sum_c h_t^{(c)}$ is the aggregate expert hedging mandate across all countries. The volatility-linked claims are derivatives contracts with zero net supply, so experts and arbitrageurs stand on opposite sides of every contract. The sign of each λ_k is therefore the

[17] I use annualized monthly excess returns for the test assets and the risk factors.

opposite of the sign of the aggregate mandate, while its magnitude is determined by the maturity-specific supply friction W_{kk}, intermediary tightness κ_t and the hedge efficiency φ_H.

Experts collectively demand protection against a rise in the whole volatility curve ($H_L > 0$), for example because institutional portfolios lose value broadly when volatility rises, and they pay for that protection. Arbitrageurs absorb the other side of this trade, are left net short the level bucket, and since the arbitrageur is the marginal pricer in this economy the equilibrium condition delivers a negative price of level risk, $\lambda_L < 0$, with magnitude governed by intermediary tightness κ_t, the maturity-specific friction W_{LL} and the hedge efficiency φ_H. This is the volatility market's counterpart of the negative variance risk premium documented by Dew-Becker et al. (2017). A negative λ_L identifies level volatility as a hedge factor: currencies with a positive level beta pay off when volatility rises and therefore provide insurance to the Home investor who holds them. As in Lustig et al. (2011) and Drechsler and Yaron (2010), investors are willing to accept a negative expected excess return in exchange for that insurance, and by Equation 26 the currencies earning the most negative premia are those whose own hedging share exceeds Home's, since with $\lambda_L < 0$ the beta sort places them in the top basket.

Experts also demand protection against curvature shocks ($H_C > 0$), for instance because institutional portfolios carry nonlinear sensitivities to volatility that concentrate at intermediate horizons. When arbitrageurs face high costs of absorbing convexity risk (W_{CC} large), the market-clearing condition produces a negative price of curvature risk, $\lambda_C < 0$. Curvature therefore acts as a hedge factor analogously to the level, and currencies positively exposed to curvature shocks provide convexity insurance to their holders, who accept a negative expected return for that protection.

The slope factor, by contrast, earns a positive price of risk, and its interpretation requires one further piece of information that is a property of the empirical construction rather than of the model, namely the orientation of the factor. The estimated loadings on the second principal component increase with maturity, and the component steps down in stress episodes as short-dated implied volatility rises relative to long-dated. Demand for protection against a spike in short-horizon volatility is therefore a claim that pays when the slope factor falls, which is a negative position in the slope bucket, so $H_S < 0$ is exactly experts demanding short-horizon volatility protection. Arbitrageurs take the other side, are left long the slope bucket, and given the relatively high cost of absorbing short-dated volatility exposure (W_{SS} large) require positive compensation, which is $\lambda_S > 0$. Slope volatility is therefore a risk factor rather than a hedge: currencies exposed to slope shocks

bear uninsured risk in the states where short-horizon volatility spikes and experts most value protection, and must be compensated with positive expected premia.

These risk prices are economically large across all four cross-sections, ranging from roughly 60% of the Carry price of risk in the portfolio and advanced-country cross-sections to about 30% higher than the Carry price of risk among all countries. They are statistically significant in every specification that includes a single term-structure factor. The one exception arises in the joint specification for the all-country cross-section, where the level and slope factors absorb most of the common variation and the curvature price of risk falls to -0.041 and is no longer statistically significant. Taken together, preferred-habitat hedging motives across volatility horizons, combined with maturity-dependent intermediary frictions, generate scarcity-based prices of risk through the aggregate mandate H_t and the arbitrageur cost structure $\kappa_t W$. Currency premia arise because countries differ in their relative hedging mandates $\omega_{c,t}$, which determine how much each currency's excess return loads on the priced global volatility term-structure shocks, and the cross-sectional dispersion in expected returns is driven by heterogeneity in $\omega_{c,t}$ across countries. By Equation 26 a currency earns a higher premium the smaller its own country's share of aggregate hedging demand relative to Home's, and this holds in every bucket whatever the sign of λ_k, because the premium depends on the price of risk only through its square. What the sign of λ_k settles is not the ordering of premia but whether the bucket is a hedge or a risk factor. The three signs together say that experts buy protection at all three horizons, directly in the level and curvature buckets and, because of the orientation of the second principal component, through a negative position in the slope bucket, and that arbitrageurs are compensated for the side they are left holding in each. That is a statement about the direction of net aggregate volatility hedging demand, which is not directly observable, recovered from currency data alone. It is a stronger claim than the paper could make without the premium in Equation 26, and it is checkable against evidence from outside the currency market, since the direction of hedging demand at the front of the volatility curve has been measured directly in the variance risk premium literature.

4.4. Bayesian selection of the stochastic discount factor

The estimates in Table 3 and Table 4 answer how large the prices of risk are, conditional on a model. They do not answer whether the term-structure factors belong in the pricing kernel at all once they compete directly with Carry and Dollar, which is a question about the model rather than about a parameter inside it. Two Bayesian exercises address the two questions in turn. Appendix G reports the first, in which the Bayesian Fama–MacBeth estimator of Bryzgalova et al. (2024) replaces the asymptotic Shanken (1992) correction

Table 3: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the portfolio-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of Carry, Momentum, and Value portfolios, while columns (6)–(10) use only the Momentum and Value portfolios. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Carry + Momentum + Value					Momentum + Value				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.075^{***}			-0.070^{***}		-0.072^{***}			-0.070^{***}
		(0.021)			(0.021)		(0.021)			(0.021)
β_{fPC2}			0.075^{***}		0.072^{***}			0.074^{***}		0.072^{***}
			(0.022)		(0.022)			(0.023)		(0.022)
β_{fPC3}				-0.070^{***}	-0.068^{***}				-0.069^{***}	-0.067^{***}
				(0.019)	(0.019)				(0.020)	(0.019)
β_{CAR}	0.114^{***}	0.109^{***}	0.110^{***}	0.109^{***}	0.109^{***}	0.161^{**}	0.110^{***}	0.112^{***}	0.111^{***}	0.111^{***}
	(0.028)	(0.026)	(0.026)	(0.026)	(0.025)	(0.065)	(0.025)	(0.025)	(0.025)	(0.025)
β_{DOL}	-0.022	-0.022	-0.022	-0.022	-0.022	-0.023	-0.022	-0.022	-0.022	-0.022
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.018)	(0.017)	(0.017)	(0.017)	(0.017)
R^2	0.890	0.965	0.976	0.976	0.982	0.792	0.966	0.976	0.975	0.982
GRS	0.871	0.865	0.894	0.908	0.893	0.808	0.808	0.789	0.84	0.784

Table 4: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the country-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of all country excess returns, while columns (6)–(10) use only advanced countries currency excess returns. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Country level - all					Country level - advanced				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.082^{***}			-0.062^{**}		-0.065^{***}			-0.069^{***}
		(0.027)			(0.026)		(0.022)			(0.022)
β_{fPC2}			0.080^{***}		0.086^{***}			0.071^{***}		0.073^{***}
			(0.029)		(0.028)			(0.023)		(0.023)
β_{fPC3}				-0.069^{**}	-0.041				-0.059^{***}	-0.058^{***}
				(0.027)	(0.027)				(0.020)	(0.020)
β_{CAR}	0.075^{**}	0.058^{**}	0.060^{**}	0.056^{**}	0.063^{**}	0.114^{**}	0.112^{***}	0.111^{***}	0.112^{***}	0.113^{***}
	(0.032)	(0.029)	(0.028)	(0.028)	(0.028)	(0.052)	(0.026)	(0.026)	(0.026)	(0.026)
β_{DOL}	-0.021	-0.021	-0.021	-0.021	-0.021	-0.014	-0.014	-0.016	-0.016	-0.017
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)
R^2	0.669	0.834	0.900	0.839	0.903	0.520	0.894	0.908	0.886	0.922
GRS	0.018	0.017	0.025	0.007	0.01	0.196	0.201	0.259	0.22	0.263

with the exact finite-sample posterior of each price of risk, and confirms the sign pattern and the improvement in cross-sectional fit documented above. Its credible intervals support the classical inference outside the advanced-country cross-section, where they include zero. Below, I explore the second question.

I use the continuous spike-and-slab estimator of Bryzgalova et al. (2023), which places on each factor's SDF risk price the mixture prior

$$\lambda_j \mid \gamma_j, \sigma^2 \sim N\Big(0,\, r(\gamma_j)\,\psi_j\,\sigma^2\Big)\,, \tag{38}$$

where $\gamma_j \in \{0,1\}$ is a latent inclusion indicator, $r(1) = 1$, and $r(0) = 0.001$, so that an excluded factor has a risk price tightly concentrated at zero while an included one is free to be large. The indicators carry a symmetric Beta–Bernoulli prior, $\Pr(\gamma_j = 1 \mid \omega_j) = \omega_j$ with $\omega_j \sim \text{Beta}(1,1)$, which gives every factor a prior inclusion probability of one half and lets the posterior be read directly as how far the data move that probability. The prior scale ψ_j is calibrated so that the prior Sharpe ratio of the SDF equals 0.10, following the standard mapping in Bryzgalova et al. (2023), and the factors are treated as traded assets that the model must price. The candidate SDF contains the three VIX term-structure factors together with Carry and Dollar. The test assets in each cross-section are the same as in the asset pricing tests, with the corner Carry portfolios excluded to remove the exact linear dependence between the Carry factor and the corner forward-premium-sorted portfolios.

Table 5 reports the results next to their values under the prior, which is the relevant benchmark for judging how much the data move the model probabilities. Panel A shows that the posterior inclusion probabilities of the three term-structure factors rise from the prior value of one half to between 0.64 and 0.76 across the four cross-sections, comparable to the Carry factor and clearly above the Dollar factor, which stays at its prior. The evidence for any individual factor is thus positive but moderate, which is unsurprising. Level, slope, and curvature innovations are correlated snapshots of the same term-structure, so the sampler can substitute one for another and no single factor becomes indispensable. Panel B asks the question that matches the paper's claim, namely whether the term-structure as a block belongs in the pricing kernel, and there the evidence is considerably sharper. The posterior probability that at least one term-structure factor enters the SDF is 0.98 to 0.99, against a prior of 0.875, and the posterior probability that two or more enter is 0.78 to 0.81, against a prior of one half. The expected number of included term-structure factors rises from 1.5 under the prior to slightly above two in every cross-section. The multi-horizon dimension of this result matters for the mechanism in Section 2. Level,

slope, and curvature load on hedging pressures at different volatility horizons, and the posterior concentrating on models with two or more of them indicates that the cross-section of currency returns compensates more than one of these horizons rather than a single summary measure of volatility risk.

Table 5: Bayesian SDF model selection for the VIX futures term-structure factors. This table reports posterior summaries from the continuous spike-and-slab SDF estimator of Bryzgalova et al. (2023), applied to the candidate stochastic discount factor containing the three VIX futures term-structure currency factors ($fPC1$, $fPC2$, $fPC3$) together with the Carry and Dollar factors, for each of the four test-asset cross-sections used in the asset pricing tests. Panel A reports the posterior probability that each factor is included in the SDF, i.e., the posterior mean of the inclusion indicator γ_j. Panel B aggregates the three term-structure factors and reports the posterior probability of including none, at least one, exactly one, two or more, or all three of them, together with the expected number of included term-structure factors and the expected total model size. The Prior column reports the value of each quantity under the symmetric Beta–Bernoulli prior, which assigns each factor an inclusion probability of one half. The prior scale ψ is calibrated so that the prior Sharpe ratio of the SDF is 0.10. Factors are treated as traded and priced by the model.

	Prior	Carry + Mom + Value	Mom + Value	Country - all	Country - advanced
Panel A: posterior factor inclusion probabilities					
$fPC1$	0.500	0.760	0.709	0.755	0.735
$fPC2$	0.500	0.687	0.668	0.723	0.685
$fPC3$	0.500	0.678	0.676	0.670	0.644
CAR	0.500	0.741	0.726	0.781	0.737
DOL	0.500	0.596	0.538	0.536	0.508
Panel B: term-structure block probabilities and model size					
Pr(no fPC)	0.125	0.015	0.022	0.015	0.024
Pr(at least one fPC)	0.875	0.985	0.978	0.985	0.977
Pr(exactly one fPC)	0.375	0.179	0.203	0.172	0.189
Pr(two or more fPC)	0.500	0.806	0.775	0.813	0.787
Pr(all three fPC)	0.125	0.335	0.299	0.350	0.300
Expected number of fPCs	1.500	2.125	2.053	2.148	2.063
Expected model size	2.500	3.463	3.317	3.465	3.309

5. Premium news or expected-volatility news? What the factors are made of

The tests so far establish that currencies loading on the VIX futures term-structure earn systematically different returns, and that the term-structure belongs in the pricing kernel. A movement in the VIX futures curve bundles together two economically different pieces of news, and the distinction between them is the distinction between the mechanism proposed in Section 2 and the leading alternative. Part of any movement is a revision in how much volatility the market expects the contracts to settle on. The rest is a revision in the compensation demanded for bearing volatility risk. If the currency premia documented

above are compensation for exposure to volatility as a macroeconomic state variable, in the spirit of the intertemporal hedging demands of Drechsler and Yaron (2010), then the expected-volatility part of the factor returns should carry the price of risk. If instead they reflect the intermediated hedging-demand mechanism of the model, in which the equilibrium price of risk vector $\lambda_{\Delta TS,t} = -\kappa_t W\left(\frac{H_t}{\varphi_H}\right)$ of Equation 10 is generated by mandate pressure intermediated at a balance-sheet cost, then the premium part should carry it. I therefore split the factors into those two components and ask which one the cross-section pays for.

5.1. Measuring news about expected volatility and splitting the factors

To isolate revisions in expected volatility I need a model of expected volatility. I build one from the object the VIX futures ultimately settle on, namely S&P 500 realized volatility. Monthly realized volatility is the annualized square root of the sum of squared daily log returns within the month, and the analysis works with its log, $v_m = \log RV_m$, because volatility is approximately log-normal and linear forecasting models behave far better in logs.

The forecasting model is a heterogeneous autoregressive (HAR) specification in the spirit of Corsi (2009), which forecasts volatility using its own averages over windows of different lengths and thereby captures the fact that volatility has both fast-moving and slow-moving components. Although simple, this model class is widely considered the benchmark in realized volatility forecasting. The original specification works at daily frequency, with realized volatility built from intraday returns and averages over one day, one week and one month. I use a monthly version of it, with realized volatility built from daily returns as above and averages over one, three and twelve months. For each forecast horizon $h \in \{1, 3, 6\}$ months I estimate

$$\bar{v}_{m+1:m+h} \;=\; c_h + \beta_{1,h}\, v_m + \beta_{3,h}\, \bar{v}_{m-2:m} + \beta_{12,h}\, \bar{v}_{m-11:m} + u_{m+h}, \tag{39}$$

where $\bar{v}_{a:b}$ is the average of v over months a through b. The three backward-looking averages on the right-hand side, over one, three, and twelve months, are the same in all three regressions, and what varies across them is how far ahead the target looks. Writing $E_m^{(h)}$ for the fitted value of (39), the month-m forecast of average log volatility over the next h months, the news at horizon h is the revision of that forecast,

$$n_{h,m} \;=\; E_m^{(h)} - E_{m-1}^{(h)}. \tag{40}$$

The move from forecast to revision is the one that matters. Returns respond to surprises rather than to levels. The expected volatility prevailing at the start of month m is already known and cannot generate an unexpected return during that month, and what can move prices within the month is the update to that expectation as information arrives. Equation (40) is precisely that update. If the factor returns were compensation for expected-volatility risk, they would co-move with these revisions.

Two further choices are deliberately generous to the expected-volatility interpretation, so that the test is stacked against the conclusion I ultimately reach. First, I add the contemporaneous change in realized log volatility, Δv_m, as a fourth news series, which allows pure realized-volatility surprises to count as expected-volatility news as well. Second, the HAR regressions are estimated on the full sample, so the forecasts are as good as this model class can be. The news vector is therefore $N_m = (n_{1,m},\, n_{3,m},\, n_{6,m},\, \Delta v_m)'$, with three horizons included so that news about short-run, medium-run and long-run volatility are separately available and the data are free to pick the combination that best explains each factor.

Each factor return is then projected on the news vector over the common sample,

$$f_{k,t} \;=\; a_k + b_k' N_t + e_{k,t}, \qquad f_{k,t}^{exp} \equiv \hat{a}_k + \hat{b}_k' N_t, \qquad f_{k,t}^{prem} \equiv \hat{e}_{k,t}, \tag{41}$$

so that the fitted value is the expected-volatility-news component and the residual is the premium-news component. Ordinary least squares makes this a clean split for three reasons. The two components add back to the factor in every month, so nothing is lost or double counted. They are orthogonal in sample, so the variances add and the projection R^2 is the share of the factor's variance attributable to expected-volatility news. And the residual has a clear reading, because under the maintained identity that a volatility-curve innovation is expected-volatility news plus premium news, whatever a generous measure of expected-volatility news fails to explain is premium news. The reading is a bound rather than an equality. The residual also keeps any expected-volatility news that the HAR forecast misses, since its information set is past realized volatility alone, and any variation in the factor that is unrelated to volatility. The premium component is therefore an upper bound on premium news, and the expected-volatility component a lower bound on expected-volatility news. This is a full-sample variance decomposition, a spanning exercise rather than a real-time forecasting one, which is the appropriate design for asking what the factors are made of.

Table 6 reports the projection. Expected-volatility news spans only 5.8%, 6.4%, and 5.7% of the variance of the level, slope, and curvature factors. The factors are therefore

at most about 94% premium news by variance, and the premium components correlate at 0.97 with the original factors. The shares are nearly uniform across the three factors. Even for the slope factor, where calendar-spread positions make trading revisions in the expected volatility path most natural, the premium component dominates by an order of magnitude.

Table 6: Decomposition of the VIX futures term-structure factor returns into expected-volatility news and premium news. Each factor return is regressed on the month-on-month revisions of full-sample HAR forecasts of future SPX realized log volatility at horizons of 1, 3, and 6 months and on the contemporaneous change in realized log volatility. The fitted value is the expected-volatility-news component and the residual is the premium-news component. The table reports the share of the variance of each factor spanned by expected-volatility news (the R^2 of the projection), the annualized standard deviation of the factor and of the two components, and the correlation between the premium component and the original factor.

	Share expected-vol news	SD factor	SD exp	SD $prem$	Corr($prem$, factor)
$fPC1$	0.058	0.260	0.063	0.252	0.971
$fPC2$	0.064	0.277	0.070	0.268	0.967
$fPC3$	0.057	0.235	0.056	0.228	0.971

5.2. Pricing the components with mimicking portfolios

Knowing what the factors are made of is not the same as knowing which part is paid for, and the natural shortcut fails. The average of f^{prem} is zero by construction because it is an ordinary least squares residual, while f^{exp} inherits the factor's entire mean through the projection intercept. Averages of the raw components therefore say nothing about pricing and are also not traded returns.

Mimicking portfolios are an option for turning every series, traded or not, into an implementable position whose average return can be read directly as a premium. Fix a cross-section with test-asset excess returns R_t. For each series $s_t \in \{f_k,\, f_k^{prem},\, f_k^{exp}\}$ I run the full-sample regression

$$s_t \;=\; a + w' R_t + \varepsilon_t, \qquad m_t \;\equiv\; \hat{w}' R_t, \tag{42}$$

and define the mimicking portfolio m_t as the fitted combination of test assets, dropping the intercept. Because the weights $\hat{w}$ are fixed numbers and each element of R_t is a tradable currency portfolio, m_t is the excess return on a position an investor could actually hold, and its sample mean is a risk premium in the most direct sense available, namely the average payoff to holding it. It is also the best replication of s_t that the cross-section can produce, with the regression R^2 reporting how much of the series those assets span. The

original factor and both of its components pass through exactly the same regression on exactly the same assets and sample, so nothing in the comparison depends on which series happens to be traded.

The construction has a property that makes the results readable at a glance. The regression coefficient $\hat{w}(s) = \widehat{\mathrm{Var}}(R)^{-1}\widehat{\mathrm{Cov}}(R, s)$ is linear in s, and the decomposition is additive month by month, so covariances add and the weights add,

$$\bar{m}^{orig} \;=\; \bar{m}^{prem} + \bar{m}^{exp}. \tag{43}$$

The factor's mimicking premium splits exactly, with no residual term, into a premium-news part and an expected-volatility part. The share reported in the table is $\bar{m}^{prem}/\bar{m}^{orig}$, and because the premium component is an upper bound on premium news, so is this share of the factor's premium. Inference is Newey–West with three lags on the mean of each mimicking-portfolio return.

Table 7 reports the mimicking premia for all four cross-sections. Take the slope factor in the Carry, Momentum, and Value cross-section as a worked example. The currency portfolio that mimics $fPC2$ earns 5.91% per year, the portfolio mimicking its premium-news component earns 5.62% per year, which is 95% of it, and the portfolio mimicking its expected-volatility news earns 0.29% and is indistinguishable from zero. The same split holds throughout the portfolio cross-sections, where the premium-news component earns 95% to 100% of the original factor's mimicking premium. The premium portfolios earn −7.1%, 5.6%, and −5.2% per year for the level, slope, and curvature factors in the Carry, Momentum, and Value cross-section, with Sharpe ratios between 0.98 and 1.27 in absolute value, against at most 0.3% per year, never statistically distinguishable from zero, for the expected-volatility portfolios. The Momentum and Value cross-section shows the same pattern, with premium shares of 97% to 100%. In the all-country cross-section, where the 34 currencies replicate the factors closely with mimicking R^2 of 0.69 to 0.80, the premium component earns 88% to 96% of the factor's premium and the expected-volatility portfolios earn at most 0.6% per year, again never distinguishable from zero. The advanced-country cross-section remains the weakest, but the split is unchanged wherever there is a premium to split. The level and slope factors' mimicking premia of −2.7% and 3.4% per year are 93% and 90% premium news, while the curvature factor's premium of −1.3% per year is significant only at the 10% level. What sets that cross-section apart is how little of the factors it can replicate at all. Its mimicking R^2 of 0.20 to 0.31, against 0.36 to 0.62 in the portfolio cross-sections, indicates that the advanced currencies on their own contain limited variation in volatility-premium exposure, which is consistent with that cross-section being

the weakest for the original factors throughout the pricing tests.

Johnson (2017) reaches the same conclusion about the VIX term-structure from a different direction. He works with the curve itself and tests the expectations hypothesis, asking whether curve-implied forecasts predict subsequent realized changes with a coefficient of one, as they would if the shape of the term-structure reflected the expected path of volatility. He rejects that restriction throughout and concludes that variation in the VIX term-structure is compensation for variance risk rather than a forecast of the volatility path, with a single slope component summarizing most of the priced information in the curve. The exercise in this section never looks at the curve directly. It works with the returns on currency portfolios sorted on exposure to the curve, and asks how much of those returns is spanned by revisions in a physical forecast of volatility. The two designs share no statistical structure and answer questions posed in different spaces, there about the curve and here about the cross-section of currencies, and they nonetheless agree that what moves the VIX term-structure, and what currencies are compensated for loading on, is the premium rather than the expectation. The agreement matters here because the decomposition depends on the HAR model being an adequate proxy for the market's expectation, whereas an expectations-hypothesis test requires no forecasting model at all. Where it differs is in which part of the curve carries the pricing. He finds that only the slope component predicts variance asset returns and that the level component does not, while in the cross-section of currencies all three factors carry a significant price of risk. The mechanism in Section 2 accommodates this difference, because what is priced in currencies is hedging pressure at particular horizons rather than the overall quantity of volatility risk.

Taken together, these results sharpen the interpretation of the main findings. The prices of risk documented in Section 4 are compensation for exposure to revisions in the price of volatility exposure, of the kind the mandate pressure H_t and the intermediation cost $\kappa_t W$ of Equation 10 generate, and not for exposure to news about the future quantity of volatility. Currencies earn or pay their term-structure premia because they co-move with shocks to the cost of hedging volatility at different horizons, which is exactly the channel through which the preferred-habitat mechanism operates, rather than because they hedge revisions in expected volatility itself. Read through the model, the result is also evidence for time variation in that compensation, since were the price of risk constant the revision in it would be identically zero, every curve movement would be news about expected volatility, and the decomposition would find the opposite corner of what it finds. What it cannot do is attribute the compensation to a particular source, because premium

Table 7: Tracking-portfolio premia of the term-structure factors and their components. For each cross-section of test assets, each series s, namely the original term-structure factor (fPC), its premium-news component ($prem$), and its expected-volatility-news component (exp), is regressed on the test-asset excess returns R_t over the full sample, $s_t = a + w^\top R_t + \varepsilon_t$, and the estimated weights define the tracking portfolio $m_t = \hat{w}^\top R_t$: the traded combination of the cross-section's own assets that best replicates the series. Mean is the annualized mean excess return of the tracking portfolio in percent per year, with Newey–West t-statistics (3 lags) in brackets, SR is its annualized Sharpe ratio, R^2 is the tracking regression R^2, the share of the series the test assets can replicate. Because the decomposition is additive, the two components' tracking portfolios sum to the factor's, so their means split the factor's tracking premium; Share *prem* is the premium-news fraction of that premium, not reported when the original tracking premium is below 0.5% per year. Reading Panel A, slope factor, as an example: the currency portfolio that tracks $fPC2$ earns 5.91% per year; the portfolio tracking its premium-news component earns 5.62% of it (a share of 0.95), while the portfolio tracking its expected-volatility news earns 0.29% and is indistinguishable from zero.

	Original factor			Premium news			Expected-vol news			
	Mean	SR	R^2	Mean	SR	R^2	Mean	SR	R^2	Share *prem*
Panel A: Carry + Mom + Value (N = 15)										
$fPC1$	-7.36	-1.25	0.61	-7.09	-1.27	0.59	-0.27	-0.31	0.23	0.96
	[-3.11]			[-3.19]			[-1.00]			
$fPC2$	5.91	0.94	0.62	5.62	0.98	0.55	0.29	0.29	0.24	0.95
	[2.27]			[2.38]			[0.87]			
$fPC3$	-5.47	-1.06	0.58	-5.22	-1.11	0.51	-0.25	-0.32	0.23	0.95
	[-2.55]			[-2.64]			[-0.97]			
Panel B: Mom + Value (N = 10)										
$fPC1$	-3.31	-0.74	0.36	-3.29	-0.78	0.34	-0.02	-0.02	0.16	1.00
	[-1.98]			[-2.12]			[-0.07]			
$fPC2$	2.39	0.48	0.40	2.33	0.53	0.33	0.06	0.08	0.16	0.97
	[1.19]			[1.31]			[0.23]			
$fPC3$	-2.55	-0.58	0.43	-2.50	-0.63	0.37	-0.06	-0.09	0.16	0.98
	[-1.48]			[-1.60]			[-0.26]			
Panel C: Country - all (N = 34)										
$fPC1$	-4.68	-0.69	0.80	-4.48	-0.71	0.76	-0.19	-0.17	0.38	0.96
	[-1.94]			[-1.98]			[-0.62]			
$fPC2$	5.11	0.73	0.77	4.52	0.69	0.71	0.59	0.48	0.37	0.88
	[1.96]			[1.89]			[1.54]			
$fPC3$	-2.55	-0.45	0.69	-2.27	-0.43	0.65	-0.28	-0.28	0.37	0.89
	[-1.17]			[-1.11]			[-0.97]			
Panel D: Country - advanced (N = 13)										
$fPC1$	-2.69	-0.64	0.31	-2.50	-0.65	0.28	-0.19	-0.21	0.26	0.93
	[-2.33]			[-2.37]			[-0.87]			
$fPC2$	3.35	0.76	0.31	3.03	0.77	0.26	0.32	0.34	0.22	0.90
	[2.82]			[2.88]			[1.33]			
$fPC3$	-1.34	-0.44	0.20	-1.11	-0.42	0.16	-0.22	-0.28	0.25	0.83
	[-1.68]			[-1.57]			[-1.12]			

news is measured as what a deliberately generous volatility forecast fails to span, so it identifies neither H_t nor $\kappa_t W$ nor their product and cannot exclude other mechanisms that move the compensation.

6. Predictive power for currency excess returns

Having shown in the cross-sectional tests that the tradable VIX term-structure currency factors carry significant prices of risk, it is natural to ask whether the underlying innovations themselves contain information about the time-series dynamics of currency excess returns. The Fama–MacBeth analysis evaluates whether returns on factor-mimicking portfolios are compensated in the cross section, whereas the panel regressions examine whether the raw shocks to the VIX term-structure components forecast next-month currency excess returns. This provides a complementary perspective: the cross-sectional tests assess whether factor-mimicking portfolios span priced dimensions of global risk, while the panel regressions test whether the shocks generating those dimensions have predictive content for individual currencies, in light of the known difficulty in predicting exchange rates (Meese and Rogoff, 1983).

6.1. Panel data regressions

Table 8 reports the predictive regressions, which follow the specification that Londono and Zhou (2017) use to study the predictive power of the world currency variance risk premium for exchange rates. One-month-ahead currency excess returns (columns 1–2) and log spot exchange-rate returns (columns 3–4) are regressed on the interest-rate differential between the United States and country i, measured under covered interest parity as minus the log forward premium, the month-t innovations to the level ($\Delta PC1$), slope ($\Delta PC2$), and curvature ($\Delta PC3$) components, and the lagged excess return, in pooled OLS and currency fixed-effects specifications. Across all models, the innovations display statistically significant predictive power. The coefficients are uniformly positive, indicating that upward shocks to any part of the VIX term-structure are followed, on average, by higher subsequent currency excess returns and a corresponding appreciation against the dollar. Among the three, $\Delta PC2$ delivers the strongest and most precisely estimated effect, highlighting the importance of shocks to the slope for near-term currency payoffs. In economic terms, a positive $\Delta PC2$ innovation is a surprise steepening of the VIX futures curve, in which near-term volatility pricing recedes relative to long-horizon pricing. As documented in Figure 6, this steepening pattern characterizes improving global risk conditions. A one-standard-deviation steepening innovation predicts next-month currency returns higher

by roughly 26 basis points (3.2% annualized). The corresponding per-standard-deviation effects of the level and curvature innovations are about 12 and 10 basis points. To put these magnitudes in context, in the risk rebound of April 2020 $\Delta PC2$ rose by almost eight points, a four-standard-deviation steepening, as immediate pandemic stress was priced out of the front of the curve faster than the back. The estimates imply an appreciation of the average currency against the dollar of about 1.1% over the following month. $\Delta PC3$ also yields economically meaningful forecasts, and $\Delta PC1$ contributes modest but significant predictive power. Because the interest-rate differential is held constant, the ΔPC coefficients are identical across the excess-return and spot-return specifications: the predictive content of the innovations operates entirely through the expected spot-rate adjustment rather than through the interest-rate component of the excess return.

The control variables behave as the exchange-rate literature would predict. The coefficient on the interest-rate differential is close to -1 in the excess-return regressions and close to zero in the spot-return regressions: interest differentials pass through essentially one-for-one into excess returns because spot rates fail to adjust in the direction implied by uncovered interest parity, the classic forward-premium puzzle and consistent with the deviations from UIP documented by Londono and Zhou (2017). Conditional on the interest-rate differential, the lagged excess return is not significant, so the predictive content of the VIX-based innovations is absorbed by neither return persistence nor the interest-rate differential. The fixed-effects results further show that the predictive coefficients survive after removing all time-invariant currency characteristics, implying that the forecasting power reflects within-currency variation.

The incremental fit attributable to the volatility innovations is modest: the adjusted R^2 of the spot-return regressions is below one percent, while the larger values in the excess-return columns are driven mechanically by the interest-rate differential, itself a component of the excess return. Such magnitudes are typical of short-horizon exchange-rate forecasting and are of the same order as the gains Londono and Zhou (2017) report for the world variance risk premium. The results align closely with the economic interpretation developed earlier: the same shocks that produce priced, tradable risk factors in the cross-section also transmit forward-looking information about expected currency returns.

Read through the model, this is a statement about the price of volatility term-structure risk rather than about the quantity of it. Expected currency returns in Equation 26 move only through that price and through the relative hedging mandates, so a curve innovation forecasts returns only as far as it shifts intermediary tightness or aggregate hedging demand in the month that follows, which a volatility spike that tightens balance sheets or raises

demand for protection would do. The model permits that correlation without delivering it, since it leaves the dynamics of those two objects unrestricted, and it does not pin the sign of the predictive coefficient. The evidence here is therefore consistent with the mechanism.

Table 8: Predictive panel regressions. The dependent variable is the one-month-ahead currency excess return (rx_{t+1}, columns 1–2) or the one-month log change in the U.S. dollar price of currency i (Δs_{t+1}, columns 3–4, where an increase is an appreciation of currency i against the dollar). The regressors are the month-t interest-rate differential between the United States and country i ($y_t^{US} - y_t^i$, measured under covered interest parity as minus the log forward premium), the month-t innovations to the level ($\Delta PC1$), slope ($\Delta PC2$), and curvature ($\Delta PC3$) of the VIX futures term-structure, and the lagged excess return rx_t. Columns (1) and (3) report pooled OLS estimates for all currencies, and columns (2) and (4) report corresponding within estimations with individual (currency) fixed effects. Panel Newey–West standard errors (one lag) are shown in parentheses. Constant terms are included in the pooled OLS specifications but their coefficients are omitted from the table. This is similar to the specification of Londono and Zhou (2017). Returns and the interest-rate differential are annualized. Adjusted R^2 values are reported in percent. Statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	rx_{t+1}		Δs_{t+1}	
	(1)	(2)	(3)	(4)
$y_t^{US} - y_t^i$	-0.989^{***}	-1.043^{***}	0.011	-0.043
	(0.106)	(0.105)	(0.106)	(0.105)
$\Delta PC1$	0.005^{**}	0.005^{*}	0.005^{**}	0.005^{*}
	(0.003)	(0.003)	(0.003)	(0.003)
$\Delta PC2$	0.016^{***}	0.016^{***}	0.016^{***}	0.016^{***}
	(0.003)	(0.003)	(0.003)	(0.003)
$\Delta PC3$	0.010^{**}	0.009^{**}	0.010^{**}	0.009^{**}
	(0.004)	(0.004)	(0.004)	(0.004)
rx_t	0.034	0.024	0.034	0.024
	(0.022)	(0.021)	(0.022)	(0.021)
FE	No	Yes	No	Yes
R^2 (%)	14.02	13.77	0.63	0.14

7. Conclusion

This paper examines the pricing implications of the VIX futures term-structure on currency excess returns. In a simple two-agent model where experts have preferred-habitat demand for volatility-linked contracts and arbitrageurs supply these contracts subject to maturity-dependent capital costs, I show that each country's stochastic discount factor

is driven by local and global shocks, and that sorting currencies by their exposure to innovations in the volatility term-structure yields pricing factors related to hedging demand through volatility-linked contracts. The model also delivers a conditional currency risk premium, which rises with the wedge between a country's hedging mandate share and Home's.

I build currency risk factors from long-short beta-sorted portfolios based on each currency's exposure to the principal components of the VIX futures term-structure. These term-structures summarize market-implied pricing of volatility risk across different horizons and exhibit a strong factor structure: the first three principal components (level, slope, and curvature) account for over 99% of their variation. Sorting currencies by their conditional beta with respect to these components, as suggested by the model, captures the global component of volatility term-structure risk.

I then provide empirical evidence across four sets of test assets showing that these currency factors explain a significant portion of the cross-sectional variation in excess returns. This explanatory power is incremental to the well-established Carry and Dollar factors of Lustig et al. (2011), and supports the view that innovations in VIX futures term-structure reflect priced global SDF shocks. Bayesian model selection points the same way. The posterior probability that at least one term-structure factor belongs in the stochastic discount factor rises well above its prior value in every cross-section, and the posterior concentrates on specifications containing more than one of them, which indicates that currency returns compensate volatility risk at more than one horizon rather than a single summary measure of it. Additionally, the innovations in the components of the VIX futures term-structure provide incremental information for currency excess return predictability.

Finally, I decompose the factor returns into news about expected volatility and news about the premium demanded for bearing it, which separates the mechanism of the model from the interpretation of volatility as a macroeconomic state variable. Revisions in forecasts of realized volatility span less than a tenth of the variation in the term-structure factors, and the mimicking portfolio of the premium component earns almost the entire premium of the factor it replicates, an upper bound on the share due to premium news, while the expected-volatility component earns nothing distinguishable from zero. The prices of risk documented in this paper are therefore compensation for shocks to the cost of hedging volatility across horizons, of the kind the preferred-habitat mechanism generates, rather than for news about its future quantity. The exercise does not identify that compensation with any particular source, since premium news is measured as what a

deliberately generous volatility forecast fails to span, but it is evidence for time variation in it, because a constant price of risk would leave no revision for the decomposition to find.

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Appendix

This appendix collects the derivations, the data details and the robustness exercises that support the main text. The first two sections complete the model. Appendix A derives the country loading of Equation 12 from a household and a manager who runs its volatility hedge under a mandate, and relaxes the assumption that hedging demand is inelastic. Appendix B derives the conditional currency risk premium step by step, from the Euler equation to its form in mandate shares.

The next two sections describe the data. Appendix C lists the currencies in the sample with their coverage and country group, the Datastream sources of the spot and forward rates, and the periods of large deviations from covered interest rate parity during which a currency is excluded. Appendix D reports summary statistics for the individual currencies and for the Carry, Momentum and Value portfolios used as test assets.

The remaining sections check the results. Appendix E builds the factors from three portfolios (Subsection E.1) and from ten portfolios (Subsection E.2) instead of five, and reports the rolling betas, the portfolio statistics, the Carry regressions, the factor composition and the asset pricing tests for each. Appendix F repeats the analysis, from the principal components to the asset pricing tests and the predictive regressions, on a longer sample of synthetic VIX futures that I build from S&P500 index options, which are available from 1996. Appendix G estimates the cross-sectional tests with the Bayesian Fama–MacBeth estimator of Bryzgalova et al. (2024), which gives the exact finite-sample posterior of each price of risk, and reports how often the posterior keeps the sign pattern of the main text. Appendix H uses index options from three European markets to test whether the US curve contains a global component and whether the currency factors load on it.

A. Microfoundation of the country loadings and elastic hedging demand

The model section posits the country loading in Equation 12 and takes hedging demand to be inelastic. This appendix derives the first and relaxes the second. Both follow from one structure, a household that owns the wealth and a manager who runs its volatility hedge under a mandate, and the two results are connected. The loading is recovered up to a positive scale, and that scale is where the elasticity sits, so intermediary tightness reaches currency premia only to the extent that hedging demand responds to price.

A.1. Households and managers

Each country has a household and a manager. The household owns the wealth, has constant absolute risk aversion a over it, and its marginal utility is the country kernel of Equation 13. Its first-stage portfolio choice, which I do not model, leaves it with a background exposure $b_t^{(c)} \in \mathbb{R}^3$ to the global volatility shock, so that its wealth loads on ΔTS_{t+1}^g as $-b_t^{(c)\prime}\Delta TS_{t+1}^g$ and falls when volatility rises. The manager runs the household's volatility hedge under a mandate that covers a fraction ρ of that exposure when protection is free,

$$h_t^{(c)} = \rho\, b_t^{(c)}, \qquad \rho \in (0,1), \tag{44}$$

and is the only route through which the household holds the Home claims.

Market access is as in the model section. Households trade currencies and bonds internationally and that market is complete, which is what delivers the asset market view of the exchange rate. Households do not trade the volatility claims. Managers trade the volatility claims and nothing else, and do so under the mandate rather than as free portfolio choosers. Arbitrageurs trade the volatility claims and nothing else. No agent stands in both markets, and the manager's position is the only channel running from the claims market to the currency market. Positions and mandates are measured in units of global exposure, as $h_t^{(c)}$ is in the model section, so a position of q global units is held through q/φ_H claims and earns $\lambda'_{\Delta TS,t} q/\varphi_H$ in expectation.

A.2. Elastic demand and the equilibrium price of risk

The manager chooses the position to maximise the expected excess payoff on the hedge, net of a quadratic penalty for running it away from the mandate,

$$\begin{aligned} &\max_{q_t^{(c)}} \ \frac{1}{\varphi_H}\,\lambda'_{\Delta TS,t}\, q_t^{(c)} - \frac{1}{2}\Big(q_t^{(c)} - h_t^{(c)}\Big)'\Big[\varphi_H \operatorname{diag}(\omega_{c,t})\,\mathcal{E}\Big]^{-1}\Big(q_t^{(c)} - h_t^{(c)}\Big), \\ &\qquad q_t^{(c)*} = h_t^{(c)} + \operatorname{diag}(\omega_{c,t})\,\mathcal{E}\,\lambda_{\Delta TS,t}, \end{aligned} \tag{45}$$

where $\mathcal{E} \succeq 0$ is a diagonal elasticity matrix. The penalty matrix is diagonal and positive definite, since $\mathcal{E}$ and the shares are, so the problem is strictly concave and the first-order condition is its solution. The elasticity is proportional to the country's share, so that a country which hedges more in a bucket also adjusts more in it, and the scaling by φ_H is what makes the aggregate response in global units equal to $\mathcal{E}\lambda_{\Delta TS,t}$. As $\mathcal{E} \to 0$ the penalty becomes infinite, the manager holds exactly $h_t^{(c)}$ whatever protection costs, and Equation 45 collapses to the tracking-error problem Equation 3. The inelastic baseline is

that limit rather than a separate assumption about behaviour, which is also how Vayanos and Vila (2021) treat it. The manager's objective is not the household's utility, and that is the point. A household able to trade the claims itself would equate its marginal utility to the claims' price. The manager instead follows a mandate with a penalty, and the gap between the two is the delegation friction of Gârleanu and Pedersen (2011).

Summing Equation 45 across countries and using $\sum_c \omega_{c,t} = 1$, aggregate demand in global units is the mandate plus the elastic response, and clearing sets the arbitrageurs' position against it,

$$\sum_c q_t^{(c)*} = H_t + \mathcal{E}\,\lambda_{\Delta TS,t}, \qquad x_t = -\,\frac{H_t + \mathcal{E}\,\lambda_{\Delta TS,t}}{\varphi_H}. \tag{46}$$

Demand falls with the cost of protection, and the plus sign is what that means here. Under Equation 8 a costly hedge carries $\lambda_{k,t} < 0$, since the buyer of protection is the one who pays, so demand that recedes when protection is expensive enters with a positive sign.

Combining Equation 46 with the arbitrageur supply function Equation 6 gives the equilibrium price of risk,

$$\lambda_{\Delta TS,t} = -\left(I + \frac{\kappa_t}{\varphi_H} W\mathcal{E}\right)^{-1} \frac{\kappa_t}{\varphi_H} W H_t, \quad \lambda_{k,t} = -\,\frac{\kappa_t W_{kk} H_{k,t}}{\varphi_H\,(1 + g_{k,t})}, \quad g_{k,t} \equiv \frac{\kappa_t W_{kk}\mathcal{E}_{kk}}{\varphi_H} \geq 0, \tag{47}$$

the second expression using diagonal W and $\mathcal{E}$. The single object $g_{k,t}$ is tightness times elasticity in dimensionless units, and it separates the two cases. The dampening matrix has eigenvalues in $(0, 1]$, so elasticity lowers the price of risk in absolute value without changing its sign, and at $\mathcal{E} = 0$ the expression is Equation 10. The realised aggregate position is $H_{k,t}/(1 + g_{k,t})$, smaller than the mandate because managers economise on costly protection, so the price of risk is set by the realised position rather than by the mandate, while the sign of $\lambda_{k,t}$ is the sign of $-H_{k,t}$ whatever the elasticity.[18]

A.3. The country loading derived

With constant absolute risk aversion the log kernel loads on a shock as a times the household's wealth exposure to it. The household carries the background exposure $-b_t^{(c)}$ to the global shock and, through the manager's position of $q_t^{(c)*}/\varphi_H$ claims, the claims' payoff, which by Equation 1 is φ_H units of the global shock and ψ_H units of the Home

[18] Diagonality of W and $\mathcal{E}$ does real work rather than simplifying. With an off-diagonal $\mathcal{E}$ the dampening matrix can reverse the sign of a bucket's price of risk while every eigenvalue stays in $(0, 1]$, since eigenvalue bounds do not constrain individual components. Every bucket-level sign statement in the paper is therefore conditional on both matrices being diagonal, or on an own-bucket dominance condition that would have to be verified.

local shock per claim. The kernel loads on each accordingly,

$$\theta^{g}_{c,t} = -\,a\left(b^{(c)}_{t} - q^{(c)*}_{t}\right), \qquad \theta^{\ell}_{c,t} = a\,\frac{\psi_H}{\varphi_H}\,q^{(c)*}_{t}. \tag{48}$$

The global loading is minus risk aversion times the unhedged residual, so a country left exposed to a rise in volatility has marginal utility that rises when volatility rises, which is the sign a hedge factor needs. The local loading is the basis risk the hedge carries, and it is positive, since holding the claims means holding the Home local component.

Substituting the mandate Equation 44, the position Equation 45, the share definition $h^{(c)}_{t} = \omega_{c,t} \odot H_t$ and Equation 47 solved for H_t,

$$\theta^{g}_{c,t} = \xi_t \odot \omega_{c,t} \odot \lambda_{\Delta TS,t}, \quad \xi_{k,t} = \frac{a}{\rho}\left[\mathcal{E}_{kk} + (1-\rho)\,\frac{\varphi_H}{\kappa_t W_{kk}}\right] > 0, \quad \theta^{\ell}_{c,k,t} = a\,\frac{\psi_H}{\varphi_H}\,\frac{\omega_{c,k,t} H_{k,t}}{1+g_{k,t}}. \tag{49}$$

The global loading is Equation 12 exactly, up to the positive scale $\xi_{k,t}$, which is common to all countries within a bucket. One global price of risk per bucket, one country-specific share per bucket, and the separation of global pricing from local institutional structure, all preserved and now with a reason. The scale passes through every sign and every ranking in the paper unchanged.

Where it carries content is the magnitude, and specifically the role of intermediary tightness. The second term in $\xi_{k,t}$ is the unhedged residual and contains $1/\kappa_t$, which cancels the κ_t inside $\lambda_{\Delta TS,t}$. The first term is the elasticity and does not cancel. The cancellation is clearest in levels, where substituting H_t back into Equation 49 gives

$$\theta^{g}_{c,k,t} = -\,\frac{a}{\rho}\,h^{(c)}_{k,t}\,\frac{(1-\rho)+g_{k,t}}{1+g_{k,t}}. \tag{50}$$

A country's global loading is its own mandate, scaled by risk aversion and the hedge ratio, multiplied by a fraction that rises from $(1-\rho)$ when hedging is inelastic to one when it is perfectly elastic. In the inelastic case $g_{k,t} = 0$ and the loading is $-a((1-\rho)/\rho)h^{(c)}_{k,t}$, which contains neither the aggregate nor κ_t. That is the sharpest form of the statement. With inelastic mandates, intermediary tightness sets the price of the claims and touches no country's marginal utility, so currency premia are what they would be with no arbitrageurs at all. I keep the exposition in shares and in the price of risk rather than in levels, because the cross-sectional driver of currency premia is the mandate wedge and the time-series driver is the price of risk, and both are visible in that form and hidden in Equation 50.

A.4. Why a mandate is needed

The derivation also shows why the cross-section exists at all. A household with no mandate, choosing $q_t^{(c)}$ by mean-variance optimisation over its global exposure, would satisfy $\lambda_{\Delta TS,t}/\varphi_H = a\,\Sigma_{\Delta TS^g}(q_t^{(c)} - b_t^{(c)})$ and hence carry $\theta^g_{c,t} = (\varphi_H \Sigma_{\Delta TS^g})^{-1}\lambda_{\Delta TS,t}$, which is identical for every country. Unconstrained agents equalise marginal rates of substitution on a traded shock, so heterogeneity in the loadings can only come from something that stops them, and the mandate is that something.

This settles what would otherwise be an open question about the Home claim, which two agents value. The arbitrageur's first-order condition prices it at $\lambda_{k,t}$, while the Home household's kernel, which prices currencies, would assign it

$$\mathrm{Cov}_t\Big(-m_{H,t+1},\, \Delta TS^{(H)}_{k,t+1}\Big) = \varphi_H\,\theta^g_{H,k,t}\,\sigma_k^2 + \psi_H\,\theta^\ell_{H,k,t}\,\sigma^2_{\mathrm{loc},k}. \tag{51}$$

These are not equal, and under the structure above they must not be. The Home household does not trade the claims. Its manager holds them under Equation 45 rather than by equating marginal utility to price, so the household's Euler equation for the claims does not hold and there is no arbitrage to be had, because the household cannot trade its way to one. The difference between $\lambda_{k,t}$ and Equation 51 is the shadow price of the mandate. Were the constraint slack, every country's loading would collapse to the common value above and the cross-section of currency premia would disappear with it. What rules out the arbitrage is participation rather than spanning, since an arbitrage needs both legs and no agent in the model can take both. Restricted asset menus of this kind are the standard primitive in this literature, as in Vayanos and Vila (2021) and Gourinchas et al. (2025), and the completeness assumed in Subsection 2.2 should be read accordingly, as completeness within the currency and bond space rather than over every payoff in the economy.

A.5. What the derivation changes in the results

Substituting the derived loadings into the premium of the model section, and using the diagonality of the two covariance matrices, the expected log currency excess return is

$$E_t[rx_{c,t+1}] = \Pi^{\mathrm{other}}_{c,t} + \tfrac{1}{2}\sum_k \left(\sigma_k^2\,\xi^2_{k,t}\lambda^2_{k,t} + \sigma^2_{\mathrm{loc},k}\left[a\tfrac{\psi_H}{\varphi_H}\tfrac{H_{k,t}}{1+g_{k,t}}\right]^2\right)\Big(\omega_{H,k,t} - \omega_{c,k,t}\Big)\Big(\omega_{H,k,t} + \omega_{c,k,t}\Big), \tag{52}$$

with $\Pi^{\mathrm{other}}_{c,t}$ collecting the non-volatility shocks exactly as in Equation 26. The Home basis risk enters with the same sign and the same cross-sectional shape as the global term,

since it multiplies the same difference of squared shares by a positive coefficient, so in the premium the two add rather than compete and every statement about signs and orderings holds for them together. The chain from the model's price of risk to the estimated one therefore picks up exactly one new positive scale, $\xi_{k,t}\lambda_{k,t}$ in place of $\lambda_{k,t}$, which leaves the sign result untouched and joins the list of unmeasured positive quantities standing between the model and the estimate.

The observable beta is the one case in which the two components compete. Measured against the Home curve innovation rather than the global shock, and using both loadings of Equation 49, the beta is the mandate wedge multiplied by a common factor whose global piece carries the sign of $\lambda_{k,t}$ and whose local piece carries the sign of $H_{k,t}$. Those are opposite, so the sort ranks currencies as the model intends if and only if

$$\varphi_H^2 \left(1 - \rho + g_{k,t}\right) \sigma_k^2 \;>\; \rho\, \psi_H^2\, \sigma_{\mathrm{loc},k}^2. \tag{53}$$

When Equation 53 fails the ranking inverts and the long-short factor carries the sign opposite to $\lambda_{k,t}$. This is a restriction on the measurement of the curve rather than on the mechanism, and it is the price of using a single country's volatility curve as the proxy for the global one, relaxed by a larger φ_H. It is also the condition that carries the rank-invariance argument of the model section, which is stated for the unobservable global beta, across to the beta the sort actually uses.

A.6. Inelastic against elastic

The two cases agree on everything the paper reports. In both the global loading is $\xi_{k,t}\omega_{c,k,t}\lambda_{k,t}$ with $\xi_{k,t} > 0$, the premium is Equation 52, the sort orders on the mandate wedge under Equation 53, the traded factor recovers the sign of $\lambda_{k,t}$, and Equation 8 maps that sign to the direction of aggregate hedging demand. The reading of the three estimated signs holds in both cases word for word, and no estimate, table or sample depends on which case the world is in.

They differ on magnitudes, and on one thing that matters for what the paper can claim. In the inelastic case κ_t enters the price of the claims and nothing else, and the intermediary sector explains the variance risk premium on those claims while playing no part in the currency cross-section. In the elastic case a tighter intermediary raises the cost of protection, managers hedge less, households' residual exposure grows, their kernels load more heavily on the shock, and the global part of currency premia rises in magnitude by the factor $[(1-\rho) + g_{k,t}]^2/(1+g_{k,t})^2$, which increases from $(1-\rho)^2$ at $g_{k,t} = 0$ to one as $g_{k,t}$ grows. The effect is therefore bounded, and it works through exactly one door. The

local part moves the other way, falling as $1/(1+g_{k,t})^2$ because elastic managers hold fewer claims and carry less basis risk, so the time-series prediction needs Equation 53 as well.

The two cases make the same cross-sectional predictions and different time-series ones, so the place to separate them is the conditional premium, where the coefficient in Equation 52 rises with intermediary tightness when $\mathcal{E}_{kk} > 0$. The tests reported in this paper cannot do it. The factor return carries the sign of $\lambda_{k,t}$, so the dependent variable would have to be the sign-adjusted premium, the same coefficient moves with $H^2_{k,t}$ and with the basket gap in squared shares, so hedging demand and portfolio composition would have to be controlled for, and intermediary tightness is itself endogenous to volatility, so identification needs variation in it that the volatility state does not drive. That is a new exercise rather than a reading of the tables here, and either outcome is consistent with every result reported. What the derivation establishes is the conditional claim. A claim that intermediary frictions in the volatility market price currencies requires $\mathcal{E} > 0$ and should be stated together with the fact that it is untested here, while a claim that horizon-specific hedging demand prices currencies and intermediaries price the hedges needs only the inelastic case.

B. Derivation of the conditional currency risk premium

This appendix derives the conditional currency risk premium of the model section in full, from the Euler equation to the mandate-wedge form.

B.1. Objects

Let $M_{H,t+1} > 0$ price payoffs denominated in Home currency and $m_{H,t+1} = \log M_{H,t+1}$. All moments carry a t subscript and are conditional on the date-t information set $\mathcal{F}_t$.

Two assets, both held from t to $t+1$ and both measured in Home currency. The Home riskless bond, with gross return $R_{H,t} = \exp(i_{H,t})$ set at t. The country c bond, bought with Home currency and converted back at $t+1$, with gross return $R_{c,t+1}$ and log return

$$r_{c,t+1} \;\equiv\; \log R_{c,t+1} \;=\; i_{c,t} - \Delta s_{c,t+1}, \qquad rx_{c,t+1} \;\equiv\; r_{c,t+1} - i_{H,t}, \tag{54}$$

where $s_{c,t}$ is the log exchange rate in units of country c currency per unit of Home currency. The empirical counterpart is $f_{c,t} - s_{c,t} - \Delta s_{c,t+1}$, the same quantity under covered interest parity.

The log kernel of country c is the one stated in the model section,

$$-m_{c,t+1} = i_{c,t} + a_{c,t} + \delta_{c,t}\, u_{c,t+1} + \gamma_{c,t}\, u_{g,t+1} + \theta'_{c,t}\, \Delta TS^g_{t+1}, \qquad \theta_{c,t} = \omega_{c,t} \odot \lambda_{\Delta TS,t}, \tag{55}$$

with Home the case $c = H$, and the excess return implied by it is

$$rx_{c,t+1} = \overbrace{a_{H,t} - a_{c,t}}^{\text{known at } t} + \overbrace{\delta_{H,t} u_{H,t+1} - \delta_{c,t} u_{c,t+1} + \big(\gamma_{H,t} - \gamma_{c,t}\big) u_{g,t+1} + \big(\theta_{H,t} - \theta_{c,t}\big)' \Delta TS^g_{t+1}}^{\text{realises at } t+1}. \tag{56}$$

In a date-t conditional moment only the second group survives.

B.2. Assumptions

(E1) **No arbitrage in Home-currency payoffs.** There exists a strictly positive $M_{H,t+1}$ with $E_t[M_{H,t+1} R_{j,t+1}] = 1$ for every gross return $R_{j,t+1}$ on a Home-currency payoff. No identification of $M_{H,t+1}$ with any agent's marginal utility is required.

(E2) **Measurability.** The rates $i_{H,t}$, $i_{c,t}$ and the loadings $a_{c,t}$, $\delta_{c,t}$, $\gamma_{c,t}$, $\theta_{c,t}$ are $\mathcal{F}_t$-measurable, and the shocks $u_{c,t+1}$, $u_{g,t+1}$, ΔTS^g_{t+1} realise at $t+1$ with zero conditional mean. This is (A4) of the model section.

(E3) **Conditional joint lognormality.** $(m_{H,t+1}, r_{c,t+1})$ is jointly normal conditional on $\mathcal{F}_t$, with conditional moments that may vary with the date-t state. This is (A1).

(E4) **Completeness within the traded currency and bond space.** $\Delta s_{c,t+1} = m_{H,t+1} - m_{c,t+1}$, which is what delivers (56) from (55). This is (A2).

(E5) **Conditional orthogonality.** $u_{c,t+1}$, $u_{g,t+1}$ and ΔTS^g_{t+1} are mutually orthogonal at every date, and country-specific shocks are orthogonal across countries, so $E_t[u_{H,t+1} u_{c,t+1}] = 0$. This is (A3).

(E6) **Second moments.** $\Sigma_{\Delta TS^g} = \mathrm{Var}_t(\Delta TS^g_{t+1})$ is diagonal with entries σ_k^2, and conditional second moments do not depend on the date-t state. This is (A5).

Steps 1 to 3 use (E1) to (E3) and nothing about the model. Steps 4 to 7 use (E2) and (E4) to (E6).

B.3. Step 1. The Euler equation in levels

By (E1), applied to the two assets,

$$E_t[M_{H,t+1}\, R_{c,t+1}] = 1, \qquad E_t[M_{H,t+1}\, R_{H,t}] = 1. \tag{57}$$

By (E2) the riskless condition gives

$$R_{H,t}\, E_t[M_{H,t+1}] = 1 \qquad \Longleftrightarrow \qquad E_t[M_{H,t+1}] = \frac{1}{R_{H,t}} = \exp(-i_{H,t}). \tag{58}$$

Expanding the risky condition as $E_t[M_{H,t+1}]\,E_t[R_{c,t+1}] + \mathrm{Cov}_t(M_{H,t+1}, R_{c,t+1}) = 1$ and substituting (58),

$$E_t[R_{c,t+1}] = R_{H,t}\Big[1 - \mathrm{Cov}_t(M_{H,t+1}, R_{c,t+1})\Big] = R_{H,t} - \frac{\mathrm{Cov}_t(M_{H,t+1}, R_{c,t+1})}{E_t[M_{H,t+1}]}. \tag{59}$$

Differencing the two conditions in (57) gives the exact excess-return form,

$$E_t\Big[M_{H,t+1}\Big(R_{c,t+1} - R_{H,t}\Big)\Big] = 0, \tag{60}$$

which holds with no distributional assumption and is the last step free of (E3).

B.4. Step 2. Logs

Write $M_{H,t+1}R_{c,t+1} = \exp(m_{H,t+1} + r_{c,t+1})$, so that (57) reads $E_t[\exp(m_{H,t+1} + r_{c,t+1})] = 1$. Under (E3), for X conditionally normal,

$$E_t[\exp(X)] = \exp\Big(E_t[X] + \tfrac{1}{2}\mathrm{Var}_t(X)\Big) \qquad \Longleftrightarrow \qquad \log E_t[\exp(X)] = E_t[X] + \tfrac{1}{2}\mathrm{Var}_t(X), \tag{61}$$

the moment generating function of the normal evaluated at one. With $X = m_{H,t+1} + r_{c,t+1}$, and $\mathrm{Var}_t(m + r) = \mathrm{Var}_t(m) + \mathrm{Var}_t(r) + 2\mathrm{Cov}_t(m, r)$,

$$E_t[m_{H,t+1}] + E_t[r_{c,t+1}] + \tfrac{1}{2}\mathrm{Var}_t(m_{H,t+1}) + \tfrac{1}{2}\mathrm{Var}_t(r_{c,t+1}) + \mathrm{Cov}_t(m_{H,t+1},\, r_{c,t+1}) = 0. \tag{62}$$

With $X = m_{H,t+1} + i_{H,t}$, where $i_{H,t}$ contributes neither variance nor covariance by (E2),

$$E_t[m_{H,t+1}] + i_{H,t} + \tfrac{1}{2}\mathrm{Var}_t(m_{H,t+1}) = 0, \qquad \text{that is} \qquad -E_t[m_{H,t+1}] = i_{H,t} + \tfrac{1}{2}\mathrm{Var}_t(m_{H,t+1})\,. \tag{63}$$

B.5. Step 3. The conditional log excess return

Substituting (63) into (62), the kernel's own first and second moments cancel,

$$E_t[r_{c,t+1}] = i_{H,t} - \mathrm{Cov}_t(m_{H,t+1},\, r_{c,t+1}) - \tfrac{1}{2}\mathrm{Var}_t(r_{c,t+1})\,. \tag{64}$$

By (E2) the rate $i_{H,t}$ is a conditional constant, so

$$E_t[rx_{c,t+1}] = E_t[r_{c,t+1}] - i_{H,t}, \qquad \mathrm{Var}_t(r_{c,t+1}) = \mathrm{Var}_t(rx_{c,t+1})\,,$$
$$\mathrm{Cov}_t(m_{H,t+1},\, r_{c,t+1}) = \mathrm{Cov}_t(m_{H,t+1},\, rx_{c,t+1})\,,$$

which gives

$$E_t[rx_{c,t+1}] = \underbrace{\mathrm{Cov}_t(-m_{H,t+1},\, rx_{c,t+1})}_{\text{risk premium}} - \underbrace{\tfrac{1}{2}\,\mathrm{Var}_t(rx_{c,t+1})}_{\text{Jensen term}}. \tag{65}$$

B.6. Step 4. The covariance term

By (E2) only the date-$(t+1)$ terms of (55) and (56) enter,

$$\begin{aligned} -m_{H,t+1} &= \Big[i_{H,t} + a_{H,t}\Big] + \delta_{H,t}u_{H,t+1} + \gamma_{H,t}u_{g,t+1} + \theta'_{H,t}\Delta TS^g_{t+1}, \\ rx_{c,t+1} &= \Big[a_{H,t} - a_{c,t}\Big] + \delta_{H,t}u_{H,t+1} - \delta_{c,t}u_{c,t+1} + \Big(\gamma_{H,t} - \gamma_{c,t}\Big)u_{g,t+1} \\ &\quad + \Big(\theta_{H,t} - \theta_{c,t}\Big)'\Delta TS^g_{t+1}. \end{aligned} \tag{66}$$

Under (E5) every cross-shock product drops, including $\mathrm{Cov}_t(\delta_{H,t}u_{H,t+1},\, -\delta_{c,t}u_{c,t+1}) = 0$, so

$$\begin{aligned} \mathrm{Cov}_t(-m_{H,t+1},\, rx_{c,t+1}) = \underbrace{\delta^2_{H,t}\,\mathrm{Var}_t(u_{H,t+1})}_{\text{Home country shock}} + \underbrace{\gamma_{H,t}\Big(\gamma_{H,t} - \gamma_{c,t}\Big)\mathrm{Var}_t(u_{g,t+1})}_{\text{other global shocks}} \\ + \underbrace{\theta'_{H,t}\,\Sigma_{\Delta TS^g}\Big(\theta_{H,t} - \theta_{c,t}\Big)}_{\text{volatility term structure}}. \end{aligned} \tag{67}$$

The local component of the Home curve does not appear here. It is treated as noise in the observable proxy rather than as a priced shock, which is the modelling choice recorded with (55) in the model section.

B.7. Step 5. The Jensen term

The same orthogonality kills every cross term in the variance,

$$\begin{aligned} \mathrm{Var}_t(rx_{c,t+1}) = \delta^2_{H,t}\,\mathrm{Var}_t(u_{H,t+1}) + \delta^2_{c,t}\,\mathrm{Var}_t(u_{c,t+1}) + \Big(\gamma_{H,t} - \gamma_{c,t}\Big)^2\mathrm{Var}_t(u_{g,t+1}) \\ + \Big(\theta_{H,t} - \theta_{c,t}\Big)'\Sigma_{\Delta TS^g}\Big(\theta_{H,t} - \theta_{c,t}\Big). \end{aligned} \tag{68}$$

B.8. Step 6. The premium

Subtracting half of (68) from (67), block by block,

$$\begin{aligned}
&\delta_{H,t}^2 \mathrm{Var}_t(u_{H,t+1}) - \tfrac{1}{2}\Big[\delta_{H,t}^2 \mathrm{Var}_t(u_{H,t+1}) + \delta_{c,t}^2 \mathrm{Var}_t(u_{c,t+1})\Big] \\
&\qquad\qquad = \tfrac{1}{2}\Big[\delta_{H,t}^2 \mathrm{Var}_t(u_{H,t+1}) - \delta_{c,t}^2 \mathrm{Var}_t(u_{c,t+1})\Big], \\
&\gamma_{H,t}\big(\gamma_{H,t} - \gamma_{c,t}\big) - \tfrac{1}{2}\big(\gamma_{H,t} - \gamma_{c,t}\big)^2 \;=\; \tfrac{1}{2}\big(\gamma_{H,t}^2 - \gamma_{c,t}^2\big), \\
&\theta_{H,t}'\Sigma_{\Delta TS^g}\big(\theta_{H,t} - \theta_{c,t}\big) - \tfrac{1}{2}\big(\theta_{H,t} - \theta_{c,t}\big)'\Sigma_{\Delta TS^g}\big(\theta_{H,t} - \theta_{c,t}\big) \\
&\qquad\qquad = \tfrac{1}{2}\Big[\theta_{H,t}'\Sigma_{\Delta TS^g}\theta_{H,t} - \theta_{c,t}'\Sigma_{\Delta TS^g}\theta_{c,t}\Big],
\end{aligned} \tag{69}$$

the third line using the symmetry of $\Sigma_{\Delta TS^g}$. Hence

$$\begin{aligned}
E_t[rx_{c,t+1}] = \tfrac{1}{2}\Big[\delta_{H,t}^2 \mathrm{Var}_t(u_{H,t+1}) - \delta_{c,t}^2 \mathrm{Var}_t(u_{c,t+1})\Big] + \tfrac{1}{2}\big(\gamma_{H,t}^2 - \gamma_{c,t}^2\big)\mathrm{Var}_t(u_{g,t+1}) \\
+ \tfrac{1}{2}\Big[\theta_{H,t}'\Sigma_{\Delta TS^g}\theta_{H,t} - \theta_{c,t}'\Sigma_{\Delta TS^g}\theta_{c,t}\Big].
\end{aligned} \tag{70}$$

Each bracket pairs a Home term with the matching country c term, and by (E2) and (E5) the three Home terms are exactly $\mathrm{Var}_t(m_{H,t+1})$ and the three country terms exactly $\mathrm{Var}_t(m_{c,t+1})$, so

$$E_t[rx_{c,t+1}] = \tfrac{1}{2}\Big[\mathrm{Var}_t(m_{H,t+1}) - \mathrm{Var}_t(m_{c,t+1})\Big], \tag{71}$$

the half-variance-difference form of Backus et al. (2001).

B.9. Step 7. The premium in mandate shares

Collect the shocks the model takes as given into

$$\Pi_{c,t}^{\text{other}} \;\equiv\; \tfrac{1}{2}\Big[\delta_{H,t}^2 \mathrm{Var}_t(u_{H,t+1}) - \delta_{c,t}^2 \mathrm{Var}_t(u_{c,t+1})\Big] + \tfrac{1}{2}\big(\gamma_{H,t}^2 - \gamma_{c,t}^2\big)\mathrm{Var}_t(u_{g,t+1}), \tag{72}$$

and substitute the country loading $\theta_{c,t} = \omega_{c,t} \odot \lambda_{\Delta TS,t}$ of (55),

$$\begin{aligned}
E_t[rx_{c,t+1}] = \Pi_{c,t}^{\text{other}} + \tfrac{1}{2}\Big\{\big[\omega_{H,t} \odot \lambda_{\Delta TS,t}\big]'\Sigma_{\Delta TS^g}\big[\omega_{H,t} \odot \lambda_{\Delta TS,t}\big] \\
- \big[\omega_{c,t} \odot \lambda_{\Delta TS,t}\big]'\Sigma_{\Delta TS^g}\big[\omega_{c,t} \odot \lambda_{\Delta TS,t}\big]\Big\},
\end{aligned} \tag{73}$$

a difference between two quadratic forms and not between two levels. With $\Sigma_{\Delta TS^g}$ diagonal and $\sigma_k^2 = \mathrm{Var}_t(\Delta TS^g_{k,t+1})$ by (E6),

$$\begin{aligned} E_t[rx_{c,t+1}] &= \Pi^{\text{other}}_{c,t} + \tfrac{1}{2}\sum_k \sigma_k^2\,\lambda_{k,t}^2\left(\omega_{H,k,t}^2 - \omega_{c,k,t}^2\right) \\ &= \Pi^{\text{other}}_{c,t} + \tfrac{1}{2}\sum_k \sigma_k^2\,\lambda_{k,t}^2\,\underbrace{\left(\omega_{H,k,t} - \omega_{c,k,t}\right)}_{\text{mandate wedge}}\underbrace{\left(\omega_{H,k,t} + \omega_{c,k,t}\right)}_{\text{combined share}}, \end{aligned} \tag{74}$$

which is the premium used in the model section. The price of risk enters squared and is common to every currency, so it scales the cross-section without generating it. The mandate wedge carries the country subscript and is the cross-sectional driver, and the combined share is positive under the maintained sign restriction on the shares, so the premium is increasing in the wedge in every bucket whatever the sign of $\lambda_{k,t}$.

C. FX data

This appendix documents the exchange rate data behind the currency excess returns. Table C1 lists the currencies in the sample, the period over which each is available and whether the country is classified as advanced or emerging, the split used for the advanced-country cross-section and for the composition figures. Most series start in the mid-1990s, and the legacy currencies of the euro area end when the euro replaces them, in December 1998 for the founding members and later for Greece, Slovenia, Cyprus, Slovakia and Croatia. Table C2 gives the Datastream mnemonics of the spot and one-month forward rates, from Barclays Bank International, WM/Refinitiv and Refinitiv. The forward rates are what turn spot changes into excess returns, since under covered interest rate parity the forward discount equals the interest rate differential. That equivalence breaks down in some emerging markets during episodes of capital controls or market stress, so following Kroencke et al. (2014) and Corte et al. (2016) I exclude a currency during the periods listed in Table C3. Within the baseline sample, which starts in December 2010, the exclusions affect two currencies, Egypt in 2011 to 2013 and 2016 to 2017 and Russia in 2014 to 2015. The Ukrainian series ends before its exclusion period starts.

Table C1: Sample coverage for foreign exchange series. The table lists all countries included in the sample, their respective currency codes, start and end dates of data availability, and the economic classification of each country (Advanced or Emerging).

Country	Currency code	Start	End	Group
Austria	ATS	Dec-95	Dec-98	Advanced
Australia	AUD	Dec-95	Mar-24	Advanced
Belgium	BEF	Dec-95	Dec-98	Advanced
Bulgaria	BGN	Mar-04	Mar-24	Emerging
Brazil	BRL	Mar-04	Mar-24	Emerging
Canada	CAD	Dec-95	Mar-24	Advanced
Switzerland	CHF	Dec-95	Mar-24	Advanced
Cyprus	CYP	Mar-04	Dec-07	Advanced
Czech Republic	CZK	Dec-96	Mar-24	Emerging
Germany	DEM	Dec-95	Dec-98	Advanced
Denmark	DKK	Dec-95	Mar-24	Advanced
Egypt	EGP	Mar-04	Mar-24	Emerging
Spain	ESP	Dec-95	Dec-98	Advanced
Euro Area	EUR	Jan-99	Mar-24	Advanced
Finland	FIM	Dec-96	Dec-98	Advanced
France	FRF	Dec-95	Dec-98	Advanced
United Kingdom	GBP	Dec-95	Mar-24	Advanced
Greece	GRD	Dec-96	Dec-00	Advanced
Hong Kong	HKD	Dec-95	Mar-24	Advanced
Croatia	HRK	Mar-04	Dec-22	Emerging
Hungary	HUF	Oct-97	Mar-24	Emerging
Indonesia	IDR	Dec-96	Mar-24	Emerging
Ireland	IEP	Dec-95	Dec-98	Advanced
India	INR	Oct-97	Mar-24	Emerging
Iceland	ISK	Mar-04	Mar-24	Advanced
Italy	ITL	Dec-95	Dec-98	Advanced
Japan	JPY	Dec-95	Mar-24	Advanced
South Korea	KRW	Feb-02	Mar-24	Emerging
Kuwait	KWD	Dec-95	Mar-24	Emerging
Mexico	MXN	Dec-96	Mar-24	Emerging
Malaysia	MYR	Dec-96	Mar-24	Emerging
Netherlands	NLG	Dec-95	Dec-98	Advanced
Norway	NOK	Dec-95	Mar-24	Advanced
New Zealand	NZD	Dec-95	Mar-24	Advanced
Philippines	PHP	Dec-96	Mar-24	Emerging
Poland	PLN	Aug-96	Mar-24	Emerging
Portugal	PTE	Dec-95	Dec-98	Advanced
Russia	RUB	Mar-04	Mar-24	Emerging
Saudi Arabia	SAR	Dec-95	Mar-24	Emerging
Sweden	SEK	Dec-95	Mar-24	Advanced
Singapore	SGD	Dec-95	Mar-24	Advanced
Slovenia	SIT	Mar-04	Dec-06	Advanced
Slovakia	SKK	Feb-02	Dec-08	Advanced
Thailand	THB	Dec-95	Mar-24	Emerging
Turkey	TRY	Dec-96	Mar-24	Emerging
Taiwan	TWD	Dec-96	Mar-24	Emerging
Ukraine	UAH	Mar-04	Oct-14	Emerging
South Africa	ZAR	Dec-95	Mar-24	Emerging

Table C2: FX Spot and Forward Rates. This table reports the Datastream mnemonics of the exchange rates spot and forward data used for each country. Data is sourced from Barclays Bank International (BBI), World Markets Company/Refinitiv (WMR) and Refinitiv, accessed via Datastream.

Country	Spot	1M Forward
Australia	BBAUDSP, AUSTDO$	BBAUD1F, USAUD1F, TDAUD1M
Austria	AUSTSC$	USATS1F, NBBEC1F
Belgium	BELGLU$	USBEF1F, NBBEC1F
Brazil	BRACRU$	USBRL1F, BRL$1MF
Bulgaria	BULGLV$	USBGN1F
Canada	BBCADSP, CNDOLL$	BBCAD1F, USCAD1F, TDCAD1M
Croatia	CROATK$	USHRK1F
Cyprus	CYPRUS$	USCYP1F
Czech	CZECHC$	USCZK1F, TDCZK1M
Denmark	BBDKKSP, DANISH$	BBDKK1F, USDKK1F, TDDKK1M
Egypt	EGYPTN$	USEGP1F
Euro	BBEURSP, USEURSP	BBEUR1F, USEUR1F, TDEUR1M
Finland	FINMAR$	USFIM1F
France	BBFRFSP, FRENFR$	BBFRF1F, USFRF1F, NBFRF1F
Germany	BBDEMSP, DMARKE$	BBDEM1F, USDEM1F, NBDEM1F
Greece	GREDRA$	USGRD1F
Hong Kong	BBHKDSP, HKDOLL$	BBHKD1F, USHKD1F, TDHKD1M
Hungary	HUNFOR$	USHUF1F, TDHUF1M
Iceland	ICEKRO$	USISK1F, TDISK1M
India	INDRUP$	USINR1F, TDINR1M
Indonesia	INDORU$	USIDR1F, TDIDR1M
Ireland	BBIEPSP, IPUNTE$	USIEP1F
Italy	BBITLSP, ITALIR$	BBITL1F, USITL1F, NBITL1F
Japan	BBJPYSP, JAPAYE$	BBJPY1F, USJPY1F, TDJPY1M
Kuwait	KUWADI$	USKWD1F, TDKWD1M
Malaysia	MALADL$	USMYR1F, TDMYR1M
Mexico	MEXPES$	USMXN1F, TDMXN1M
Netherlands	BBNLGSP, GUILDE$	BBNLG1F
New Zealand	BBNZDSP, NZDOLL$	BBNZD1F, USNZD1F, TDNZD1M
Norway	BBNOKSP, NORKRO$	BBNOK1F, USNOK1F, TDNOK1M
Philippines	PHILPE$	USPHP1F, TDPHP1M
Poland	POLZLO$	USPLN1F, TDPLN1M
Portugal	PORTES$	USPTE1F
Russia	CISRUB$	USRUB1F, TDRUB1M
Saudi Arabia	SAUDRI$	USSAR1F, TDSAR1M
Singapore	BBSGDSP, SINGDO$	BBSGD1F, USSGD1F, TDSGD1M
Slovakia	SLOVKO$	USSKK1F
Slovenia	SLOVTO$	USSIT1F
South Africa	BBZARSP, COMRAN$	BBZAR1F, USZAR1F, TDZAR1M
South Korea	KORSWO$	USKRW1F, TDKRW1M
Spain	SPANPE$	USESP1F
Sweden	BBSEKSP, SWEKRO$	BBSEK1F, USSEK1F, TDSEK1M
Switzerland	BBCHFSP, SWISSF$	BBCHF1F, USCHF1F, TDCHF1M
Taiwan	TAIWDO$	USTWD1F, TDTWD1M
Thailand	THABAH$	USTHB1F, TDTHB1M
Turkey	TURKLI$	USTRY1F, TDTRY1M
United Kingdom	BBGBPSP, USDOLLR	BBGBP1F, USGBP1F, TDGBP1M
Ukraine	UKRAHY$	USUAH1F, TDUAH1M

Table C3: Periods of large deviations from covered interest rate parity. This table reports the time intervals during which covered interest rate parity exhibited substantial deviations for specific countries. Following Kroencke et al. (2014) and Corte et al. (2016), among others, these currencies are excluded from the empirical analysis during these periods.

Country	Start Date	End Date
Egypt	01/01/2011	30/08/2013
	03/10/2016	28/02/2017
Indonesia	01/12/1997	31/07/1998
	01/02/2001	31/05/2005
Malaysia	01/05/1998	30/06/2005
Russia	01/12/2008	30/01/2009
	03/11/2014	27/02/2015
South Africa	01/08/1985	30/08/1985
	01/01/2002	31/05/2005
Turkey	01/11/2000	30/11/2001
Ukraine	03/11/2014	12/04/2024

D. Summary statistics of test-assets

This appendix describes the test assets priced in Table 3 and Table 4, over the baseline sample of the traded VIX futures. A cross-sectional test is only informative if the test assets differ in average returns, so the question here is how much spread there is to explain. Table D1 reports the individual currencies of the country-level cross-sections. Average excess returns range from -29.95% to 4.03% a year, but only four of the 34 currencies, the Egyptian pound, the Hong Kong dollar, the Japanese yen and the Saudi riyal, earn an average return significantly different from zero at the 5% level, so the country-level tests rest on the spread across many noisy returns rather than on a few large premia. Table D2 reports the Carry, Momentum and Value portfolios. The Carry portfolios show the familiar spread, from -8.85% a year for the low-interest-rate portfolio (t-statistic -3.83) to 2.13% for the high-interest-rate portfolio. Average returns also rise from the first to the last momentum portfolio, from -5.92% to -0.21%, while the value portfolios show little spread, with averages between -2.65% and 0.63% a year.

D.1. Individual currencies

Table D1: Currencies test-assets summary statistics. This table reports summary statistics for monthly currency excess returns of test-asset currencies. The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. *t-stat* refers to the t-statistic testing whether the average monthly excess return is significantly different from zero. The sample corresponds to the baseline analysis period using traded VIX futures.

Currency	Avg Ann. Return	t-stat	Ann. Volatility	Sharpe Ratio	Skewness
AUD	-1.84	[-0.61]	11.01	-0.17	0.08
BGN	-2.40	[-1.10]	7.98	-0.30	-0.34
BRL	-1.14	[-0.27]	15.50	-0.07	-0.19
CAD	-2.00	[-0.93]	7.84	-0.26	-0.46
CHF	-1.07	[-0.44]	8.95	-0.12	-0.33
CZK	-1.65	[-0.59]	10.21	-0.16	-0.39
DKK	-2.78	[-1.28]	7.94	-0.35	-0.29
EGP	-29.95	[-2.61]	36.43	-0.82	-2.93
EUR	-2.60	[-1.19]	7.95	-0.33	-0.32
GBP	-1.98	[-0.86]	8.38	-0.24	-0.24
HKD	-0.43	[-2.82]	0.55	-0.78	0.38
HRK	-2.22	[-0.89]	8.66	-0.26	-0.20
HUF	-1.81	[-0.54]	12.30	-0.15	-0.47
IDR	0.01	[0.00]	8.10	0.00	-0.34
INR	0.95	[0.47]	7.38	0.13	-0.08
ISK	2.00	[0.73]	10.07	0.20	-0.05
JPY	-6.00	[-2.48]	8.85	-0.68	0.04
KRW	-0.64	[-0.26]	8.86	-0.07	-0.19
KWD	-0.46	[-1.00]	1.68	-0.27	-0.79
MXN	2.54	[0.77]	12.03	0.21	-1.02
MYR	-1.56	[-0.72]	7.87	-0.20	-0.32
NOK	-4.05	[-1.23]	12.02	-0.34	-0.05
NZD	-0.40	[-0.12]	12.10	-0.03	-0.21
PHP	-0.48	[-0.34]	5.19	-0.09	-0.36
PLN	-0.32	[-0.10]	11.84	-0.03	-0.40
RUB	4.03	[0.68]	21.37	0.19	-0.10
SAR	0.20	[3.69]	0.19	1.05	1.08
SEK	-3.92	[-1.45]	9.89	-0.40	-0.01
SGD	-0.30	[-0.20]	5.57	-0.05	-0.51
THB	-1.16	[-0.59]	7.13	-0.16	-0.16
TRY	-7.18	[-1.52]	17.25	-0.42	-2.44
TWD	-1.58	[-1.15]	5.00	-0.32	0.05
UAH	-2.44	[-0.43]	11.22	-0.22	-2.94
ZAR	-2.10	[-0.51]	15.04	-0.14	-0.37

D.2. Portfolios

Table D2: Test-assets portfolios summary statistics. This table reports summary statistics for monthly currency excess returns of test-asset portfolios. Panel A sorts currencies on the previous month forward premium as in Lustig et al. (2011), Panel B on previous month excess return as in Menkhoff et al. (2012b), and Panel C on the 5-year-log-return of the real exchange rate as in Menkhoff et al. (2017). P1 contains the currencies with the lowest value of the sorting variable, and P5 those with the highest. The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. *t-stat* refers to the t-statistic testing whether the average monthly excess return is significantly different from zero. The sample corresponds to the baseline analysis period using traded VIX futures.

Portfolio	P1	P2	P3	P4	P5
Panel A: Carry					
Avg Ann. Return	-8.85	-1.71	-1.38	-1.26	2.13
t-stat	[-3.83]	[-1.01]	[-0.73]	[-0.70]	[0.88]
Ann. Volatility	8.45	6.16	6.88	6.57	8.84
Sharpe Ratio	-1.05	-0.28	-0.20	-0.19	0.24
Skewness	-0.99	0.12	-0.39	-0.55	-0.94
Panel B: Momentum					
Avg Ann. Return	-5.92	-1.45	-1.45	-1.64	-0.21
t-stat	[-2.16]	[-0.76]	[-0.81]	[-0.92]	[-0.11]
Ann. Volatility	10.04	6.98	6.49	6.51	7.19
Sharpe Ratio	-0.59	-0.21	-0.22	-0.25	-0.03
Skewness	-0.68	0.28	-0.31	-0.73	-0.48
Panel C: Value					
Avg Ann. Return	0.63	-1.86	-2.65	-1.17	-1.80
t-stat	[0.24]	[-0.79]	[-1.27]	[-0.56]	[-0.91]
Ann. Volatility	9.66	8.61	7.62	7.53	7.24
Sharpe Ratio	0.07	-0.22	-0.35	-0.16	-0.25
Skewness	0.05	-0.26	-0.31	-0.54	-0.70

E. Alternative number of portfolios in factor construction

The main text forms the factors from quintile portfolios. The number of portfolios trades off the spread in betas between the two legs against the number of currencies in each leg, and five is a convention rather than an implication of the model, so this appendix rebuilds the factors from three portfolios (Subsection E.1) and from ten (Subsection E.2) and repeats the analysis. The rolling betas keep a persistent spread across portfolios in both cases (Figure E1 and Figure E4), and the legs of the factors mix advanced and emerging currencies as in the main text (Figure E2, Figure E3, Figure E5 and Figure E6).

The results keep the sign pattern of the main text. With three portfolios the level,

slope and curvature factors earn −4.97%, 3.75% and −4.43% a year (Table E1), against −6.91%, 6.96% and −6.67% with five. With ten portfolios the spreads widen to −10.91%, 10.10% and −10.34% (Table E5), as expected when the legs hold only the currencies with the most extreme betas, at the cost of a higher volatility. All six factors are significant. As in the main text, the level and curvature factors load negatively on Carry and the slope factor positively (Table E2 and Table E6). In the asset pricing tests (Table E3, Table E4, Table E7 and Table E8), the prices of risk have the same signs as with quintiles and are significant in every single-factor specification of the four cross-sections. Their size moves with the average return of the factor, as it should for a traded factor that is included among the test assets. The only losses of significance are the level and curvature prices in the specification with all three factors on the all-country cross-section with terciles, the same specification in which the curvature price is insignificant with quintiles.

E.1. Tercile sorts

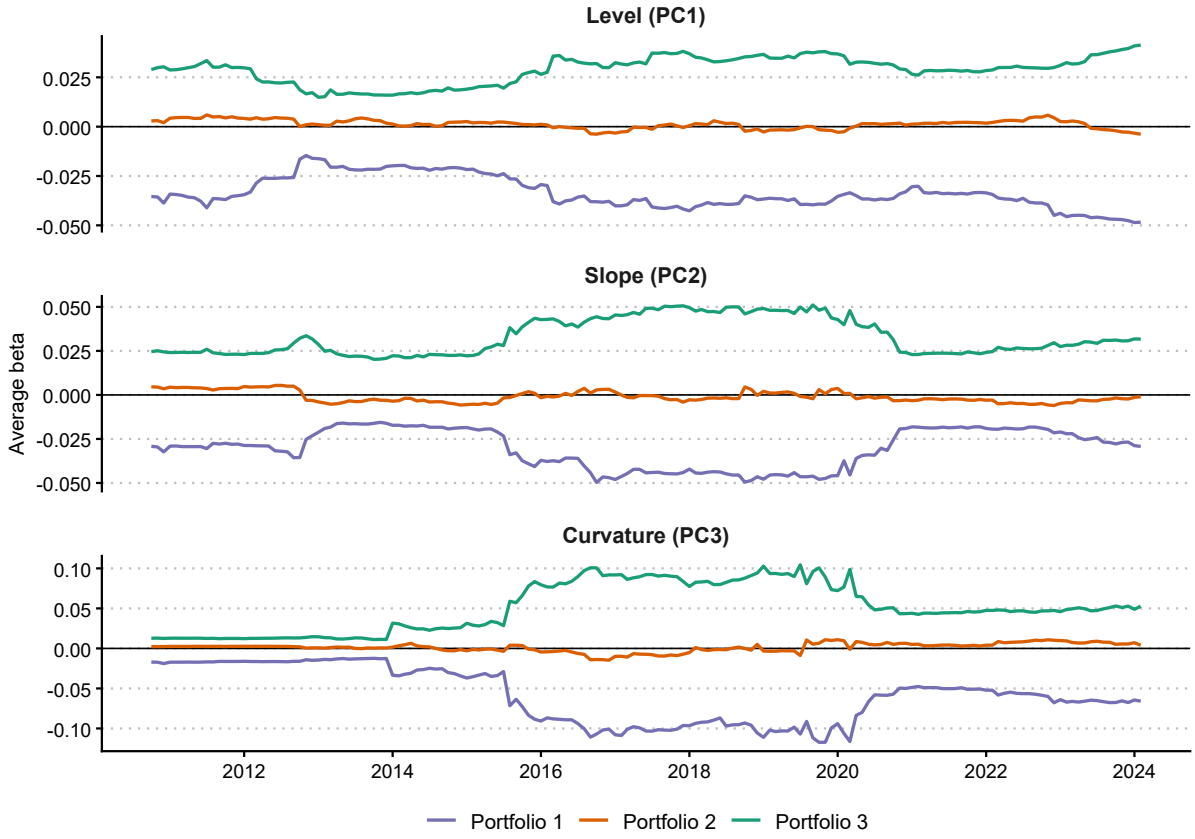


Figure E1: Rolling betas to VIX principal component innovations by currency portfolio. This figure shows the time-series of average betas for three currency portfolios sorted each month by their estimated exposure to innovations in VIX term-structure components. The principal components, capturing the level (PC1), slope (PC2), and curvature (PC3) of the VIX futures curve, are extracted monthly using an expanding-window PCA. Innovations in each component are computed as the monthly change relative to the previous observation. For each currency, rolling 48-month regressions of excess returns on these innovations are estimated following Equation 28, controlling for the Dollar factor. The resulting betas measure time-varying sensitivities to shocks in the VIX term-structure. The figure reports portfolio-average betas across the three beta-sorted groups, illustrating persistent cross-sectional heterogeneity and cyclical variation in exposures to volatility-level (PC1), slope (PC2), and curvature (PC3) innovations.

Table E1: VIX futures beta-sorted portfolios summary statistics. This table reports summary statistics for monthly currency excess returns of portfolios sorted by estimated betas with respect to VIX futures innovations. Panel A sorts currencies on β_{PC1} (level), Panel B on β_{PC2} (slope), and Panel C on β_{PC3} (curvature). P1 contains currencies with the lowest beta exposures, while P3 contains those with the highest. HML denotes the high-minus-low portfolio (P3 – P1). The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. *t-stat* refers to the t-statistic testing whether the average monthly excess return is significantly different from zero.

Portfolio	P1	P2	P3	HML
Panel A: sorts on β_{PC1}				
Avg Ann. Return	-0.39	-1.00	-5.37	-4.97
t-stat	[-0.21]	[-0.64]	[-2.49]	[-3.27]
Ann. Volatility	6.96	5.77	7.90	5.57
Sharpe Ratio	**-0.06**	**-0.17**	**-0.68**	**-0.89**
Skewness	-0.13	-0.58	-0.37	-0.69
Panel B: sorts on β_{PC2}				
Avg Ann. Return	-4.97	-0.87	-1.22	3.75
t-stat	[-2.39]	[-0.61]	[-0.58]	[2.55]
Ann. Volatility	7.63	5.21	7.66	5.38
Sharpe Ratio	**-0.65**	**-0.17**	**-0.16**	**0.70**
Skewness	-0.30	-0.41	-0.62	0.46
Panel C: sorts on β_{PC3}				
Avg Ann. Return	-0.53	-1.28	-4.96	-4.43
t-stat	[-0.26]	[-0.87]	[-2.39]	[-3.35]
Ann. Volatility	7.42	5.43	7.61	4.85
Sharpe Ratio	**-0.07**	**-0.24**	**-0.65**	**-0.91**
Skewness	-0.25	-0.59	-0.35	-0.42

Table E2: Time-series regression of VIX futures beta-sorted portfolios on the Carry factor. This table reports the results of time-series regressions of monthly excess returns of VIX futures beta-sorted currency portfolios on the Carry factor of Lustig et al. (2011). Panels A–C correspond to portfolios sorted on betas with respect to innovations to the first three principal components of VIX futures: level (PC1), slope (PC2), and curvature (PC3), respectively. P1 includes currencies with the lowest beta exposure and P3 includes those with the highest. HML denotes the high-minus-low portfolio (P3 – P1). Newey-West t-statistics are reported in brackets.

Portfolio	P1	P2	P3	HML
Panel A: sorts on β_{PC1}				
Intercept	-0.03	-0.02	-0.05	-0.02
	[-1.65]	[-1.35]	[-1.99]	[-1.32]
Carry Factor	0.40	0.17	-0.09	-0.49
	[3.36]	[1.57]	[-0.43]	[-4.81]
R^2	0.148	0.040	0.005	0.342
Panel B: sorts on β_{PC2}				
Intercept	-0.04	-0.02	-0.04	0.01
	[-1.98]	[-1.21]	[-1.84]	[0.44]
Carry Factor	-0.08	0.15	0.39	0.47
	[-0.37]	[1.66]	[2.80]	[4.40]
R^2	0.005	0.038	0.116	0.338
Panel C: sorts on β_{PC3}				
Intercept	-0.03	-0.02	-0.05	-0.02
	[-1.27]	[-1.59]	[-2.10]	[-1.64]
Carry Factor	0.31	0.18	-0.00	-0.31
	[2.64]	[1.76]	[-0.01]	[-3.19]
R^2	0.079	0.048	0.000	0.187

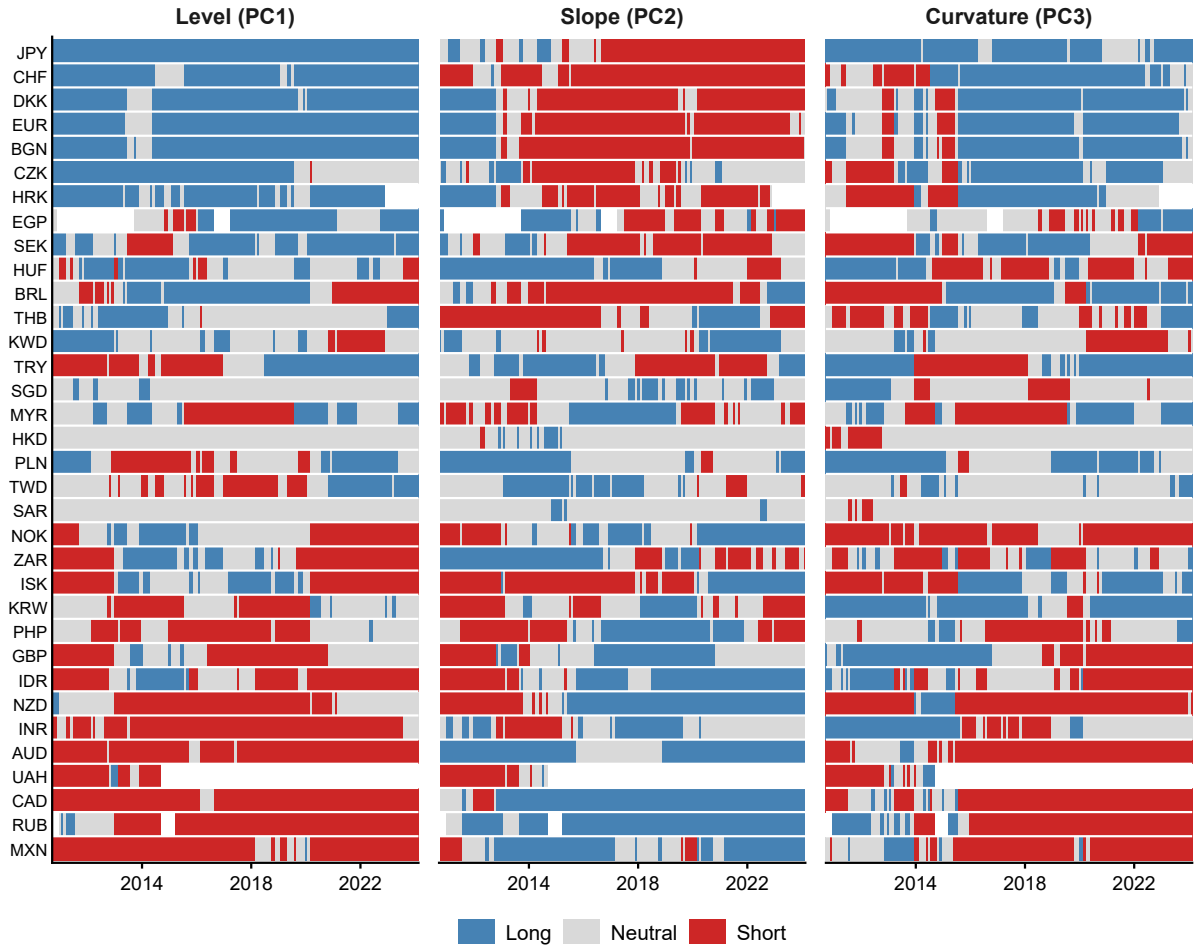


Figure E2: Composition of currency risk factors constructed from VIX principal component betas. This figure shows the time-varying composition of the high-minus-low (HML) currency risk factors associated with exposures to innovations in the first three VIX term-structure principal components: level (PC1), slope (PC2), and curvature (PC3). For each month, currencies are sorted into three portfolios based on their rolling 48-month betas with respect to the respective principal component innovation, and the factor is constructed as the return difference between the highest- and lowest-beta portfolios. The heatmap indicates, for each currency and month, whether it enters the factor with a long position (blue), short position (red), or is excluded from the factor (gray). Each panel corresponds to one VIX component. The figure highlights persistent cross-sectional asymmetries in how currencies load onto the VIX futures term-structure level, slope, and curvature shocks, as well as shifts in factor composition reflecting changing sources of volatility risk exposure across time.

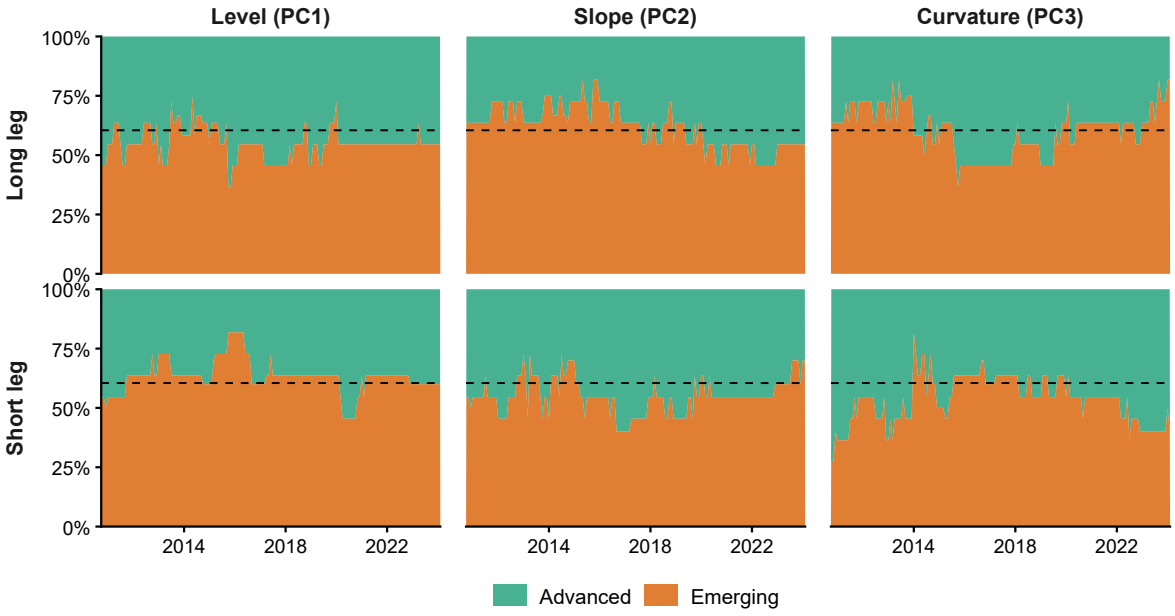


Figure E3: Emerging-market composition of the VIX term-structure factors, tercile sorts. Share of emerging-market currencies (orange) and advanced-market currencies (green) among the currencies in the long leg (top row) and the short leg (bottom row) of the high-minus-low factors built from tercile portfolios sorted on betas to innovations in the level, slope and curvature of the VIX futures term-structure, monthly. The dashed line is the average share of emerging-market currencies among all currencies in the cross-section over the sample, so a leg whose orange area lies above (below) the line holds more (fewer) emerging-market currencies than the cross-section as a whole.

Table E3: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the portfolio-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of Carry, Momentum, and Value portfolios, while columns (6)–(10) use only the Momentum and Value portfolios. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Carry + Momentum + Value						Momentum + Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.045^{***}			-0.044^{***}		-0.045^{***}			-0.044^{***}
		(0.014)			(0.014)		(0.014)			(0.014)
β_{fPC2}			0.034^{**}		0.033^{**}			0.033^{**}		0.033^{**}
			(0.014)		(0.014)			(0.014)		(0.014)
β_{fPC3}				-0.041^{***}	-0.041^{***}				-0.040^{***}	-0.040^{***}
				(0.013)	(0.013)				(0.013)	(0.013)
β_{CAR}	0.071^{***}	0.068^{***}	0.068^{***}	0.068^{***}	0.068^{***}	0.130^{**}	0.069^{***}	0.069^{***}	0.069^{***}	0.069^{***}
	(0.020)	(0.019)	(0.019)	(0.019)	(0.019)	(0.065)	(0.018)	(0.018)	(0.018)	(0.018)
β_{DOL}	-0.020	-0.021	-0.021	-0.021	-0.021	-0.021	-0.020	-0.020	-0.021	-0.021
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.018)	(0.018)	(0.017)	(0.017)	(0.017)
R^2	0.882	0.967	0.971	0.972	0.976	0.819	0.968	0.971	0.970	0.976
GRS	0.787	0.833	0.83	0.723	0.751	0.641	0.734	0.718	0.589	0.627

Table E4: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the country-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of all country excess returns, while columns (6)–(10) use only advanced countries currency excess returns. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Country level - all					Country level - advanced				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.072^{**}			-0.027		-0.048^{***}			-0.048^{***}
		(0.034)			(0.018)		(0.016)			(0.015)
β_{fPC2}			0.068^{**}		0.054^{**}			0.037^{**}		0.036^{**}
			(0.031)		(0.022)			(0.016)		(0.016)
β_{fPC3}				-0.056^{**}	-0.048				-0.037^{**}	-0.033^{**}
				(0.028)	(0.030)				(0.015)	(0.015)
β_{CAR}	0.055^{**}	0.045^{**}	0.041^{*}	0.042^{**}	0.041^{*}	0.086^{**}	0.074^{***}	0.072^{***}	0.075^{***}	0.075^{***}
	(0.024)	(0.021)	(0.021)	(0.021)	(0.021)	(0.035)	(0.020)	(0.020)	(0.020)	(0.020)
β_{DOL}	-0.021	-0.021	-0.021	-0.021	-0.021	-0.011	-0.014	-0.014	-0.016	-0.016
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.019)	(0.018)	(0.018)	(0.018)	(0.018)
R^2	0.653	0.803	0.854	0.859	0.875	0.627	0.867	0.869	0.840	0.879
GRS p-value	0.04	0.06	0.054	0.045	0.049	0.179	0.245	0.199	0.14	0.17

E.2. Decile sorts

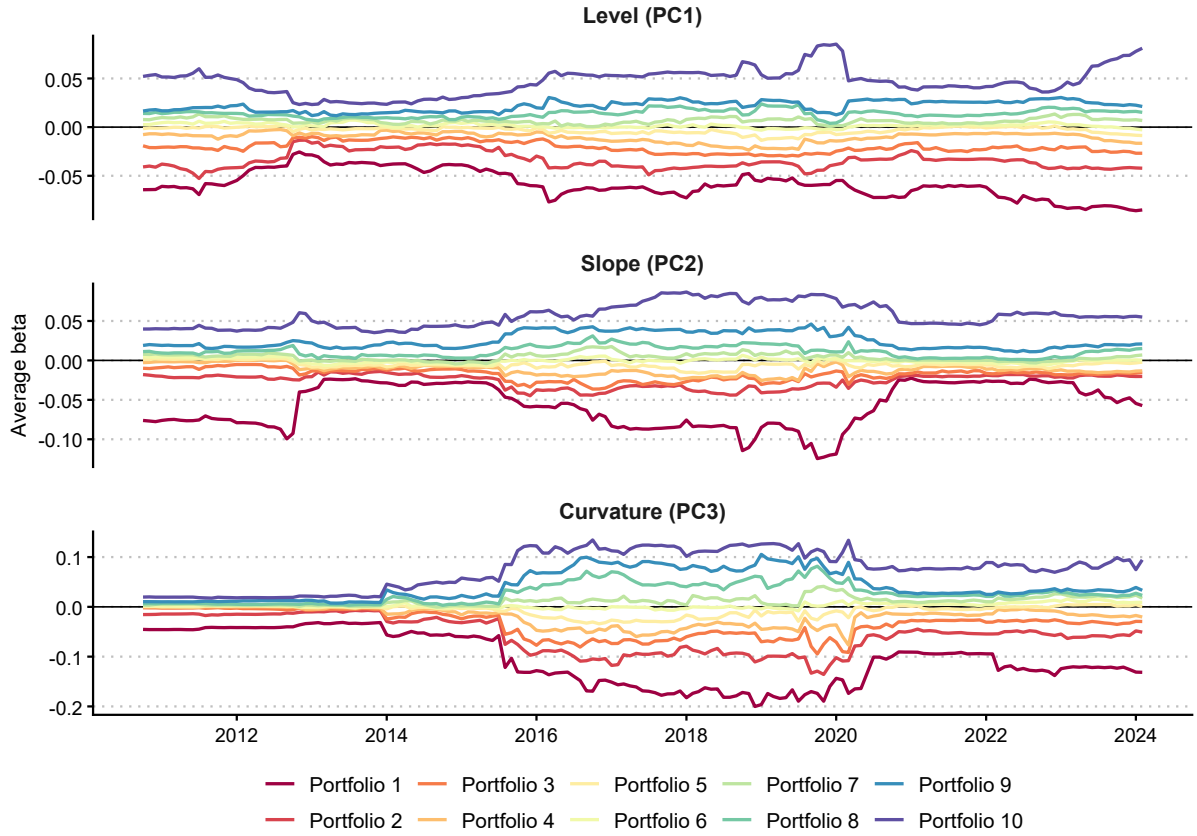


Figure E4: Rolling betas to VIX principal component innovations by currency portfolio. This figure shows the time-series of average betas for ten currency portfolios sorted each month by their estimated exposure to innovations in VIX term-structure components. The principal components, capturing the level (PC1), slope (PC2), and curvature (PC3) of the VIX futures curve, are extracted monthly using an expanding-window PCA. Innovations in each component are computed as the monthly change relative to the previous observation. For each currency, rolling 48-month regressions of excess returns on these innovations are estimated following Equation 28, controlling for the Dollar factor. The resulting betas measure time-varying sensitivities to shocks in the VIX term-structure. The figure reports portfolio-average betas across the ten beta-sorted groups, illustrating persistent cross-sectional heterogeneity and cyclical variation in exposures to volatility-level (PC1), slope (PC2), and curvature (PC3) innovations.

Table E5: VIX futures beta-sorted portfolios summary statistics. This table reports summary statistics for monthly currency excess returns of portfolios sorted by estimated betas with respect to VIX futures innovations. Panel A sorts currencies on β_{PC1} (level), Panel B on β_{PC2} (slope), and Panel C on β_{PC3} (curvature). P1 contains currencies with the lowest beta exposures, while P10 contains those with the highest. HML denotes the high-minus-low portfolio (P10 – P1). The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. *t-stat* refers to the t-statistic testing whether the average monthly excess return is significantly different from zero.

Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	HML
Panel A: sorts on β_{PC1}											
Avg Ann. Return	1.33	-0.77	-1.26	-1.84	1.07	-2.39	-1.21	-3.37	-2.30	-9.58	-10.91
t-stat	[0.51]	[-0.35]	[-0.58]	[-0.95]	[0.60]	[-1.56]	[-0.57]	[-1.49]	[-1.11]	[-3.18]	[-3.33]
Ann. Volatility	9.49	8.04	7.98	7.09	6.58	5.62	7.77	8.31	7.62	11.02	11.98
Sharpe Ratio	**0.14**	**-0.10**	**-0.16**	**-0.26**	**0.16**	**-0.43**	**-0.16**	**-0.41**	**-0.30**	**-0.87**	**-0.91**
Skewness	-0.05	-0.05	-0.34	-0.66	-0.84	-0.40	-0.91	-0.84	-0.19	-1.37	-1.11
Panel B: sorts on β_{PC2}											
Avg Ann. Return	-11.04	-3.22	-1.63	-1.49	-1.31	-0.28	-1.93	-1.66	-0.18	-0.95	10.10
t-stat	[-3.21]	[-1.53]	[-0.67]	[-0.77]	[-0.74]	[-0.17]	[-1.12]	[-0.86]	[-0.08]	[-0.34]	[2.69]
Ann. Volatility	12.62	7.74	8.96	7.06	6.43	5.84	6.31	7.08	8.14	10.32	13.75
Sharpe Ratio	**-0.87**	**-0.42**	**-0.18**	**-0.21**	**-0.20**	**-0.05**	**-0.31**	**-0.23**	**-0.02**	**-0.09**	**0.73**
Skewness	-2.14	-0.29	-0.72	-0.44	-0.14	0.11	-0.50	-0.66	-0.56	-0.45	1.45
Panel C: sorts on β_{PC3}											
Avg Ann. Return	0.17	0.07	-0.58	-3.68	-1.98	-0.51	0.54	-2.66	-1.30	-10.17	-10.34
t-stat	[0.06]	[0.03]	[-0.26]	[-2.02]	[-1.16]	[-0.32]	[0.29]	[-1.26]	[-0.64]	[-3.39]	[-3.38]
Ann. Volatility	10.35	8.80	8.34	6.67	6.23	5.76	6.81	7.74	7.38	10.99	11.21
Sharpe Ratio	**0.02**	**0.01**	**-0.07**	**-0.55**	**-0.32**	**-0.09**	**0.08**	**-0.34**	**-0.18**	**-0.93**	**-0.92**
Skewness	-0.37	-0.22	-0.07	-0.60	-1.15	-0.76	-0.41	-0.48	-0.47	-1.30	-1.14

Table E6: Time-series regression of VIX futures beta-sorted portfolios on the Carry factor. This table reports the results of time-series regressions of monthly excess returns of VIX futures beta-sorted currency portfolios on the Carry factor of Lustig et al. (2011). Panels A–C correspond to portfolios sorted on betas with respect to innovations to the first three principal components of VIX futures: level (PC1), slope (PC2), and curvature (PC3), respectively. P1 includes currencies with the lowest beta exposure and P10 includes those with the highest. HML denotes the high-minus-low portfolio (P10 – P1). Newey-West t-statistics are reported in brackets.

Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	HML
Panel A: sorts on β_{PC1}											
Intercept	-0.03	-0.02	-0.02	-0.03	0.01	-0.03	-0.03	-0.03	-0.02	-0.04	-0.01
	[-1.45]	[-0.85]	[-0.94]	[-1.71]	[0.26]	[-1.97]	[-1.43]	[-1.19]	[-1.08]	[-1.20]	[-0.26]
Carry Factor	0.25	0.08	0.05	0.08	0.03	0.05	0.10	-0.02	-0.01	-0.33	-0.59
	[2.04]	[1.27]	[1.05]	[1.57]	[0.67]	[1.10]	[1.78]	[-0.31]	[-0.11]	[-1.38]	[-5.32]
R^2	0.139	0.018	0.008	0.027	0.004	0.016	0.030	0.001	0.000	0.176	0.465
Panel B: sorts on β_{PC2}											
Intercept	-0.04	-0.03	-0.02	-0.02	-0.02	-0.01	-0.03	-0.03	-0.01	-0.04	-0.00
	[-1.18]	[-1.28]	[-0.75]	[-0.84]	[-0.95]	[-0.51]	[-1.66]	[-1.40]	[-0.63]	[-1.71]	[-0.10]
Carry Factor	-0.41	-0.01	0.03	0.01	0.02	0.03	0.06	0.07	0.07	0.20	0.60
	[-1.23]	[-0.29]	[0.39]	[0.33]	[0.72]	[0.95]	[1.32]	[1.21]	[1.41]	[2.11]	[3.44]
R^2	0.203	0.001	0.002	0.001	0.003	0.006	0.017	0.018	0.016	0.071	0.375
Panel C: sorts on β_{PC3}											
Intercept	-0.04	-0.00	-0.01	-0.04	-0.04	-0.01	-0.00	-0.03	-0.02	-0.05	-0.01
	[-1.61]	[-0.23]	[-0.53]	[-2.04]	[-1.74]	[-0.75]	[-0.11]	[-1.44]	[-0.71]	[-1.74]	[-0.39]
Carry Factor	0.25	0.03	0.04	0.02	0.09	0.04	0.04	0.02	0.03	-0.29	-0.54
	[2.11]	[0.56]	[0.89]	[0.36]	[1.69]	[0.95]	[0.87]	[0.33]	[0.49]	[-0.65]	[-5.77]
R^2	0.113	0.002	0.006	0.001	0.043	0.008	0.008	0.001	0.002	0.137	0.453

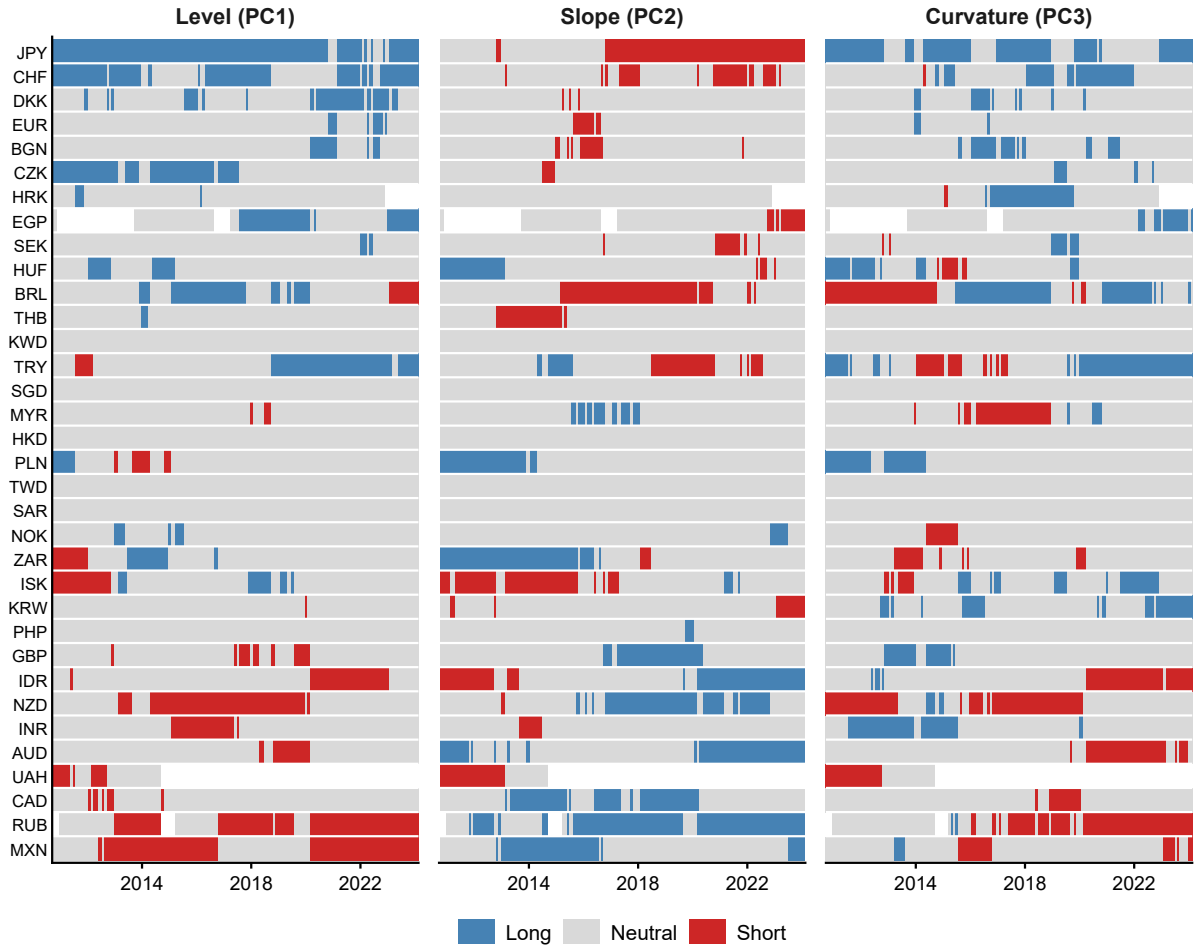


Figure E5: Composition of currency risk factors constructed from VIX principal component betas. This figure shows the time-varying composition of the high-minus-low (HML) currency risk factors associated with exposures to innovations in the first three VIX term-structure principal components: level (PC1), slope (PC2), and curvature (PC3). For each month, currencies are sorted into ten portfolios based on their rolling 48-month betas with respect to the respective principal component innovation, and the factor is constructed as the return difference between the highest- and lowest-beta portfolios. The heatmap indicates, for each currency and month, whether it enters the factor with a long position (blue), short position (red), or is excluded from the factor (gray). Each panel corresponds to one VIX component. The figure highlights persistent cross-sectional asymmetries in how currencies load onto the VIX futures term-structure level, slope, and curvature shocks, as well as shifts in factor composition reflecting changing sources of volatility risk exposure across time.

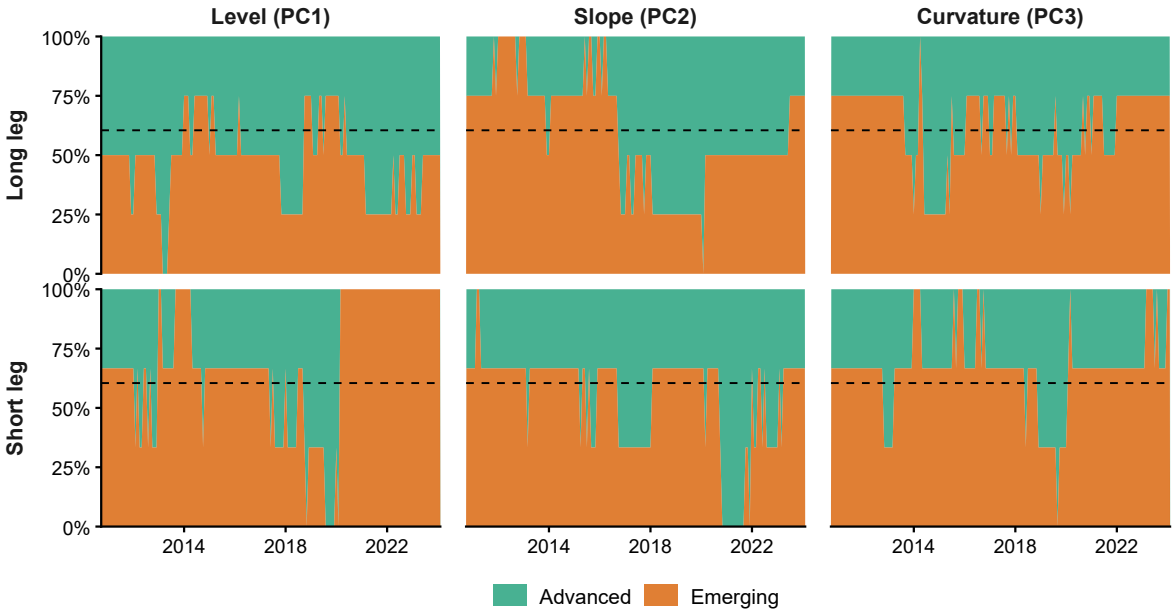


Figure E6: Emerging-market composition of the VIX term-structure factors, decile sorts. Share of emerging-market currencies (orange) and advanced-market currencies (green) among the currencies in the long leg (top row) and the short leg (bottom row) of the high-minus-low factors built from decile portfolios sorted on betas to innovations in the level, slope and curvature of the VIX futures term-structure, monthly. The dashed line is the average share of emerging-market currencies among all currencies in the cross-section over the sample, so a leg whose orange area lies above (below) the line holds more (fewer) emerging-market currencies than the cross-section as a whole.

Table E7: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the portfolio-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of Carry, Momentum, and Value portfolios, while columns (6)–(10) use only the Momentum and Value portfolios. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Carry + Momentum + Value					Momentum + Value				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.094^{***}			-0.098^{***}		-0.095^{***}			-0.099^{***}
		(0.030)			(0.030)		(0.030)			(0.030)
β_{fPC2}			0.087^{***}		0.087^{***}			0.087^{***}		0.088^{***}
			(0.034)		(0.034)			(0.034)		(0.034)
β_{fPC3}				-0.084^{***}	-0.081^{***}				-0.086^{***}	-0.081^{***}
				(0.028)	(0.028)				(0.028)	(0.028)
β_{CAR}	0.175^{***}	0.173^{***}	0.173^{***}	0.173^{***}	0.173^{***}	0.181^{***}	0.174^{***}	0.174^{***}	0.174^{***}	0.173^{***}
	(0.040)	(0.039)	(0.039)	(0.039)	(0.039)	(0.046)	(0.039)	(0.039)	(0.039)	(0.039)
β_{DOL}	-0.023	-0.023	-0.023	-0.023	-0.023	-0.023	-0.023	-0.023	-0.023	-0.023
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
R^2	0.879	0.952	0.954	0.947	0.958	0.758	0.950	0.951	0.944	0.959
GRS	0.87	0.894	0.878	0.883	0.869	0.798	0.831	0.807	0.803	0.769

Table E8: Asset pricing tests of VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the country-level test-assets using the VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of all country excess returns, while columns (6)–(10) use only advanced countries currency excess returns. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Country level - all					Country level - advanced				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.112^{***}			-0.111^{***}		-0.097^{***}			-0.101^{***}
		(0.040)			(0.032)		(0.030)			(0.030)
β_{fPC2}			0.097^{**}		0.081^{**}			0.082^{**}		0.080^{**}
			(0.043)		(0.038)			(0.034)		(0.034)
β_{fPC3}				-0.099^{**}	-0.092^{***}				-0.090^{***}	-0.089^{***}
				(0.040)	(0.035)				(0.028)	(0.028)
β_{CAR}	0.112^{**}	0.115^{***}	0.117^{***}	0.116^{***}	0.116^{***}	0.254^{**}	0.177^{***}	0.178^{***}	0.178^{***}	0.178^{***}
	(0.046)	(0.042)	(0.042)	(0.042)	(0.042)	(0.117)	(0.039)	(0.039)	(0.039)	(0.039)
β_{DOL}	-0.022	-0.021	-0.022	-0.022	-0.021	-0.018	-0.020	-0.018	-0.020	-0.020
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)
R^2	0.726	0.853	0.838	0.844	0.873	0.547	0.949	0.945	0.945	0.962
GRS	0.033	0.045	0.036	0.045	0.055	0.23	0.294	0.236	0.292	0.324

F. Synthetic VIX futures

The traded VIX futures curve has nine listed maturities only from October 2006, and the factors built on it start in December 2010. To check the results on a longer sample, this appendix builds synthetic VIX futures from S&P500 index options, which are available from January 1996, and repeats the analysis on them. The synthetic futures are the same object as the traded ones. Each one prices the 30-day implied volatility of the S&P500 at a future date, at horizons of one, two, three, six and nine months, which span the same range of maturities as the traded curve. The factors run from March 2000 to March 2024, almost twice the length of the baseline sample. Everything else, from the expanding-window principal components to the beta sorts, the asset pricing tests and the predictive regressions, is the same as in the main text.

F.1. Construction

A VIX futures contract that expires at date T settles at the value of the VIX index at T. The CBOE computes the VIX so that its square is the risk-neutral expected variance of the S&P500 over the next 30 days, measured model-free from the prices of out-of-the-money index options (Carr and Madan, 1998; Demeterfi et al., 1999; Britten-Jones and Neuberger, 2000). Let $V_t(h)$ denote the annualized risk-neutral variance of the S&P500 from t to $t+h$ and let τ be 30 days. Since $\mathrm{VIX}_T^2 = V_T(\tau)$, the law of iterated expectations gives the expected square of the VIX at T as a forward variance that can be priced at t from options that expire at T and at $T+\tau$,

$$E_t^{\mathbb{Q}}\left[\mathrm{VIX}_T^2\right] = \frac{(T-t+\tau)\,V_t(T-t+\tau) - (T-t)\,V_t(T-t)}{\tau} \equiv FV_t(T). \tag{75}$$

I define the synthetic VIX future with expiry T as $\tilde{F}_t(T) = 100\sqrt{FV_t(T)}$. The traded future is priced at $E_t^{\mathbb{Q}}[\mathrm{VIX}_T]$, so by Jensen's inequality the synthetic future is an upper bound on the traded one, and the difference is the convexity adjustment of Carr and Wu (2006). The construction has two steps. The first measures $V_t(h)$ at a set of fixed maturities with the CBOE formula, and the second turns these into forward variances at the horizons of the futures.

The first step follows the CBOE VIX methodology (Cboe Exchange, 2019). For an expiry h (in years), the CBOE computes

$$\sigma^2(h) = \frac{2}{h}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rh}\, Q(K_i) - \frac{1}{h}\left(\frac{F}{K_0} - 1\right)^2, \tag{76}$$

where $Q(K_i)$ is the midquote of the out-of-the-money option with strike K_i (the average of the call and the put at K_0), F is the forward index level implied by put-call parity at the strike where the call and put prices are closest, K_0 is the first strike equal to or below F, and ΔK_i is the spacing between strikes. The first term is a discrete version of the integral $\frac{2}{h}\int_0^\infty e^{rh}Q(K)/K^2\,dK$ that defines the model-free implied variance, and the second term corrects the discretization around K_0. I compute the same integral on the volatility surface of OptionMetrics IvyDB US, which OptionMetrics interpolates from the implied volatilities of the listed options. On each day it gives the implied volatility smile of the S&P500 at fixed maturities of 30, 60, 91, 122, 152, 182, 273, 365 and 547 days, through the out-of-the-money calls and puts with deltas between 0.10 and 0.50 in absolute value. For each maturity, I recover the forward from the strikes of the 50-delta call and put, at which $\ln K = \ln F + \sigma^2 h/2$ in the Black model. I then convert the smile into out-of-the-money option prices with the Black formula on a fine grid of strikes that spans ten standard deviations on each side of the forward, and integrate. Beyond the strikes of the 10-delta options the implied volatility is held flat. The calculation differs from the CBOE in four ways. The CBOE uses the listed quotes of the two expiries around 30 days and interpolates the two variances to 30 days, while the surface already gives the smile at each fixed maturity. The CBOE takes the forward from put-call parity, while I take it from the 50-delta strikes. The CBOE cuts the strike range after two consecutive zero bids, while I extrapolate the smile beyond the 10-delta strikes. The correction term in Equation 76 is not needed, because the integral is computed on a continuous grid.

Table F1 checks this step against the CBOE. As a sanity check, Panel A applies Equation 76 directly to the listed S&P500 option quotes in IvyDB, following each step of the CBOE methodology, and interpolates the variances of the two expiries around 30 days. This exact replication has a correlation of 0.998 with the published VIX index and is 0.33 points lower on average. Panel B compares the 30-day implied volatility from the surface with the VIX index. The correlation is 0.997, and the surface is 0.89 points lower on average. Panel C compares each maturity of the surface with the exact replication at the same maturity, interpolated between the two listed expiries around it. The correlation is between 0.956 and 0.997. The surface lies below the exact replication by 0.58 points at 30 days and by between 1.08 and 1.16 points at 122 to 273 days. This gap is consistent with the flat extrapolation of the smile beyond the 10-delta strikes, which assigns lower prices than the listed quotes to deep out-of-the-money puts.

The second step builds the forward variances. A future at a horizon of K months needs $V_t(h)$ at $h = 30K$ and $h = 30K + 30$ days, which mostly fall between the nine

maturities of the surface. I fit one smooth curve to all nine maturities on each day, so that every synthetic future uses the whole term-structure of variance rather than only the two nearest maturities. The curve is the functional form of Nelson and Siegel (1987), with the average variance $V_t(h)$ in the role of the yield and the forward variance in the role of the instantaneous forward rate,

$$V_t(h) = \beta_{0,t} + \beta_{1,t}\,\frac{1 - e^{-h/\lambda}}{h/\lambda} + \beta_{2,t}\left(\frac{1 - e^{-h/\lambda}}{h/\lambda} - e^{-h/\lambda}\right). \tag{77}$$

The implied instantaneous forward variance is $\beta_{0,t} + \beta_{1,t}e^{-h/\lambda} + \beta_{2,t}(h/\lambda)e^{-h/\lambda}$, so $\beta_{0,t}$ is the long-run level of variance and $\beta_{1,t}$ the difference between short-run and long-run variance. I estimate the three coefficients by OLS on each day, with λ fixed at 60 days, the value that minimizes the fitting error over the whole sample. The median fitting error is between 0.11 and 0.13 volatility points in every five-year period, and the fitted forward variance is positive on every day. The synthetic futures follow from Equation 75 with the fitted curve, $\tilde{F}_t(K) = 100\,\{[(30K + 30)\,V_t(30K + 30) - 30K\,V_t(30K)]/30\}^{1/2}$, for K equal to one, two, three, six and nine months, on 7,107 days from January 1996 to March 2024.

Panel D of Table F1 and Figure F1 compare the synthetic futures with the traded futures at the same maturities from October 2006, when the traded curve starts. At one to six months the average synthetic future lies within 0.2 points of the average traded future, and the correlation is between 0.98 and 0.99. A small gap is expected, since the convexity adjustment raises the synthetic future above the traded one and the lower variance of the surface in Panels B and C lowers it. The nine-month point tracks less closely, with a correlation of 0.895 and an average 1.18 points below the traded future.

F.2. Results

The synthetic curve has the same factor structure as the traded one (Figure F2 and Figure F3). The first three principal components are again a level, a slope and a curvature, and explain 96.21%, 3.57% and 0.22% of the variation (Figure F4 and Figure F5). The rolling betas keep a persistent spread across portfolios, and the legs of the factors mix advanced and emerging currencies (Figure F6, Figure F7, Figure F8 and Figure F9).

The level and slope results of the main text hold on the longer sample, and the curvature result does not. The level and slope factors earn −3.97% and 4.02% a year, with t-statistics of −2.39 and 2.48, against −6.91% and 6.96% in the baseline sample, while the curvature factor earns −0.29% and is not significant (Table F2). The three factors load on Carry with the same signs as in the main text (Table F3). In the asset pricing tests on the

portfolio cross-sections (Table F4), the prices of level and slope risk keep their signs and are significant in every specification, and the GRS test does not reject the models. On the country cross-sections (Table F5), the price of slope risk is significant in every specification. The price of level risk keeps its sign and is significant at the 10% level in three of the four specifications, but not when all three factors enter together on the all-country cross-section. The price of curvature risk is close to zero in every specification. The GRS test rejects the models on both country-level cross-sections, including the advanced countries, where it does not reject in the baseline sample.

In the predictive regressions (Table F6), the level and slope innovations enter with positive coefficients, as in the baseline, but neither is significant. The curvature innovation enters with a coefficient of -0.050, significant at the 1% level, the opposite sign to the baseline. The lagged excess return is significant here, unlike in the baseline sample. Overall, the longer sample confirms that level and slope risk are priced in the cross-section of currency returns, while the predictive results and the evidence on curvature rest on the traded futures sample.

Table F1: Synthetic VIX futures against benchmarks. Panel A compares the 30-day implied volatility from an exact replication of the CBOE VIX methodology on the listed S&P500 option quotes in OptionMetrics IvyDB US with the CBOE VIX index. Panel B compares the 30-day model-free implied volatility computed from the IvyDB US volatility surface, as described in Subsection F.1, with the CBOE VIX index. Panel C compares the model-free implied volatility from the surface with the exact replication at each maturity of the surface. The exact replication at a maturity is interpolated linearly in total variance between the two listed expiries around it, so it is available only on days when listed expiries lie on both sides of that maturity. Panel D compares the synthetic VIX futures with the traded VIX futures at the same maturity. Panels A to C cover January 1996 to March 2024, and Panel D covers October 2006 to March 2024. Means and the mean gap (own minus benchmark) are in volatility points. The correlation is computed on daily levels.

		Mean			
	Days	Own	Benchmark	Mean gap	Correlation
Panel A: CBOE replication on listed quotes and the CBOE VIX					
30 days	6,300	19.80	20.13	-0.33	0.998
Panel B: surface and the CBOE VIX					
30 days	7,107	19.42	20.30	-0.89	0.997
Panel C: surface and CBOE replication on listed quotes					
30 days	6,300	19.23	19.80	-0.58	0.997
60 days	7,107	20.01	20.74	-0.74	0.995
91 days	7,106	20.37	21.29	-0.92	0.991
122 days	7,102	20.64	21.72	-1.08	0.990
152 days	7,090	20.85	21.99	-1.14	0.988
182 days	7,090	21.01	22.18	-1.16	0.984
273 days	6,994	21.40	22.55	-1.16	0.969
365 days	2,540	21.51	23.43	-1.92	0.956
Panel D: synthetic and traded VIX futures					
1M	4,387	20.27	20.44	-0.16	0.990
2M	4,387	21.10	21.30	-0.20	0.992
3M	4,387	21.66	21.77	-0.10	0.991
6M	4,387	22.46	22.54	-0.07	0.982
9M	4,387	22.70	23.88	-1.18	0.895

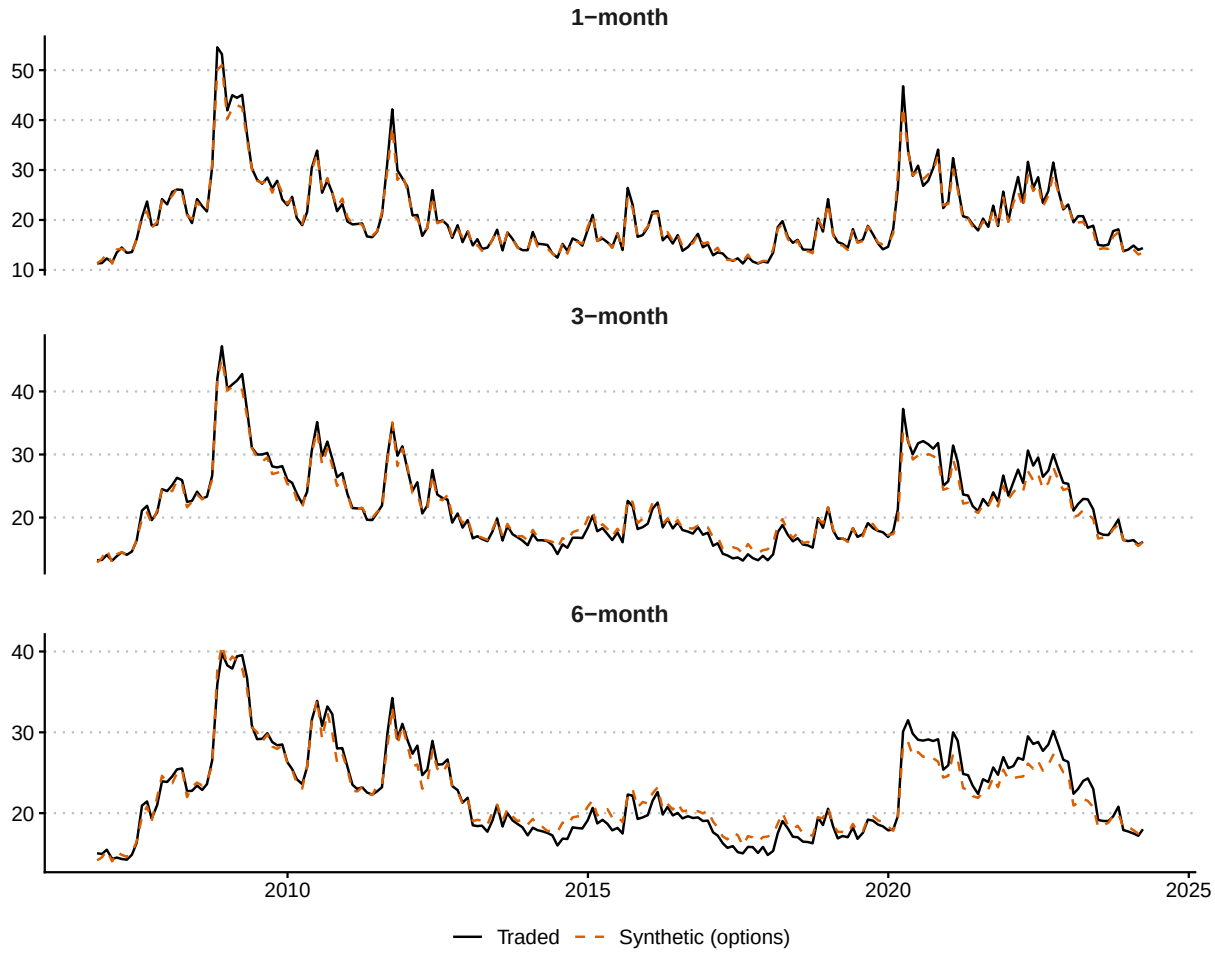


Figure F1: Synthetic and traded VIX futures. This figure shows the month-end synthetic VIX futures built from S&P500 options (dashed) and the traded VIX futures (solid) at maturities of one, three and six months, from October 2006 to March 2024.

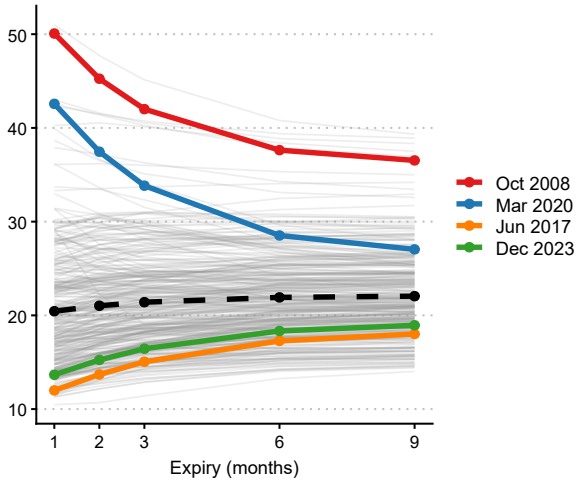


Figure F2: Selected end-of-month synthetic VIX futures term-structures. Grey lines are the term-structures at all other month-ends in the sample, the coloured lines are the selected dates and the dashed black line is the sample average.

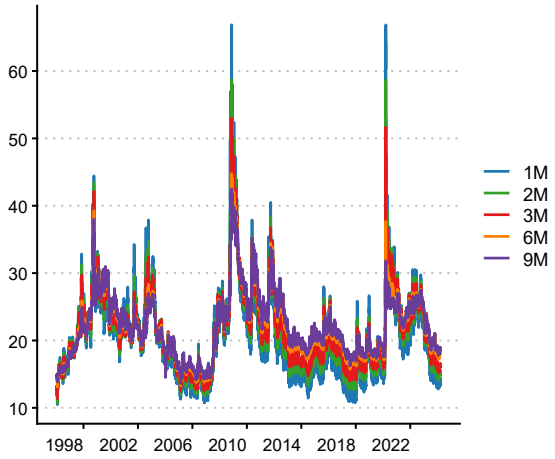


Figure F3: Time-series of synthetic VIX futures, daily

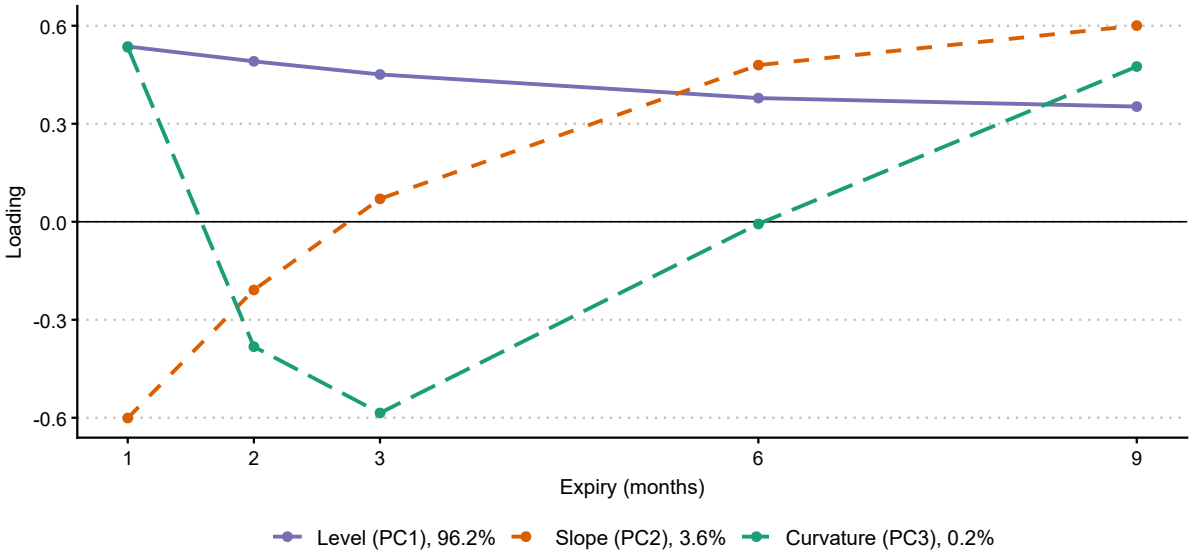


Figure F4: Factor loadings of synthetic VIX futures PCA. This figure shows the factor loadings of the full-sample principal component analysis on the level of synthetic VIX futures. Loadings are scaled such that they correspond to common measures of level, slope, and curvature. Percentage of variance explained by each principal component is between parenthesis.

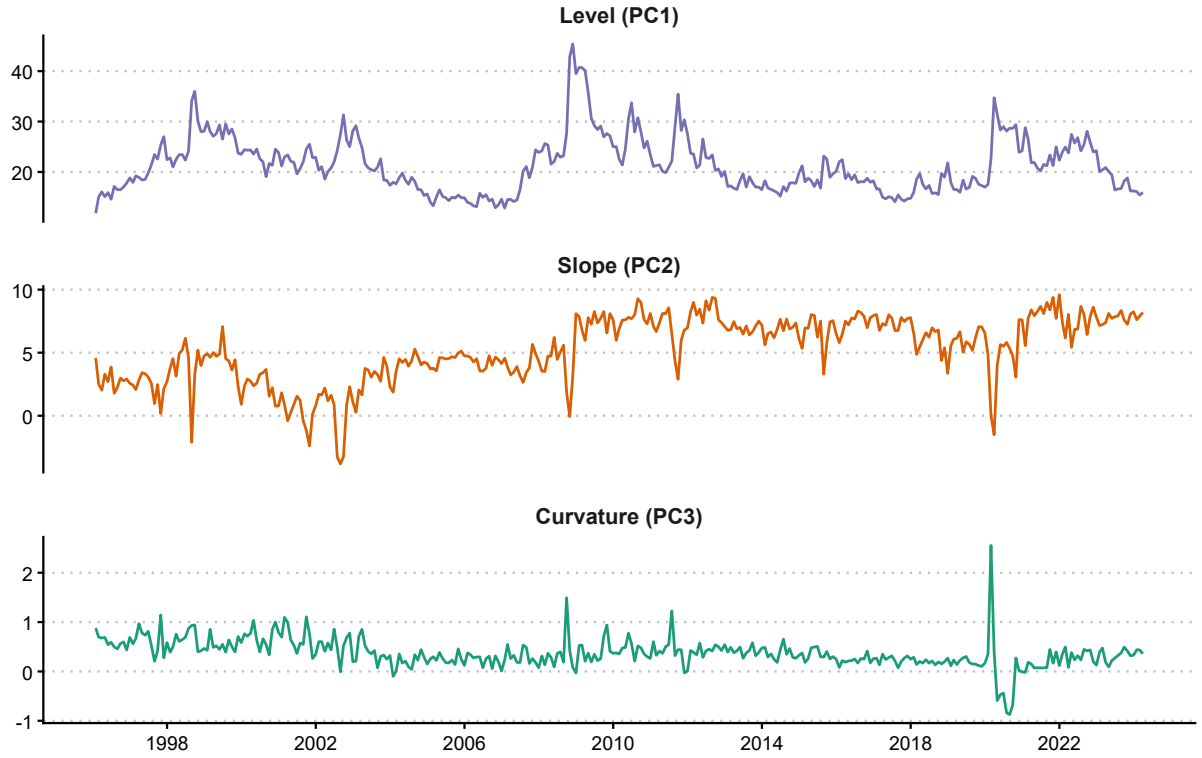


Figure F5: Principal components of expanding window PCA. This figure shows the time-series of the first three principal components extracted from synthetic VIX futures using an expanding window PCA, with a startup period of 252 days. PCA is performed on daily data, with the resulting factors sampled at month-end. The signs of the loadings are such that PC1 captures the level of the curve, PC2 captures the slope, and PC3 captures curvature. Increases in PC1 reflect upward shifts in the entire curve, typically associated with deteriorating market conditions, while increases in PC2 reflect steepening, often interpreted as improving conditions. PC3 captures changes in curvature, though its interpretation is less directly linked to macro-financial states.

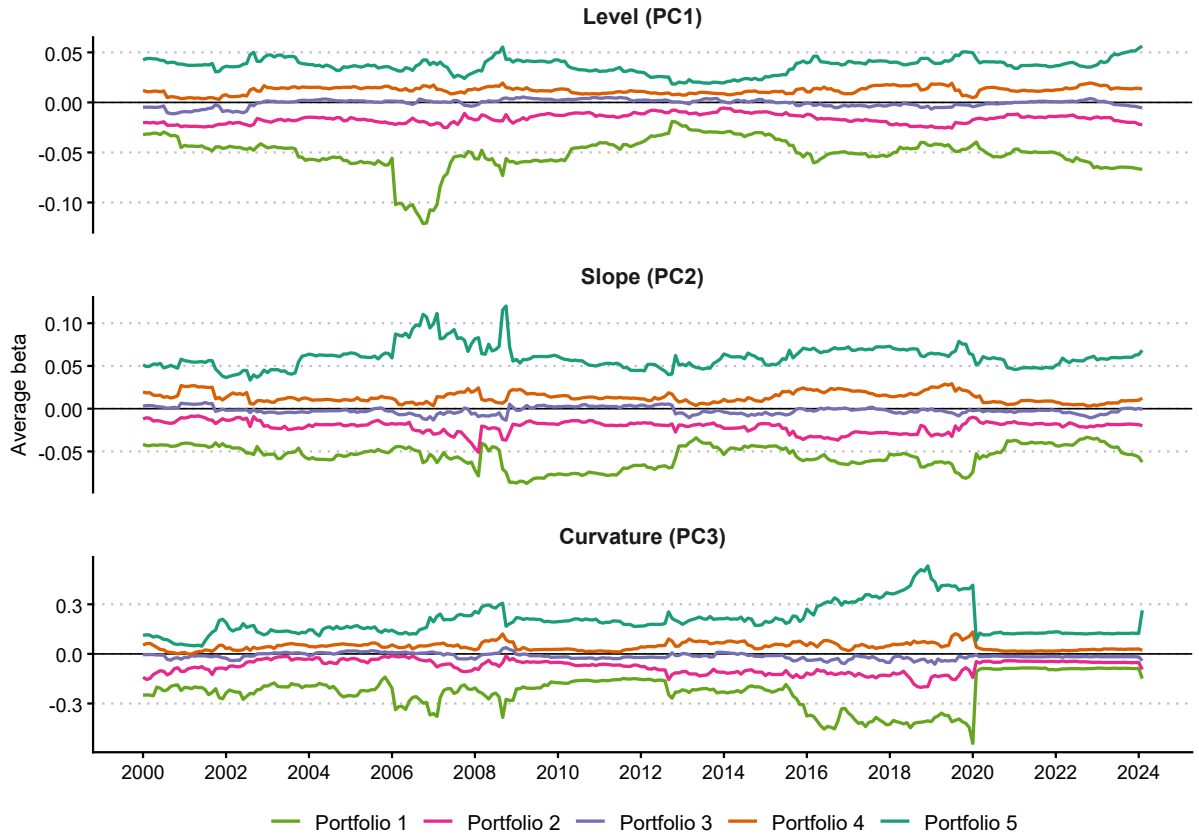


Figure F6: Rolling betas to synthetic VIX futures principal component innovations by currency portfolio. This figure shows the time-series of average betas for five currency portfolios sorted each month by their estimated exposure to innovations in synthetic VIX futures term-structure components. The principal components, capturing the level (PC1), slope (PC2), and curvature (PC3) of the synthetic VIX futures curve, are extracted monthly using an expanding-window PCA. Innovations in each component are computed as the monthly change relative to the previous observation. For each currency, rolling 48-month regressions of excess returns on these innovations are estimated following Equation 28, controlling for the Dollar factor. The resulting betas measure time-varying sensitivities to shocks in the synthetic VIX futures term-structure. The figure reports portfolio-average betas across the five beta-sorted groups, illustrating persistent cross-sectional heterogeneity and cyclical variation in exposures to volatility-level (PC1), slope (PC2), and curvature (PC3) innovations.

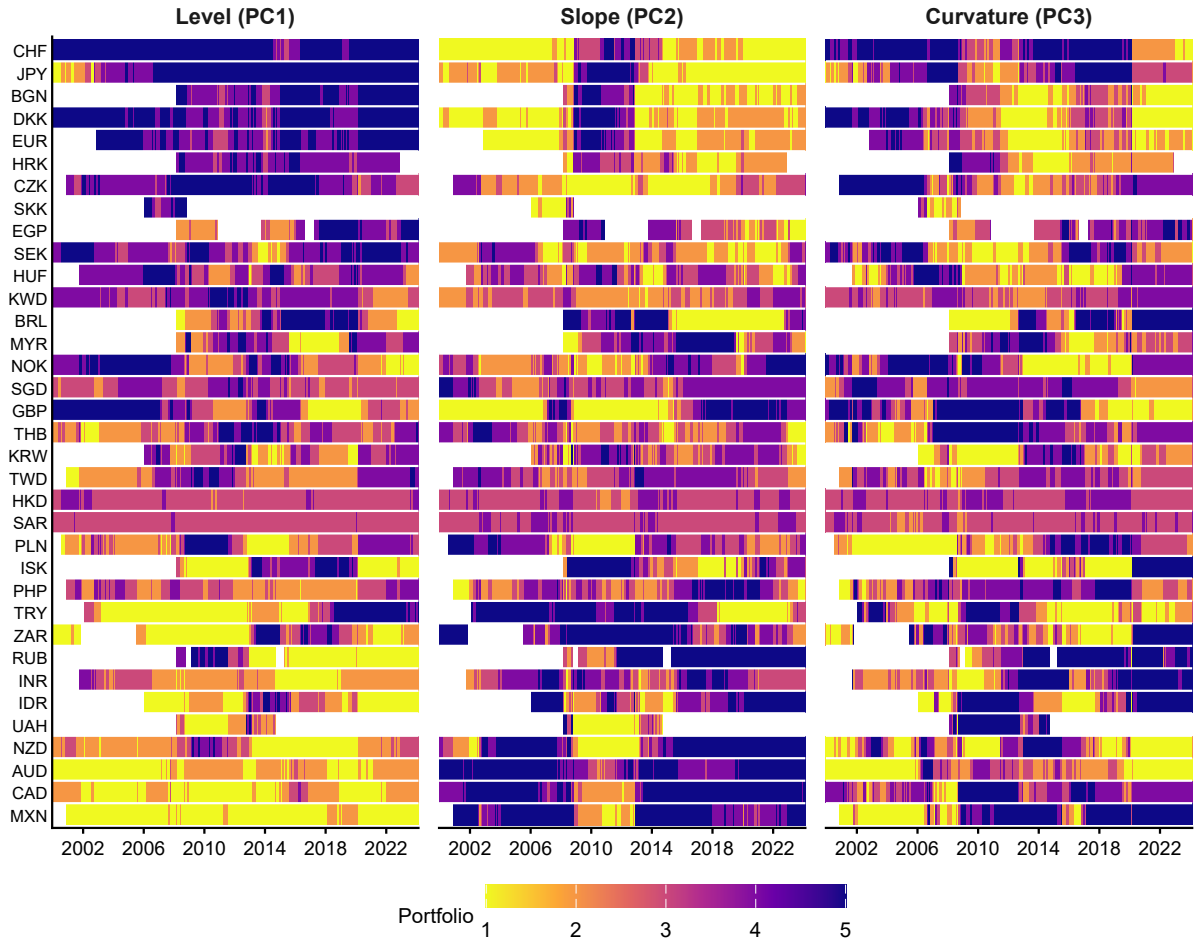


Figure F7: Composition of currency portfolios sorted on synthetic VIX futures principal component betas. This figure displays the time-varying portfolio assignments of individual currencies based on their rolling 48-month betas with respect to innovations in the first three synthetic VIX futures principal components: level (PC1), slope (PC2), and curvature (PC3). Each color represents the portfolio (from 1 to 5) to which a currency belongs in a given month, determined by its beta ranking from the previous month's estimation. Left corresponds to portfolios sorted on PC1 betas, middle panel on PC2, and right on PC3. Lighter shades indicate lower-exposure portfolios, while darker shades represent higher exposures. The figure highlights persistent clustering of currencies with similar sensitivities to volatility term-structure shocks, as well as transitions across portfolios reflecting evolving exposure patterns over time.

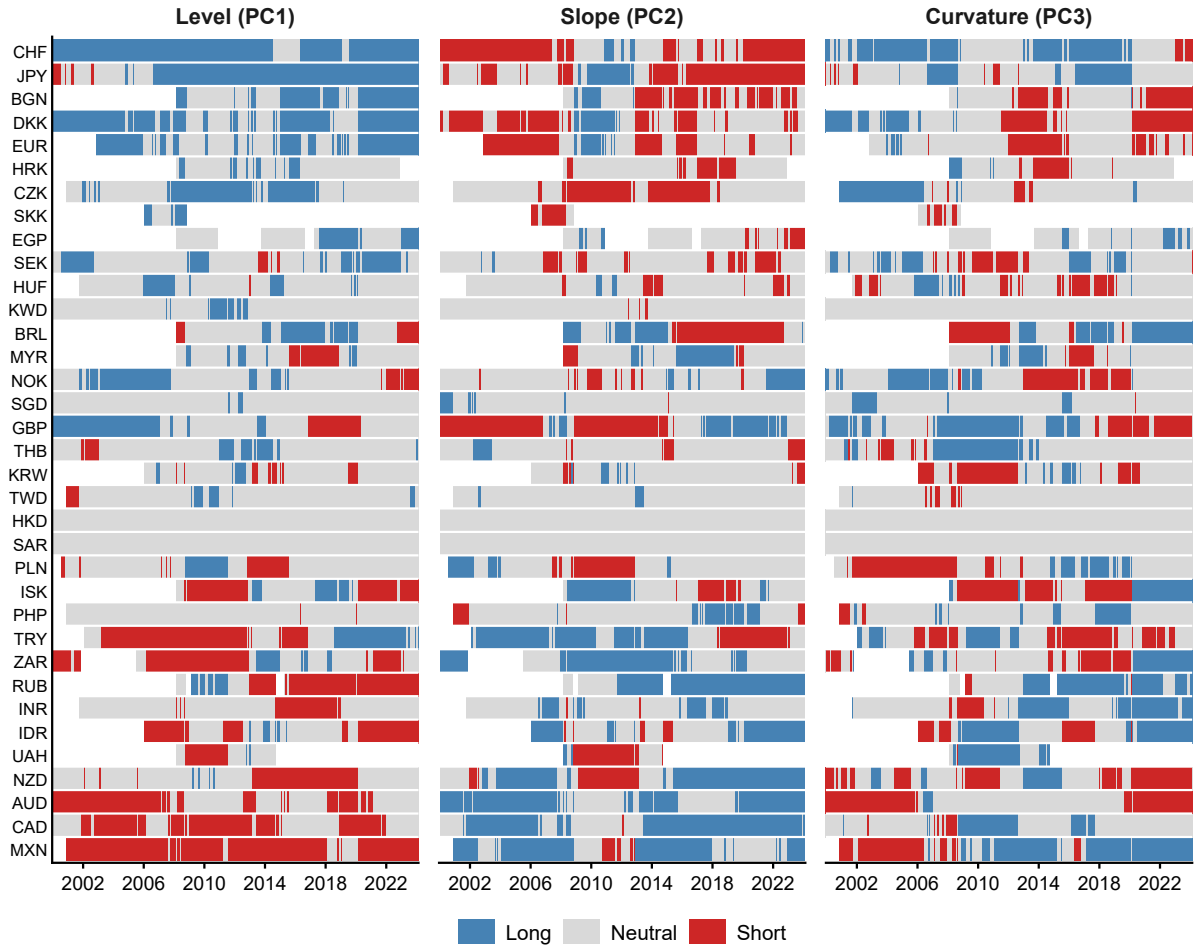


Figure F8: Composition of currency risk factors constructed from synthetic VIX futures principal component betas. This figure shows the time-varying composition of the high-minus-low (HML) currency risk factors associated with exposures to innovations in the first three synthetic VIX futures principal components: level (PC1), slope (PC2), and curvature (PC3). For each month, currencies are sorted into five portfolios based on their rolling 48-month betas with respect to the respective principal component innovation, and the factor is constructed as the return difference between the highest- and lowest-beta portfolios. The heatmap indicates, for each currency and month, whether it enters the factor with a long position (blue), short position (red), or is excluded from the factor (gray). Each panel corresponds to one VIX component. The figure highlights persistent cross-sectional asymmetries in how currencies load onto the VIX futures term-structure level, slope, and curvature shocks, as well as shifts in factor composition reflecting changing sources of volatility risk exposure across time.

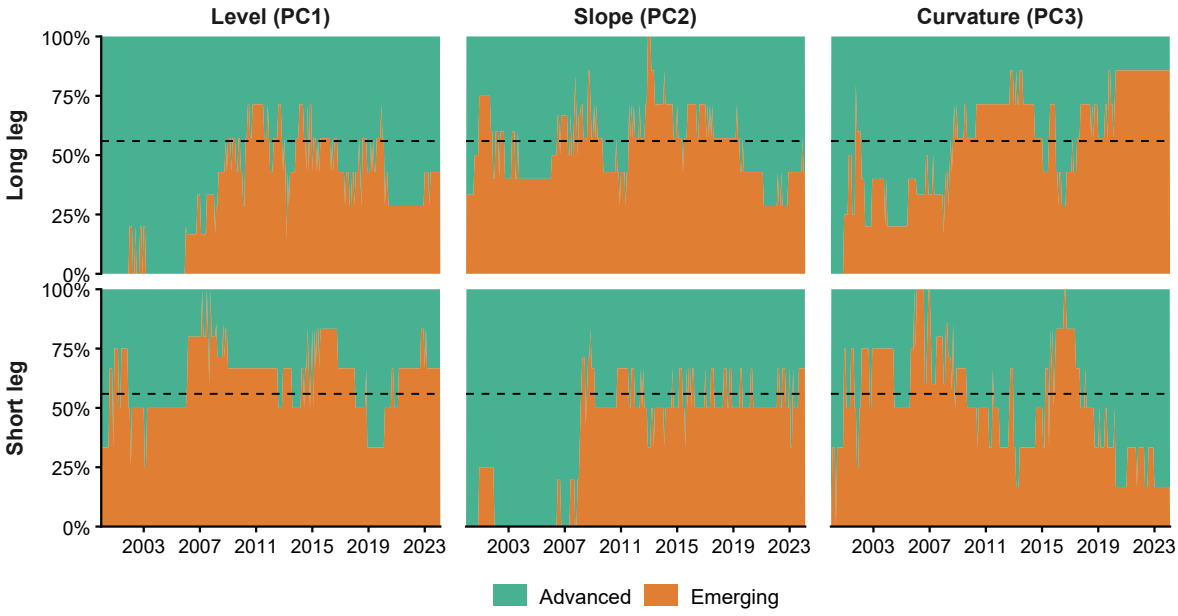


Figure F9: Emerging-market composition of the synthetic VIX futures term-structure factors. Share of emerging-market currencies (orange) and advanced-market currencies (green) among the currencies in the long leg (top row) and the short leg (bottom row) of the high-minus-low factors built on betas to innovations in the level, slope and curvature of the synthetic VIX futures term-structure, monthly. The dashed line is the average share of emerging-market currencies among all currencies in the cross-section over the sample, so a leg whose orange area lies above (below) the line holds more (fewer) emerging-market currencies than the cross-section as a whole.

Table F2: Synthetic VIX futures beta-sorted portfolios summary statistics. This table reports summary statistics for monthly currency excess returns of portfolios sorted by estimated betas with respect to synthetic VIX futures innovations. Panel A sorts currencies on β_{PC1} (level), Panel B on β_{PC2} (slope), and Panel C on β_{PC3} (curvature). P1 contains currencies with the lowest beta exposures, while P5 contains those with the highest. HML denotes the high-minus-low portfolio (P5 – P1). The Sharpe Ratio is computed as the ratio of average annualized return to annualized volatility. t-stat refers to the t-statistic testing whether the average monthly excess return is significantly different from zero.

Portfolio	P1	P2	P3	P4	P5	HML
Panel A: sorts on β_{PC1}						
Avg Ann. Return	2.42	1.85	-0.05	0.02	-1.55	-3.97
t-stat	[1.40]	[1.21]	[-0.04]	[0.01]	[-0.86]	[-2.39]
Ann. Volatility	8.49	7.52	5.58	7.46	8.86	8.17
Sharpe Ratio	**0.29**	**0.25**	**-0.01**	**0.00**	**-0.17**	**-0.49**
Skewness	-0.62	-0.42	-0.61	-0.61	-0.22	-0.01
Panel B: sorts on β_{PC2}						
Avg Ann. Return	-1.84	0.34	0.35	0.85	2.18	4.02
t-stat	[-0.96]	[0.23]	[0.32]	[0.62]	[1.19]	[2.48]
Ann. Volatility	9.45	7.25	5.42	6.76	9.04	7.99
Sharpe Ratio	**-0.19**	**0.05**	**0.06**	**0.13**	**0.24**	**0.50**
Skewness	-0.34	-0.63	-0.55	-0.33	-0.89	0.18
Panel C: sorts on β_{PC3}						
Avg Ann. Return	1.21	-0.00	-0.29	0.45	0.92	-0.29
t-stat	[0.65]	[-0.00]	[-0.26]	[0.30]	[0.59]	[-0.21]
Ann. Volatility	9.22	8.29	5.58	7.34	7.62	6.79
Sharpe Ratio	**0.13**	**-0.00**	**-0.05**	**0.06**	**0.12**	**-0.04**
Skewness	-0.63	-1.02	-0.46	-0.87	0.17	0.36

Table F3: Time-series regression of synthetic VIX futures beta-sorted portfolios on the Carry factor. This table reports the results of time-series regressions of monthly excess returns of synthetic VIX futures beta-sorted currency portfolios on the Carry factor of Lustig et al. (2011). Panels A–C correspond to portfolios sorted on betas with respect to innovations to the first three principal components of the synthetic VIX futures: level (PC1), slope (PC2), and curvature (PC3), respectively. P1 includes currencies with the lowest beta exposure and P5 includes those with the highest. HML denotes the high-minus-low portfolio (P5 – P1). Newey-West t-statistics are reported in brackets.

Portfolio	P1	P2	P3	P4	P5	HML
Panel A: sorts on β_{PC1}						
Intercept	-0.02	-0.00	-0.02	-0.01	0.01	0.03
	[-1.05]	[-0.23]	[-1.01]	[-0.33]	[0.36]	[2.00]
Carry Factor	0.43	0.22	0.14	0.06	-0.22	-0.65
	[3.98]	[2.25]	[2.02]	[0.71]	[-1.51]	[-10.08]
R^2	0.180	0.059	0.045	0.005	0.046	0.453
Panel B: sorts on β_{PC2}						
Intercept	-0.00	0.00	-0.00	-0.01	-0.02	-0.02
	[-0.01]	[0.04]	[-0.30]	[-0.55]	[-0.95]	[-1.38]
Carry Factor	-0.17	0.02	0.07	0.16	0.42	0.59
	[-0.87]	[0.24]	[1.14]	[2.42]	[3.22]	[5.62]
R^2	0.023	0.001	0.014	0.039	0.156	0.391
Panel C: sorts on β_{PC3}						
Intercept	-0.02	-0.01	-0.01	-0.00	0.00	0.02
	[-0.59]	[-0.38]	[-0.86]	[-0.23]	[0.19]	[1.09]
Carry Factor	0.26	0.09	0.09	0.08	0.05	-0.20
	[2.16]	[0.57]	[1.29]	[0.90]	[0.66]	[-2.20]
R^2	0.055	0.008	0.020	0.009	0.004	0.063

Table F4: Asset pricing tests of synthetic VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the portfolio-level test-assets using the synthetic VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of Carry, Momentum, and Value portfolios, while columns (6)–(10) use only the Momentum and Value portfolios. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Carry + Momentum + Value						Momentum + Value			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.039^{**}			-0.036^{**}		-0.039^{**}			-0.038^{**}
		(0.017)			(0.017)		(0.017)			(0.017)
β_{fPC2}			0.042^{**}		0.043^{***}			0.041^{**}		0.042^{**}
			(0.016)		(0.016)			(0.016)		(0.017)
β_{fPC3}				-0.005	-0.004				-0.005	-0.004
				(0.014)	(0.014)				(0.014)	(0.014)
β_{CAR}	0.107^{***}	0.108^{***}	0.107^{***}	0.107^{***}	0.108^{***}	0.123^{**}	0.109^{***}	0.108^{***}	0.108^{***}	0.109^{***}
	(0.020)	(0.018)	(0.018)	(0.018)	(0.018)	(0.052)	(0.017)	(0.017)	(0.017)	(0.017)
β_{DOL}	0.005	0.005	0.005	0.005	0.005	0.004	0.005	0.005	0.005	0.005
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
R^2	0.869	0.957	0.941	0.940	0.958	0.456	0.949	0.927	0.922	0.951
GRS	0.175	0.231	0.224	0.205	0.271	0.152	0.175	0.204	0.179	0.209

Table F5: Asset pricing tests of synthetic VIX futures term-structure factors - second stage. This table reports the second-stage cross-sectional asset pricing results for the country-level test-assets using the synthetic VIX futures term-structure currency risk factors. The dependent variables are the annualized average excess returns of the test assets, while the regressors are the betas estimated in the first-stage time-series regressions. Columns (1)–(5) correspond to a cross-section of all country excess returns, while columns (6)–(10) use only advanced countries currency excess returns. Each specification includes the Dollar and Carry factors from the literature, along with different combinations of the three VIX-based currency risk factors: level ($fPC1$), slope ($fPC2$), and curvature ($fPC3$). The λ estimates represent the price of risk associated with each factor. I also report the p-values from the test of Gibbons et al. (1989), which tests the joint hypothesis that all pricing errors (alphas) are equal to zero, i.e., that the model prices all assets correctly. A failure to reject this null suggests that the pricing model performs well in explaining cross-sectional variation in returns. Standard errors are corrected following Shanken (1992), and statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	Test assets									
	Country level - all					Country level - advanced				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.030^{*}			-0.019		-0.032^{*}			-0.033^{*}
		(0.018)			(0.018)		(0.018)			(0.018)
β_{fPC2}			0.043^{**}		0.052^{***}			0.033^{*}		0.041^{**}
			(0.017)		(0.017)			(0.018)		(0.017)
β_{fPC3}				-0.009	-0.005				0.002	-0.0003
				(0.014)	(0.014)				(0.014)	(0.014)
β_{CAR}	0.058^{***}	0.073^{***}	0.066^{***}	0.067^{***}	0.074^{***}	0.056^{**}	0.101^{***}	0.098^{***}	0.092^{***}	0.102^{***}
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.026)	(0.018)	(0.018)	(0.019)	(0.018)
β_{DOL}	0.007	0.006	0.007	0.007	0.006	0.008	0.009	0.011	0.010	0.009
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
R^2	0.674	0.744	0.719	0.709	0.787	0.550	0.884	0.852	0.829	0.898
GRS	0.004	0.012	0.008	0.004	0.011	0.007	0.026	0.010	0.008	0.022

Table F6: Predictive panel regressions. The dependent variable is the one-month-ahead currency excess return (rx_{t+1}, columns 1–2) or the one-month log change in the U.S. dollar price of currency i (Δs_{t+1}, columns 3–4, where an increase is an appreciation of currency i against the dollar). The regressors are the month-t interest-rate differential between the United States and country i ($y_t^{US} - y_t^i$, measured under covered interest parity as minus the log forward premium), the month-t innovations to the level ($\Delta PC1$), slope ($\Delta PC2$), and curvature ($\Delta PC3$) of the synthetic VIX futures term-structure, and the lagged excess return rx_t. Columns (1) and (3) report pooled OLS estimates for all currencies, and columns (2) and (4) report corresponding within estimations with individual (currency) fixed effects. Panel Newey–West standard errors (one lag) are shown in parentheses. Constant terms are included in the pooled OLS specifications but their coefficients are omitted from the table. This is similar to the specification of Londono and Zhou (2017). Returns and the interest-rate differential are annualized. Adjusted R^2 values are reported in percent. Statistical significance is denoted by *, **, and *** for the 10%, 5%, and 1% levels, respectively.

	rx_{t+1}		Δs_{t+1}	
	(1)	(2)	(3)	(4)
$y_t^{US} - y_t^i$	-0.898^{***}	-0.985^{***}	0.102	0.015
	(0.095)	(0.090)	(0.095)	(0.090)
$\Delta PC1$	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
$\Delta PC2$	0.006	0.006	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)
$\Delta PC3$	-0.050^{***}	-0.050^{***}	-0.050^{***}	-0.050^{***}
	(0.018)	(0.018)	(0.018)	(0.018)
rx_t	0.058^{***}	0.048^{***}	0.058^{***}	0.048^{***}
	(0.017)	(0.016)	(0.017)	(0.016)
FE	No	Yes	No	Yes
R^2 (%)	11.17	11.00	0.56	0.02

G. Bayesian Fama–MacBeth regressions

This appendix revisits the cross-sectional asset pricing tests of Table 3 and Table 4 using Bayesian methods. The two-pass Fama–MacBeth procedure treats the first-stage betas as known when estimating the second-stage prices of risk, and the Shanken (1992) correction addresses this errors-in-variables problem only asymptotically. The Bayesian Fama–MacBeth estimator of Bryzgalova et al. (2024) instead delivers the exact finite-sample posterior distribution of each price of risk, propagating the uncertainty about the joint distribution of test-asset returns and factors through both stages of the procedure. The purpose of this exercise is therefore to ask whether the economic message of the baseline tests survives when no asymptotic approximation is imposed: negative prices of risk for the level and curvature factors, a positive price of risk for the slope factor, and a sizable improvement in cross-sectional fit relative to the Carry and Dollar benchmark. I use the same VIX futures term-structure currency risk factors, the same Carry and Dollar controls, and the same four test-asset cross-sections as in the main text.

G.1. The estimator

The estimator replaces the two sequential regressions with a single model for the joint distribution of factors and test-asset returns. Let $Y_t = (f_t', rx_t')'$ stack the factors and the test-asset excess returns, and take Y_t to be multivariate normal with mean μ and covariance Σ. Under a diffuse Jeffreys prior the posterior of (μ, Σ) is available in closed form and is centred at the sample moments,

$$\Sigma \;\sim\; \text{inverse-Wishart}\big(T-1,\; T\widehat{\Sigma}\big), \qquad \mu \mid \Sigma \;\sim\; \mathcal{N}\big(\widehat{\mu},\; \Sigma/T\big), \tag{78}$$

so each draw of (μ, Σ) is one mean and covariance consistent with the observed sample.

Both passes are then carried out inside each draw. Partitioning the drawn covariance into the factor block Σ_f and the asset-factor block Σ_{rf}, and writing a for the asset block of the drawn mean, the betas are read directly from the drawn covariance and the prices of risk from a cross-sectional projection on them,

$$\beta = \Sigma_{rf}\,\Sigma_f^{-1}, \qquad \lambda = (H'H)^{-1}H'a, \qquad \alpha = a - H\lambda, \tag{79}$$

with $H = [\mathbf{1},\, \beta]$. Every draw therefore delivers one vector of prices of risk, one vector of pricing errors and one cross-sectional R^2, and the posterior of each object is the collection of draws. I use 50,000 draws after a burn-in of 10,000 and report the posterior based on the OLS second stage, which is the closest analogue to the Shanken-corrected estimates in

the main text.

Two features of this construction govern how the results should be read. The first is the source of the exactness. Because β and a are read from the same draw of (μ, Σ), uncertainty about the betas is carried into the posterior of λ by construction rather than corrected for afterwards, which is what removes the need for an asymptotic adjustment. The second is that the prior does no work on λ itself. The Jeffreys prior is uninformative, so the posterior median lands on the classical point estimate and what the exercise contributes is the shape of the distribution around it rather than a movement of its centre. That distinction is worth making because the other Bayesian exercise in the paper, the model selection of Subsection 4.4, uses a deliberately informative spike-and-slab prior, so the two results are not the same kind of statement. What the posterior supports here is a set of direct probability statements: a 90% credible interval for each price of risk taken from the draw quantiles, the posterior probability that a price of risk carries the sign of the baseline estimate, and the posterior distribution of the cross-sectional R^2.

One implementation detail deserves mention. Because the Carry factor is the return spread between the corner forward-premium-sorted portfolios and the Dollar factor is their cross-sectional average, the joint covariance matrix of factors and test assets is singular when all five Carry portfolios are included as test assets. In the Carry, Momentum, and Value cross-section I therefore exclude the two corner Carry portfolios from the test-asset matrix, which removes the exact linear dependence while leaving the pricing information of the corner portfolios represented through the Carry factor itself.

G.2. Results

Table G1 reports the posterior summaries for the portfolio cross-sections and Table G2 for the country-level cross-sections. Three results stand out. First, the posterior medians are close to the Shanken-corrected point estimates and preserve, in every specification of the portfolio and all-country cross-sections, the sign pattern documented in the main text: negative prices of risk for the level (PC1) and curvature (PC3) factors and a positive price of risk for the slope (PC2) factor. The advanced-country cross-section is the one place where the pattern frays: there the curvature posterior in the all-factor specification is centered essentially at zero. Second, the finite-sample posterior corroborates the classical inference where the paper's evidence is concentrated. The 90% credible intervals for the term-structure premia exclude zero in every single-factor specification of the portfolio and all-country cross-sections, and continue to exclude it in the all-factor specification for the level and curvature factors in the portfolio cross-sections and for the level and slope factors in the all-country cross-section. They include zero throughout the advanced-country

cross-section, where beta uncertainty is largest, and for the slope factor when all three term-structure factors enter jointly in the portfolio cross-sections. Table G3 quantifies the agreement between the Bayesian and the classical results: for every factor, specification, and cross-section, it reports the posterior probability that the price of risk has the sign of the Shanken-corrected baseline estimate. The posterior median has the sign of the baseline estimate in thirty-nine of the forty cases, so all but one entry exceeds one half. The exception is again the curvature factor in the advanced-country all-factor specification, where the posterior is an even split (0.48). In the single-factor specifications the sign probabilities of the term-structure premia range from 0.97 to 0.99 in the portfolio and all-country cross-sections, which is the range of the Carry factor itself, whose premium is not in question, and clearly above the 0.72 to 0.83 of the Dollar factor. The finite-sample posterior is therefore as decisive about the signs of the term-structure premia as it is about the Carry premium. The probabilities decline in the advanced-country cross-section and, for the slope factor, in the all-factor specification: the three term-structure factors are correlated snapshots of the same curve, so when they enter jointly they share the explanatory burden, the same substitution effect that reappears in the model-selection results of Subsection 4.4. Third, the posterior distribution of the cross-sectional R^2 reproduces the pattern emphasized in the main text: adding any of the VIX term-structure factors shifts the entire posterior of the R^2 upward relative to the Carry and Dollar benchmark, with the largest improvement in the all-country cross-section, where the posterior median rises from 0.50 to 0.76, against 0.27 to 0.42 for advanced countries and 0.40 to 0.56 in the Carry, Momentum, and Value cross-section.

Table G1: Bayesian Fama–MacBeth asset pricing tests of VIX futures term-structure factors – portfolio cross-sections. This table reports posterior summaries of the second-stage risk premia estimated with the Bayesian Fama–MacBeth procedure of Bryzgalova et al. (2024), for the same factor specifications and test assets as Table 3. Columns (1)–(5) correspond to the cross-section of Carry, Momentum, and Value portfolios, while columns (6)–(10) use only the Momentum and Value portfolios. Because the Carry and Dollar factors are exact linear combinations of the five forward-premium-sorted portfolios, the two corner Carry portfolios are excluded from the test-asset matrix in columns (1)–(5) to keep the joint covariance matrix of factors and test assets non-singular. Each cell reports the posterior median of the annualized price of risk with the 90% posterior credible interval in brackets. The table footer reports the posterior median of the cross-sectional R^2 with its 90% credible interval. Significance markers are the Bayesian analogue of the classical stars: *, **, and *** denote that the two-sided posterior tail probability, $2\min\{\Pr(\lambda > 0), \Pr(\lambda < 0)\}$, is below 10%, 5%, and 1%, respectively.

	Test assets									
	Carry + Momentum + Value					Momentum + Value				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.106^{**} [−0.186, −0.032]			-0.093^{*} [−0.182, −0.008]		-0.110^{**} [−0.195, −0.033]			-0.101^{*} [−0.202, −0.007]
β_{fPC2}			0.078^{*} [0.012, 0.147]		0.059 [−0.013, 0.133]			0.089^{**} [0.017, 0.164]		0.074 [−0.014, 0.166]
β_{fPC3}				-0.078^{**} [−0.141, −0.019]	-0.079^{*} [−0.153, −0.009]				-0.080^{**} [−0.143, −0.020]	-0.076^{*} [−0.153, −0.002]
β_{CAR}	0.112^{**} [0.032, 0.195]	0.092^{*} [0.006, 0.181]	0.105^{**} [0.024, 0.190]	0.098^{**} [0.017, 0.181]	0.090^{*} [0.003, 0.181]	0.140^{**} [0.045, 0.244]	0.117^{*} [0.004, 0.235]	0.135^{**} [0.037, 0.243]	0.123^{*} [0.018, 0.236]	0.115 [−0.003, 0.243]
β_{DOL}	−0.034 [−0.094, 0.024]	−0.032 [−0.092, 0.026]	−0.022 [−0.074, 0.030]	−0.025 [−0.080, 0.029]	−0.031 [−0.083, 0.020]	−0.031 [−0.099, 0.035]	−0.031 [−0.100, 0.035]	−0.019 [−0.078, 0.038]	−0.024 [−0.087, 0.036]	−0.029 [−0.088, 0.031]
R^2	0.404 [−0.006, 0.729]	0.496 [0.049, 0.796]	0.475 [0.008, 0.789]	0.516 [0.049, 0.815]	0.559 [0.039, 0.849]	0.487 [0.012, 0.807]	0.542 [0.031, 0.847]	0.544 [0.015, 0.849]	0.560 [0.029, 0.854]	0.565 [−0.084, 0.891]

Table G2: Bayesian Fama–MacBeth asset pricing tests of VIX futures term-structure factors – country-level cross-sections. This table reports posterior summaries of the second-stage risk premia estimated with the Bayesian Fama–MacBeth procedure of Bryzgalova et al. (2024), for the same factor specifications and test assets as Table 4. Columns (1)–(5) correspond to the cross-section of all country excess returns, while columns (6)–(10) use only advanced countries. Each cell reports the posterior median of the annualized price of risk with the 90% posterior credible interval in brackets. The table footer reports the posterior median of the cross-sectional R^2 with its 90% credible interval. Significance markers are the Bayesian analogue of the classical stars: *, **, and *** denote that the two-sided posterior tail probability, $2\min\{\Pr(\lambda > 0), \Pr(\lambda < 0)\}$, is below 10%, 5%, and 1%, respectively.

	Test assets									
	Country level - all					Country level - advanced				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
β_{fPC1}		-0.080^{**} [−0.135, −0.026]			-0.059^{*} [−0.113, −0.005]		−0.052 [−0.132, 0.030]			−0.050 [−0.144, 0.045]
β_{fPC2}			0.080^{**} [0.021, 0.139]		0.083^{**} [0.022, 0.144]			0.049 [−0.051, 0.152]		0.053 [−0.061, 0.164]
β_{fPC3}				-0.069^{**} [−0.124, −0.016]	−0.036 [−0.096, 0.022]				−0.004 [−0.112, 0.110]	0.004 [−0.118, 0.132]
β_{CAR}	0.073^{**} [0.014, 0.133]	0.047 [−0.011, 0.106]	0.051 [−0.004, 0.107]	0.043 [−0.011, 0.099]	0.053 [−0.004, 0.110]	0.101^{*} [0.012, 0.194]	0.110^{*} [0.008, 0.219]	0.103^{*} [0.014, 0.198]	0.111^{**} [0.019, 0.210]	0.116^{*} [0.006, 0.230]
β_{DOL}	−0.019 [−0.055, 0.018]	−0.016 [−0.052, 0.020]	−0.018 [−0.053, 0.017]	−0.018 [−0.053, 0.018]	−0.018 [−0.054, 0.017]	−0.012 [−0.047, 0.023]	−0.011 [−0.046, 0.025]	−0.012 [−0.049, 0.024]	−0.004 [−0.042, 0.036]	−0.004 [−0.044, 0.037]
R^2	0.504 [0.085, 0.791]	0.661 [0.229, 0.869]	0.727 [0.269, 0.901]	0.672 [0.211, 0.874]	0.760 [0.368, 0.914]	0.273 [−0.109, 0.689]	0.296 [−0.156, 0.717]	0.291 [−0.158, 0.722]	0.342 [−0.134, 0.758]	0.416 [−0.213, 0.830]

Table G3: Posterior probability that each price of risk has the sign of the classical baseline estimate. For every factor and test-asset cross-section, this table reports the Bayesian Fama–MacBeth posterior probability that the annualized price of risk lies on the same side of zero as the Shanken-corrected two-pass estimate of Table 3 and Table 4, i.e., $\Pr(\lambda_j > 0)$ when the baseline estimate is positive and $\Pr(\lambda_j < 0)$ when it is negative. In Panel A each term-structure factor is taken from the specification containing that factor together with Carry and Dollar, while Carry and Dollar are taken from the benchmark specification with the two factors alone; Panel B uses the specification including all five factors. These probabilities map into the significance markers of Table G1 and Table G2 through the two-sided posterior tail probability, which equals $2(1-p)$ whenever the reported probability p exceeds one half.

	Carry + Mom + Value	Mom + Value	Country - all	Country - advanced
Panel A: single-factor specifications				
$fPC1$	0.990	0.989	0.993	0.857
$fPC2$	0.974	0.981	0.988	0.791
$fPC3$	0.986	0.986	0.984	0.523
CAR	0.990	0.992	0.978	0.969
DOL	0.830	0.778	0.803	0.723
Panel B: all-factor specification				
$fPC1$	0.964	0.961	0.963	0.815
$fPC2$	0.911	0.917	0.987	0.789
$fPC3$	0.967	0.954	0.848	0.477
CAR	0.956	0.946	0.936	0.958
DOL	0.841	0.787	0.802	0.558

G.3. Posterior densities of the risk premia

Figure G1 to Figure G5 plot the full posterior density of each price of risk in the specification that includes all five factors, with one panel per test-asset cross-section. Each figure compares the Bayesian Fama–MacBeth posterior with a normal density centered at the two-pass point estimate with the Shanken-corrected standard error, so the distance between the two curves isolates what the finite-sample posterior adds to the classical approximation. The posteriors are wider and, in the smaller cross-sections, mildly asymmetric, but they remain centered near the classical estimates, which is the visual counterpart of the agreement between the posterior medians and the Shanken-corrected estimates in Table G1 and Table G2.

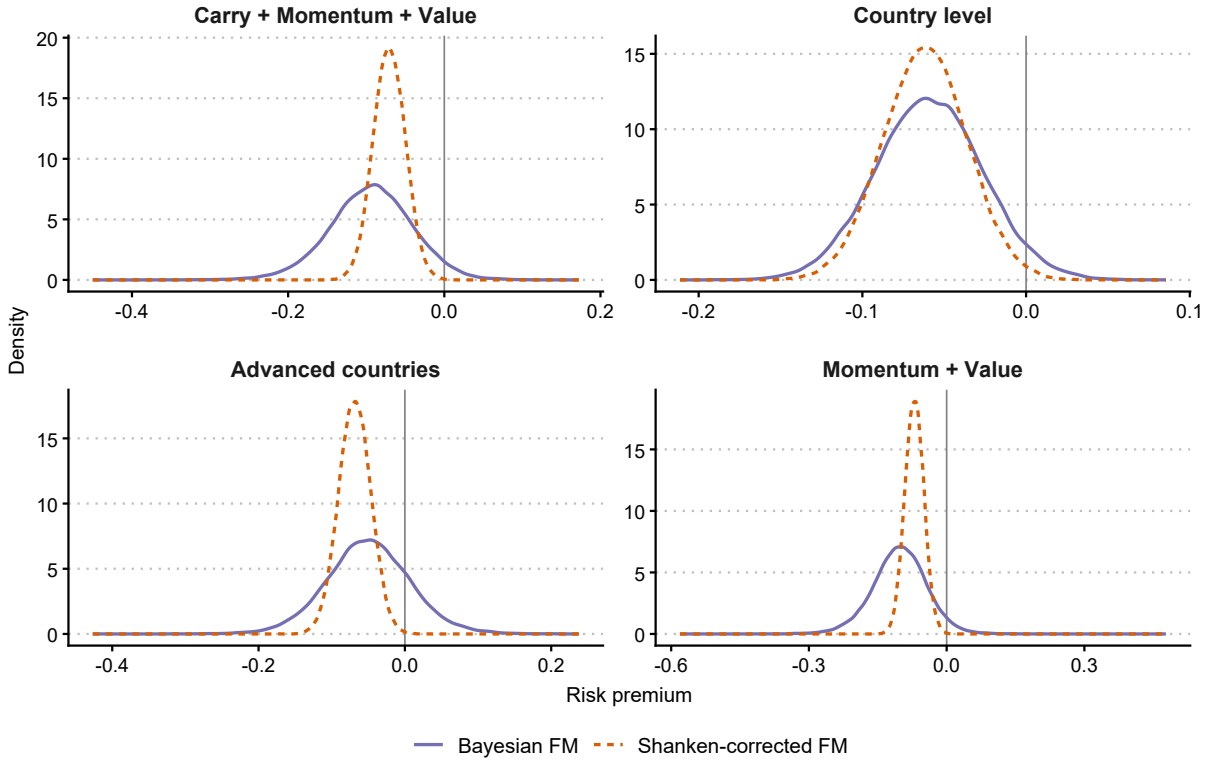


Figure G1: Posterior density of the level (PC1) risk premium. This figure plots the Bayesian Fama–MacBeth posterior distribution of the price of risk of the VIX futures level factor in the specification including all three term-structure factors together with the Carry and Dollar factors. Each panel corresponds to one cross-section of test assets. The comparison density is a normal distribution centered at the two-pass Fama–MacBeth estimate with the Shanken-corrected standard error.

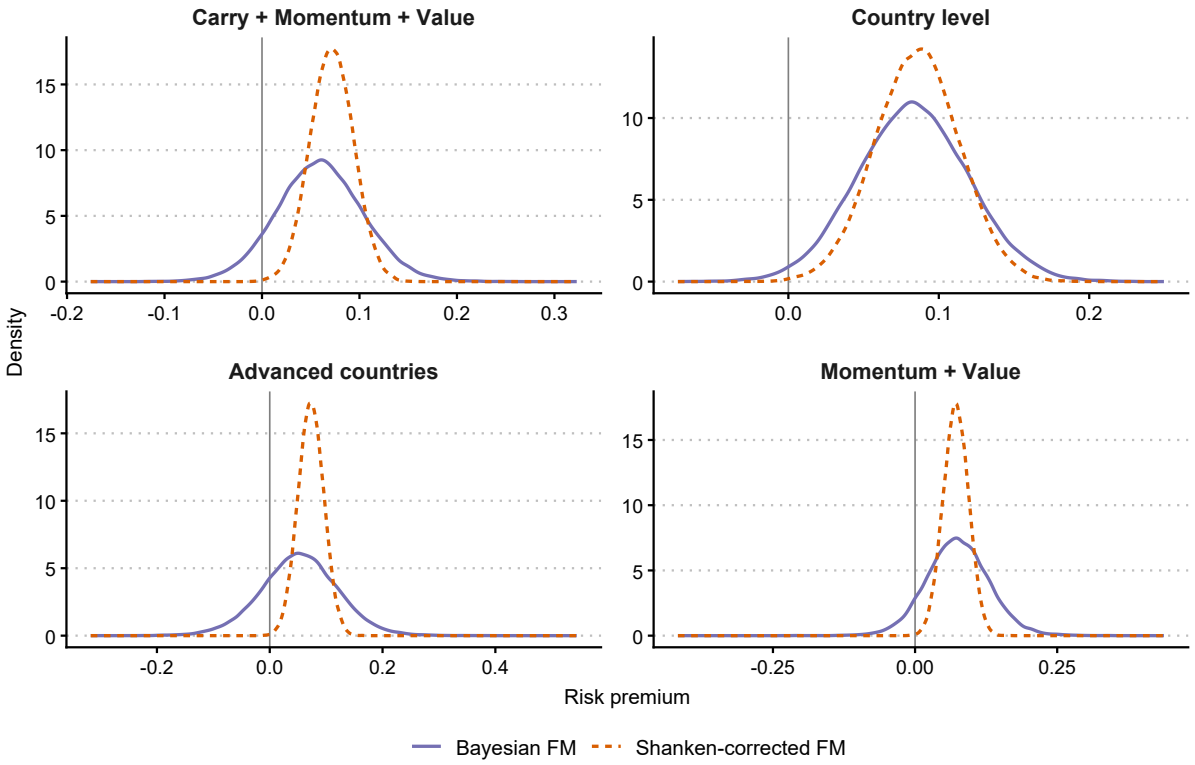


Figure G2: Posterior density of the slope (PC2) risk premium. This figure plots the Bayesian Fama–MacBeth posterior distribution of the price of risk of the VIX futures slope factor in the specification including all three term-structure factors together with the Carry and Dollar factors. Each panel corresponds to one cross-section of test assets. The comparison density is a normal distribution centered at the two-pass Fama–MacBeth estimate with the Shanken-corrected standard error.

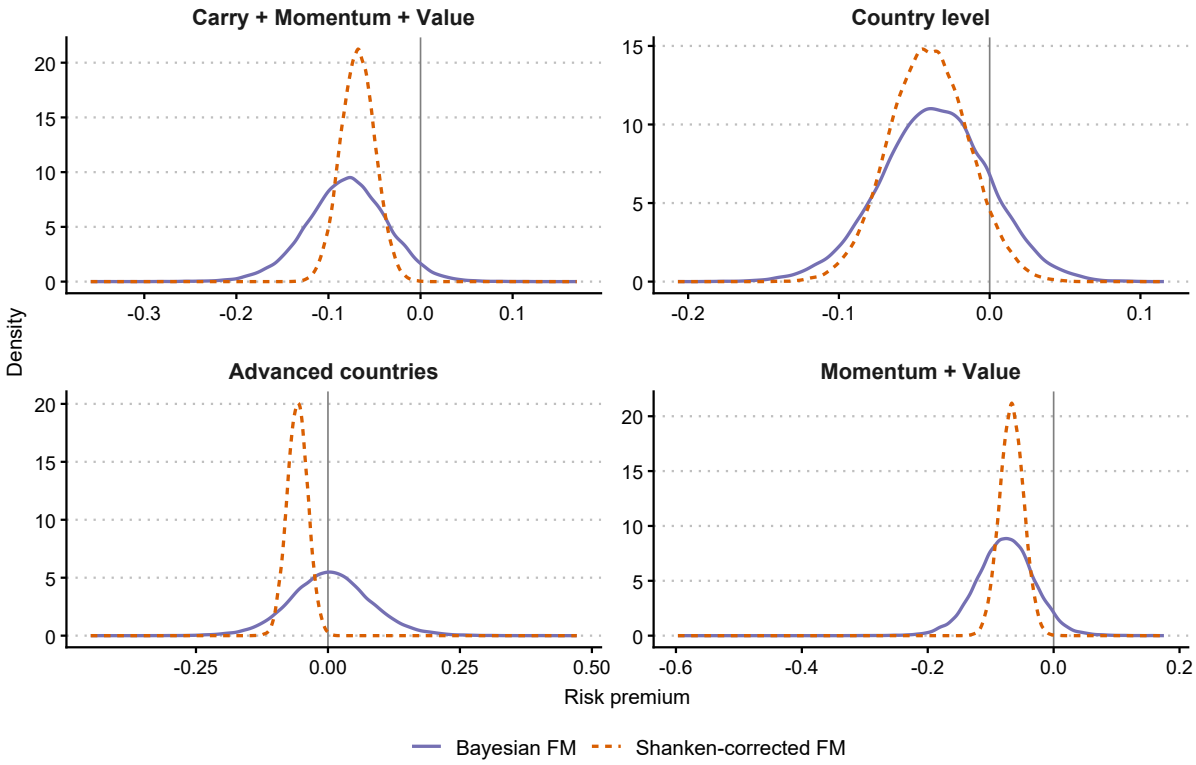


Figure G3: Posterior density of the curvature (PC3) risk premium. This figure plots the Bayesian Fama–MacBeth posterior distribution of the price of risk of the VIX futures curvature factor in the specification including all three term-structure factors together with the Carry and Dollar factors. Each panel corresponds to one cross-section of test assets. The comparison density is a normal distribution centered at the two-pass Fama–MacBeth estimate with the Shanken-corrected standard error.

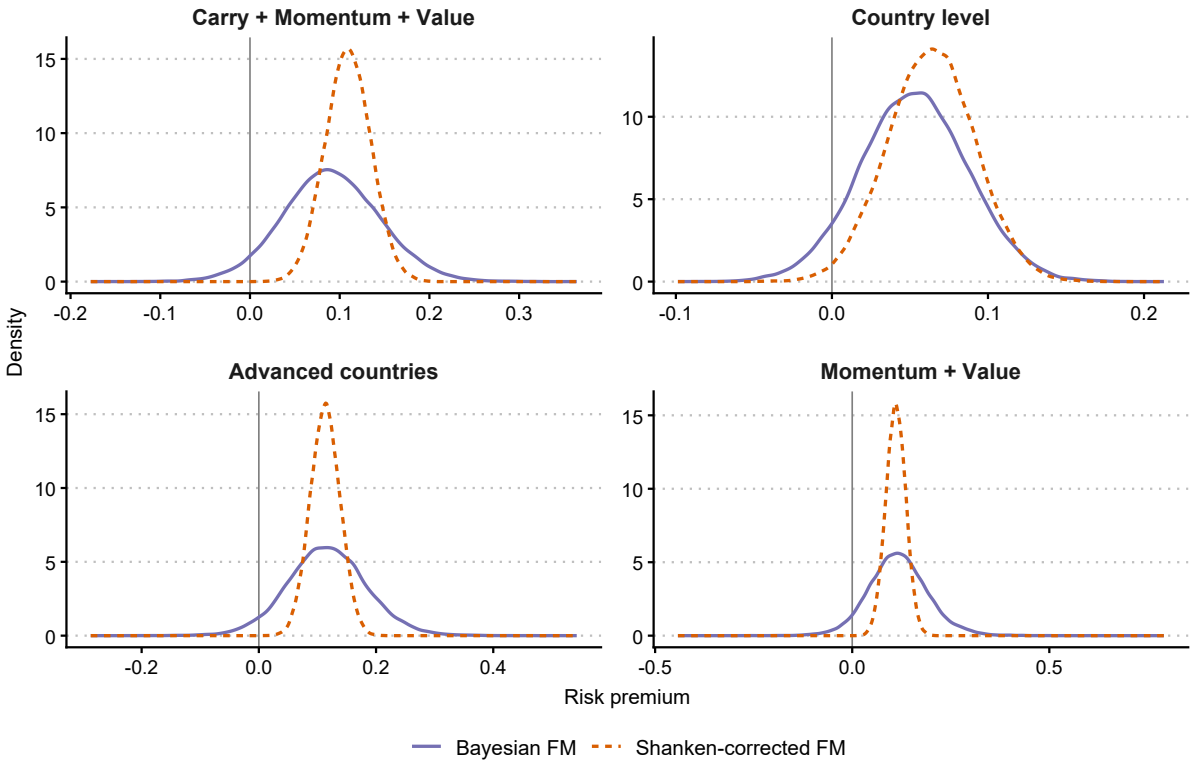


Figure G4: Posterior density of the Carry risk premium. This figure plots the Bayesian Fama–MacBeth posterior distribution of the price of risk of the Carry factor in the specification including all three term-structure factors together with the Carry and Dollar factors. Each panel corresponds to one cross-section of test assets. The comparison density is a normal distribution centered at the two-pass Fama–MacBeth estimate with the Shanken-corrected standard error.

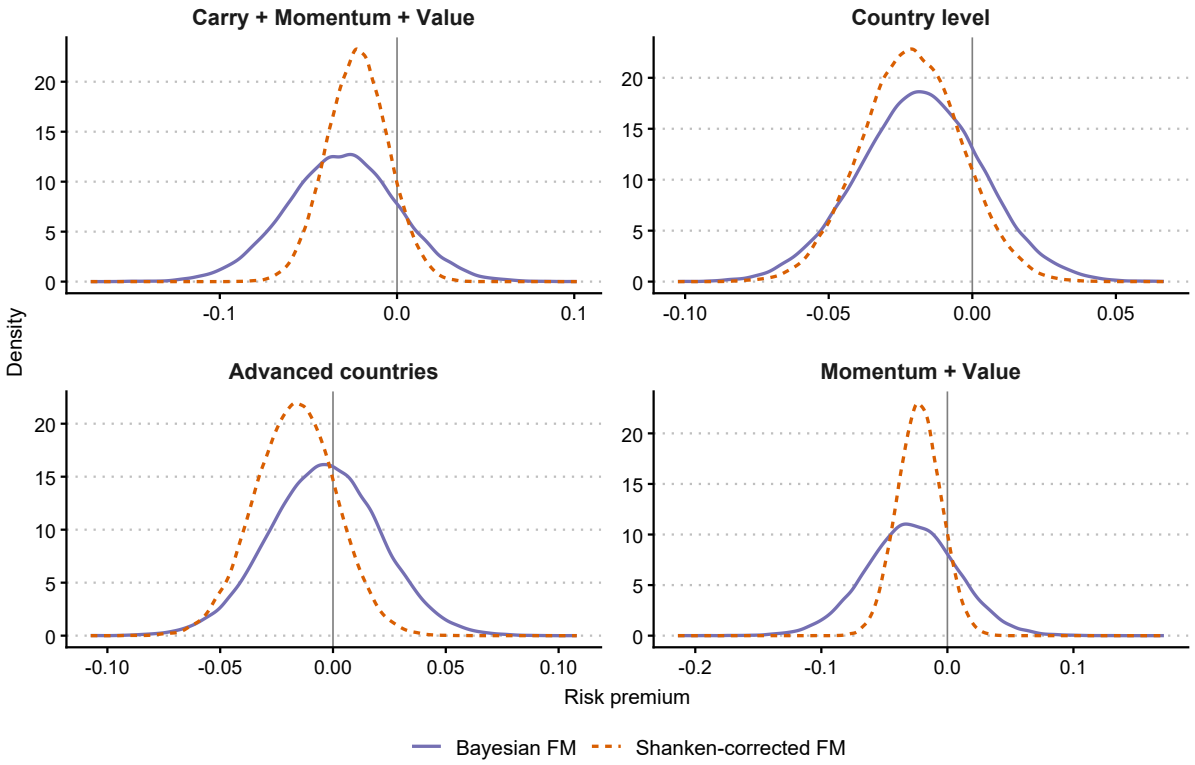


Figure G5: Posterior density of the Dollar risk premium. This figure plots the Bayesian Fama–MacBeth posterior distribution of the price of risk of the Dollar factor in the specification including all three term-structure factors together with the Carry and Dollar factors. Each panel corresponds to one cross-section of test assets. The comparison density is a normal distribution centered at the two-pass Fama–MacBeth estimate with the Shanken-corrected standard error.

H. The global component of the volatility term-structure

The model splits each country's volatility term-structure into a global and a local component, Equation 1, and sorts currencies on the Home curve as a proxy for the global one. Neither step can be checked with US data alone. Using index options from three European markets, I find that the level and slope of the volatility term-structure share a large global component, and that the level and slope currency factors load on a global component built without any US data. The curvature of the US curve does not share in a global component.

The data are the OptionMetrics IvyDB Europe volatility surfaces for the Euro Stoxx 50, the FTSE 100 and the SMI, from January 2002 to February 2023. These markets have no volatility futures over most of the sample, so I build synthetic ones from the surfaces following the VIX methodology. On each day and at each surface maturity, I compute the model-free implied variance from the out-of-the-money smile. I interpolate total variance across maturities with a monotone spline, take the forward variance over the 30 days following each horizon, and use its square root as the synthetic future at horizons of one to nine months. To compare like with like, I apply the same construction to the S&P500 surface for the US curve. It tracks the traded VIX futures closely, with a correlation of 0.99 in levels and between 0.95 and 0.98 in monthly changes at the one- to three-month horizons. Figure H1 to Figure H6 show the European curves. They are noisier than traded futures, since a forward variance amplifies small errors in the underlying surface, most visibly around the 2008 and 2020 crises. For each curve I extract level, slope and curvature with the scaled principal components of the main text and take monthly changes, from February 2002 to February 2023.

Equation 1 is a one-factor model for each bucket, so I estimate it directly. Panel A of Table H1 reports, for each curve, the maximum likelihood estimate of its global share $\varphi_c^2\sigma_k^2/(\varphi_c^2\sigma_k^2+\psi_c^2\sigma_{\text{loc},k}^2)$ in a one-factor model of the four innovations. Two simpler measures bracket it. The R^2 on the first principal component of all four curves overstates the global share, since a curve's own variance enters the component, and four independent series would still give about 0.28. The R^2 on the first principal component of the other three curves understates it, since that component carries their local noise. The level is almost entirely global. The US share is 0.83, with a bootstrap 90% interval of 0.77 to 0.89, and the European shares are 0.88 to 0.95. The slope is mostly global. The US share is 0.55, with an interval of 0.41 to 0.66, against 0.59 to 0.80 for the European curves, and the test rejects a single factor at the 5% level, so slope also has a smaller second common dimension. Curvature is different. The one-factor model fits, but its common factor is

essentially the SMI curve, and the US share is 0.09, with an interval of 0.01 to 0.24.

Panel B asks whether the currency factors load on the global component. I regress each HML factor on the factor score of a one-factor model of the three European curves alone, estimated from 2002 to 2023, so the regressor contains no information from the curve the factors were sorted on. The level and slope factors load on it with coefficients of 4.04 and 6.15 per standard deviation, t-statistics of 1.88 and 3.65, close to their loadings on the US signal itself. For the slope factor, the fit is nearly the same whether the regressor is the US curve or the European one. The covariance between currency returns and the Home curve therefore runs through its global part, which is what the observable beta derived after Equation 17 assumes. The curvature factor loads negatively on the European curvature component, which is dominated by the SMI and correlates negatively with the US curvature innovation in this subsample. Given Panel A, I do not read this as a common factor.

Table H1: Global component of the volatility term-structure. Panel A reports, for each term-structure bucket and curve, the share of the curve's monthly innovation explained by the global component. The one-factor share is the maximum likelihood estimate of the communality in a one-factor model of the four standardized innovations. The upper bound is the R^2 on the first principal component of all four innovations, and the lower bound the R^2 on the first principal component of the other three. The last column reports the p-value of the likelihood ratio test that one factor is sufficient. All four curves are synthetic VIX futures built from the S&P500 (US), Euro Stoxx 50 (EUR), FTSE 100 (GBP) and SMI (CHF) volatility surfaces of IvyDB, with level, slope and curvature extracted with the scaled principal components of the main text. The sample is February 2002 to February 2023, 253 months. Panel B reports regressions of the annualized HML factors, in percent, on the factor score of a one-factor model of the EUR, GBP and CHF innovations (global component) and, for comparison, on the US signal of the main text, both standardized. The sample is December 2010 to February 2023, 147 months. Newey-West t-statistics with three lags are in brackets.

Panel A: share of each curve explained by the global component

		US	EUR	GBP	CHF	One factor, p
Level	Upper bound	0.88	0.95	0.92	0.91	
	One-factor share	0.83	0.95	0.90	0.88	0.04
	Lower bound	0.80	0.91	0.86	0.85	
Slope	Upper bound	0.69	0.76	0.72	0.81	
	One-factor share	0.55	0.71	0.59	0.80	0.02
	Lower bound	0.50	0.58	0.53	0.65	
Curvature	Upper bound	0.21	0.60	0.50	0.78	
	One-factor share	0.09	0.40	0.26	0.96	0.69
	Lower bound	0.07	0.27	0.21	0.48	

Panel B: loading of the currency factors

	Global component		US signal		Correlation
	Coefficient	R^2	Coefficient	R^2	global, US
$fPC1$	4.04 [1.88]	0.03	5.44 [2.67]	0.05	0.91
$fPC2$	6.15 [3.65]	0.07	7.26 [3.92]	0.09	0.69
$fPC3$	−3.25 [−3.19]	0.03	3.24 [1.45]	0.01	−0.38

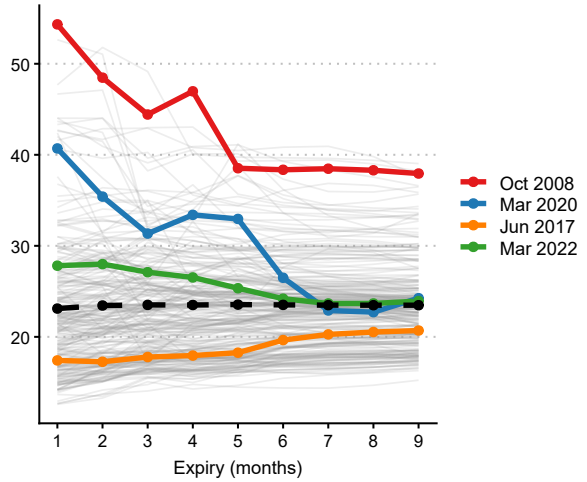


Figure H1: Selected end-of-month synthetic volatility futures term-structures, Euro Stoxx 50. Grey lines are the term-structures at all other month-ends in the sample, the coloured lines are the selected dates and the dashed black line is the sample average.

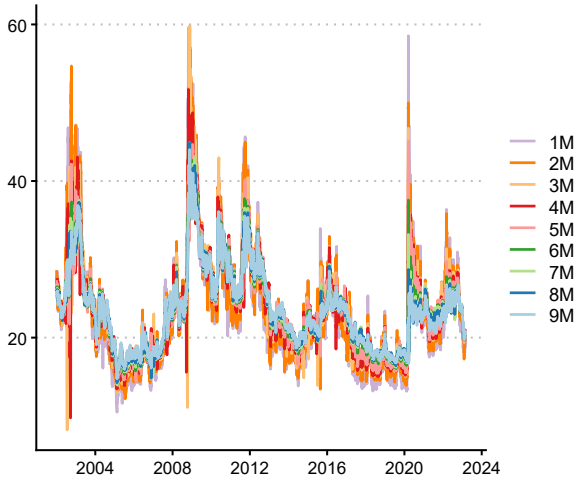


Figure H2: Time-series of synthetic volatility futures, Euro Stoxx 50, daily

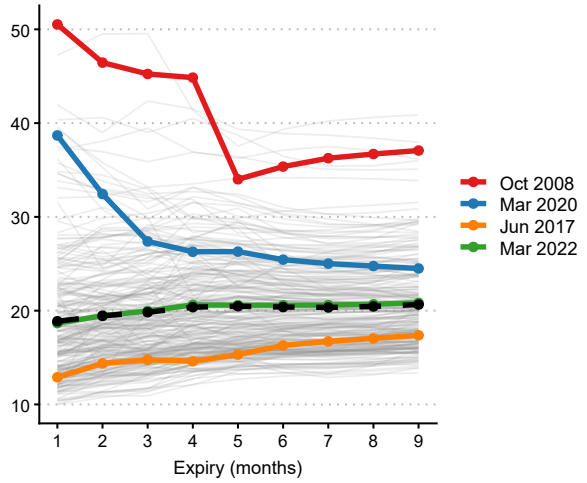


Figure H3: Selected end-of-month synthetic volatility futures term-structures, FTSE 100. Grey lines are the term-structures at all other month-ends in the sample, the coloured lines are the selected dates and the dashed black line is the sample average.

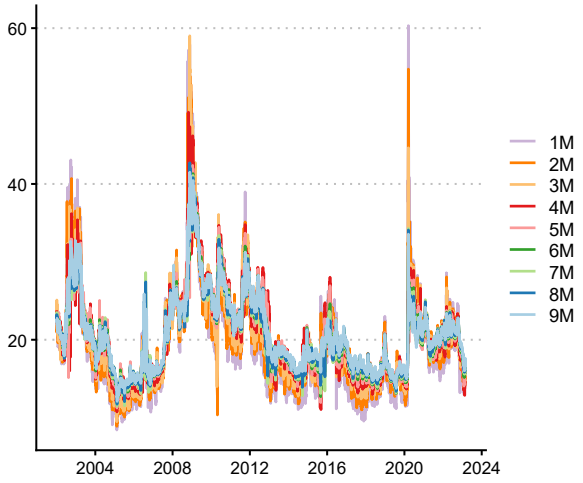


Figure H4: Time-series of synthetic volatility futures, FTSE 100, daily

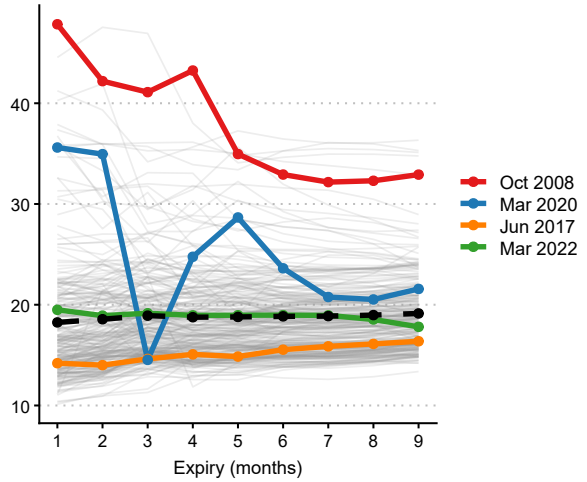


Figure H5: Selected end-of-month synthetic volatility futures term-structures, SMI. Grey lines are the term-structures at all other month-ends in the sample, the coloured lines are the selected dates and the dashed black line is the sample average.

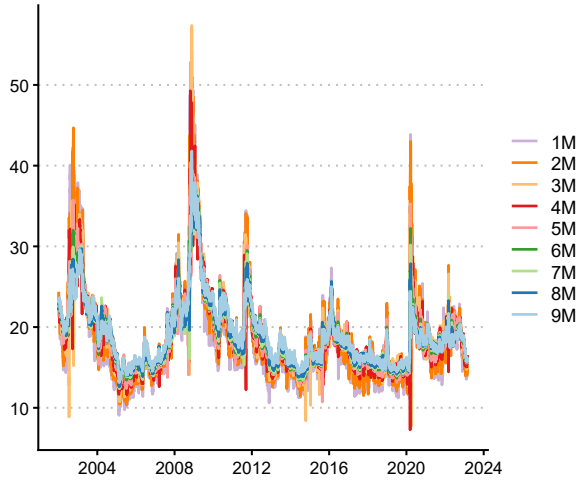


Figure H6: Time-series of synthetic volatility futures, SMI, daily